Chapter 5. Bootrom

Each RP2350 device contains 32 kB of mask ROM: a physically immutable memory resource described in Section 4.1. The RP2350 bootrom is the binary image etched into this ROM that contains the first instructions executed at reset.

The bootrom concepts section (Section 5.1) covers the following topics, which are necessary background for understanding the bootrom features and their implementation:

- Partition tables and their associated flash permissions
- Bootable images, and the block loops that store their metadata
- Versioning for images and partition tables, and A/B versions to support double-buffered upgrades
- Hashing and signing to support secure boot with public key fingerprint in OTP (see also Section 10.1.1 in the security chapter)
- Load maps for bootable images, and packaged binaries which the bootrom loads from flash into RAM according to the image's load map
- Anti-rollback protection to revoke older, compromised versions of software
- Three forms of flash boot:
  - Flash image boot, with a single binary image written directly into flash
  - Flash partition boot, with the boot image selected from the partition table
  - Partition-table-in-image boot, where the boot image is not contained in a partition table, but still embeds a partition table data structure to divide the flash address space
- Boot slots for A/B versions of partition tables
- Flash update boot, a special one-time boot mode that enables version downgrades following an image download
- Try before you buy support for phased upgrades with image self-test
- Address translation for flash images, which provides a consistent runtime address to images regardless of physical storage location
- Automatic architecture switch when attempting to run a RISC-V binary on Arm, or vice versa
- Targeting UF2 downloads to different flash partitions based on their permissions and the UF2 family ID

Besides features mentioned as concepts above, the RP2350 bootrom implements:

- The core 0 initial boot sequence (Section 5.2)
- The core 1 low-power wait and launch protocol (Section 5.3)
- Runtime APIs (Section 5.4) exported through the ROM symbol table, such as flash and OTP programming
- A subset of runtime APIs available to Non-secure code, with permission for each API entry point individually configured by Secure code
- A USB MSC class-compliant bootloader with UF2 support for downloading code/data to flash or RAM (Section 5.5), including support for versioning and A/B partitions
- The USB PICOBOOT interface for advanced operations like OTP programming (Section 5.6) and to support `picotool` or other host side tools
- Support for white-labelling all USB exposed information/identifiers (Section 5.7)
- A UART bootloader providing a minimal shell to load an SRAM binary from a host microcontroller (Section 5.8)

You should read the bootrom concepts section before diving into the features in the list above. RP2350 adds a considerable amount of new functionality compared to the RP2040 bootrom. If you are in a terrible hurry, Section 5.9.5 covers the absolute minimum requirements for a binary to be bootable on RP2350 when secure boot is not enabled.

Bootrom Source Code

All source files for the RP2350 bootrom are available under the terms of the 3-clause BSD licence:

github.com/raspberrypi/pico-bootrom-rp2350

5.1. Bootrom Concepts

Bold type in the following sections introduces a **concept**. This chapter frequently refers back to these concepts.

5.1.1. Secure and Non-secure

This datasheet uses the (capitalised) terms **Secure** and **Non-secure** to refer to the Arm execution states of the same name, defined in the Armv8-M Architecture Reference Manual. The uncapitalised term "secure" has no special meaning.

In some contexts, Secure can also refer to a RISC-V core, usually one running at the Machine privilege level. For example, the low-level flash APIs are exported to Arm Secure code and RISC-V code only, so Secure serves as a shorthand for this type of API.

A **secured RP2350** is a device where secure boot is enabled (Section 5.10.1). This is not the same as the Secure state, since the device may run a mixture of Secure and Non-secure code after completing the secure boot process.

5.1.2. Partition Tables

A **partition table** divides flash into a maximum of 16 distinct regions, known as **partitions**. Each partition defines attributes such as flash permissions for a contiguous range of flash addresses. The `PARTITION_TABLE` data structure describes a partition table, and is an example of a block. Use of partition tables is strictly optional.

Dividing flash into multiple partitions enables you to:

- Store more than one executable image on the device, e.g.:
  - For A/B boot versions (Section 5.1.7)
  - For different architectures (Arm/RISC-V) or Secure/Non-secure
  - For use with a custom bootloader
- Provision space for data, e.g.:
  - Embedded file systems
  - Shared Wi-Fi firmware
  - Application resources
- Provide different security attributes for different regions of flash (Section 5.1.3)
- Target UF2 downloads to different partitions based on family ID (Section 5.1.18), including custom-defined UF2 families specific to your platform

For more information about `PARTITION_TABLE` discovery during flash boot, see Section 5.1.5.2.

Partition tables can be versioned to support A/B upgrades. They can also be hashed and signed for security and integrity purposes. We recommend hashing partition tables to ensure that they haven't been corrupted. This is especially important when using boot slots to update your partition table, since a corrupted partition table with a higher version could be chosen over a non-corrupted partition table with a lower version.

5.1.2.1. Partition Attributes

Each partition specifies **partition attributes** for the flash addresses it encompasses, including:

- Start/end offsets within the logical 32 MB address space of the two flash windows; these offsets are specified in multiples of a flash sector (4 kB)
    - Bootable partitions must reside wholly in the first 16 MB flash window, due to limitations of the address translation hardware
- Access permissions for the partition: read/write for each of Secure (`S`), Non-secure (`NS`) and bootloader (`BL`) access
- Information on which UF2 family IDs may be dropped into the partition via the UF2 bootloader
- An optional 64-bit identifier
- An optional name (a string for human-readable identification)
- Whether to ignore the partition during Arm or RISC-V boot
- Information to group partitions together (see Section 5.1.7 and Section 5.1.18)

Section 5.9.4 documents the full list of partition attributes, along with the `PARTITION_TABLE` binary format.

If there is no partition table, the entirety of flash is considered a single region, with no restricted permissions. Without a partition table, there is no support for custom UF2 family IDs, therefore you must use one of the standard IDs specified in Table 454.

5.1.3. Flash Permissions

One of the roles of the partition tables introduced in Section 5.1.2 is to define **flash permissions**, or simply permissions. The partition table stores one set of permission flags for each partition: all bytes covered by a single partition have the same permissions. The partition table separately defines permissions for **unpartitioned space**: flash addresses which do not match any of partitions defined in the partition table.

Separate read/write permissions are specified for each of Secure (`S`), Non-secure (`NS`) and bootloader (`BL`) access. Bootloader permissions control where UF2s can be written to, and what can be accessed via `picotool` when the device is in BOOTSEL mode.

Because flash permissions may be changed dynamically at runtime, part of the partition table is resident in RAM at runtime. You can modify this table to add permissions for other areas of flash at runtime, without changing the partition table stored in flash itself. There is no bootrom API for this, however the in-memory partition table format is documented, and a pointer is available in the ROM table. The SDK provides APIs to wrap this functionality.

5.1.4. Image Definitions

An **image** is a contiguous data blob which may contain code, or data, or a mixture. An **image definition** is a block of metadata embedded near the start of an image. The metadata stored in the image definition allows the bootrom to recognise valid executable and non-executable images. The `IMAGE_DEF` data structure represents the image definition in a binary format, and is an example of a block.

For executable images, the `IMAGE_DEF` could be considered similar to an ELF header, as it can include image attributes such as architecture/chip, entry-point, load addresses, etc.

All `IMAGE_DEF`s can contain version information and be hashed or signed. Whilst the bootrom only directly boots executable images, it does provide facilities for selecting a valid (possibly signed) data image from one or more partitions on behalf of a user application.

The presence of a valid `IMAGE_DEF` allows the bootrom to discern a valid application in flash from random data. As a result, you must include a valid `IMAGE_DEF` in any executable binary that you intend to boot.

For more information about how the bootrom discovers `IMAGE_DEF`s, see the section on block loops.

For details about the IMAGE_DEF format itself, see Section 5.9.3.

For a description of the minimum requirements for a bootable image, see Section 5.9.5.

5.1.5. Blocks And Block Loops

5.1.5.1. Blocks

IMAGE_DEFs and PARTITION_TABLEs are both examples of **blocks**. A block is a recognisable, self-checking data structure containing one or more distinct data **items**. The type of the first item in a block defines the type of that entire block.

Blocks are backwards and forwards compatible; item types will not be changed in the future in ways that could cause existing code to misinterpret data. Consumers of blocks (including the bootrom) must skip items within the block whose types are currently listed as reserved; encountering reserved item types must not cause a block to fail validation.

To be considered **valid**, a block must have the following properties:

- it must begin with the 4 byte magic header, PICOBIN_BLOCK_MARKER_START (0xffffded3)
- the end of each (variably-sized) item must also be the start of another valid item
- the last item must have type PICOBIN_BLOCK_ITEM_2BS_LAST and specify the correct full length of the block
- it must end with the 4 byte magic footer, PICOBIN_BLOCK_MARKER_END (0xab123579)

The magic header and footer values are chosen to be unlikely to appear in executable Arm and RISC-V code. For more information about the block format, see Section 5.9.1.

Given a region of memory or flash (e.g. a partition), blocks are found by searching the first 4 kB of that given region (for flash boot) or the entire region (for RAM/OTP image boots) for a valid block which is part of a valid block loop.

Currently IMAGE_DEFs and PARTITION_TABLEs are the only types of block used by the RP2350 bootrom, but the block format reserves encoding space for future expansion.

5.1.5.2. Block Loops

A **block loop** is a cyclic linked list of blocks (a linked loop). Each block has a relative pointer to the next block, and the last block must link to the first. A single block can form a block loop by linking back to itself with a relative pointer of 0. The first block in a loop must have the lowest address of all blocks in the loop.

The purpose of a block loop is threefold:

- to discover which blocks belong to the same image without a brute-force search
- to allow metadata to be appended in post-link processing steps
- to detect parts of the binary being overwritten in a way that breaks the loop

For flash image boot the bootrom searches the first 4 kB of flash; the 4 kB size is a compromise between allowing flexibility for different languages' memory layouts, while avoiding scanning too much flash when trying different flash access modes and QSPI clock frequencies. flash partition boot also limits its search to the first 4 kB of the partition.

The search window may be larger, such as a RAM image boot following a UF2 SRAM download, where the search window is all of SRAM. For the fastest boot time, locate the first block as close to the beginning of the binary as possible.

Block loops support multiple blocks because:

- Signing an image duplicates the existing IMAGE_DEF and adds another (bigger) IMAGE_DEF with additional signature information.
- An image may contain multiple IMAGE_DEFs, e.g. with different signing keys.

- Placing a block at both the beginning and end of an image can detect some partial overwrites of the image (for example, due to an overly enthusiastic absolute-addressed UF2 download). The SDK does this by default. Hashing or signing the entire image is more robust, since it detects corruption in the middle of the image.
- A universal binary image might contain code for both Arm and RISC-V, including `IMAGE_DEF`s for both.
- `PARTITION_TABLE`s and `IMAGE_DEF`s are both present in the same block loop in the case of an embedded partition table.

If a block loop contains multiple `IMAGE_DEF`s or multiple `PARTITION_TABLE`s, the winner is generally the last one seen in linked-list order. The exception is the case of two `IMAGE_DEF`s for different architectures (Arm and RISC-V); an `IMAGE_DEF` for the architecture currently executing the bootrom is always preferred over one for a different architecture.

5.1.6. Block Versioning

Any block may contain a **version**. Version information consists of a tuple of either two or three 16-bit values: `(rollback).major.minor`, where the rollback part is optional. An item of type `VERSION` contains the binary data structure which defines the version of a block.

The rollback version may only be specified for `IMAGE_DEF`s and defaults to zero if not present. You cannot specify this version for partition tables. The rollback version can be used on a secured RP2350, where it, along with a current rollback verson number stored in OTP, can prevent installation of older, vulnerable code once a newer version is installed (Section 5.1.11).

The full version number can be used to pick the latest version between two `IMAGE_DEF`s or two `PARTITION_TABLE`s (see Section 5.1.7). Versions compare in lexicographic order:

1. If version *x* has a different rollback version than version *y*, then the greater rollback version determines which version is greater overall
2. Else if version *x* has a different major version than version *y*, then the greater major version determines which version is greater overall
3. Else the minor version determines which of *x* and *y* is greater

See Section 5.9.2.1 for full details on the `VERSION` item in a block.

5.1.7. A/B Versions

A pair of partitions may be grouped into an **A/B** pair. By logically grouping A and B partitions, you can keep the current executable image (or data) in one partition, and write a newer version into the other partition. When you finish writing a new version, you can safely switch to it, reverting to the older version if problems arise. This avoids partially written states that could render RP2350 un-bootable.

- When booting an A/B partition pair, the bootrom typically uses the partition with the higher version. For scenarios where this is not the case, see Section 5.1.16.
- When dragging a UF2 onto the BOOTSEL USB drive, the UF2 targets the *opposite* A/B partition to the one preferred at boot. See Section 5.1.18 for more details.

NOTE

It is also possible to have A/B versions of the partition table. For more information about this advanced topic, see Section 5.1.15.

5.1.8. Hashing and Signing

Any block may be **hashed** or **signed**. A hashed block stores the image hash value (see Section 5.9.2.3). At runtime, the bootrom calculates a hash and compares it to the stored hash to determine if the block is valid. Hashes guard against

corruption of an image, but do not provide any security guarantees.

On a secured RP2350, a hash is not sufficient for an image to be considered valid. All images must have a **signature**: a hash encrypted by a private key, plus metadata (also covered by the hash) describing how the hash was generated. This signature is stored as part of an `IMAGE_DEF` block. An image with a signature in its `IMAGE_DEF` block is called a **signed image**.

NOTE

For background on signatures and boot keys, see the introduction to secure boot in the security chapter (Section 10.1.1).

To verify a signed image, the bootrom decrypts the hash stored in the signature using a *secp256k1* public key. The bootrom also computes its own hash of the image and compares its measured hash value with the one in the signature.

The public key is also stored in the block via a `SIGNATURE` item (see Section 5.9.2.4): this key's (SHA-256) hash must match one of the boot key hashes stored in OTP locations BOOTKEY0_0 onwards. Up to four public keys can be registered in OTP, with the count defined by BOOT_FLAGS1.KEY_VALID and BOOT_FLAGS1.KEY_INVALID. A hash of a key is also referred to as a **key fingerprint**.

The data to be hashed is defined by a `HASH_DEF` item (see Section 5.9.2.2), which indicates the type of hash. It also indicates how much of the block itself is to be hashed. For a signed block, the hash *must* contain all contents of the block up to the final `SIGNATURE` item.

To be useful your hash or signature must cover actual image data in addition to the metadata stored in the block. The block's load map item specifies which data the bootrom hashes during hash or signature verification.

The above discussion mostly applies to `IMAGE_DEF`s. On a secured RP2350 with the BOOT_FLAGS0.SECURE_PARTITION_TABLE flag set, the bootrom also enforces signatures on `PARTITION_TABLE`s.

5.1.9. Load Maps

A **load map** describes regions of the binary and what to do with them before the bootrom runs the binary.

The load map supports:

- Copying portions of the binary from flash to RAM (or to the XIP cache)
- Clearing parts of RAM (either `.bss` clear, or erasing uninitialised memory during secure boot)
- Defining what parts of the binary are included in a hash or signature
- Preventing the flushing of the XIP cache when to keep loaded lines pinned up to the point the binary starts

For full details on the `LOAD_MAP` item type of `IMAGE_DEF` blocks, see Section 5.9.3.2.

When booting a signed binary from flash, it is desirable to load the signed data and code into RAM *before* checking the signature and subsequently executing it. Otherwise, an adversary could replace the flash device in between the signature check and execution, subverting the check. For this reason, the load map also serves as a convenient description of what to include in a hash or signature. The load map itself is covered by the hash or signature, and the entire metadata block is loaded into RAM before processing, so it is not itself subject to this time-of-check versus time-of-use concern.

5.1.10. Packaged Binaries

As described in Section 5.1.9, signed binaries in flash on a secured RP2350 are commonly loaded from flash into RAM, go through signature verification in RAM, and then execute from the verified version in RAM.

A **packaged binary** is a binary stored in flash that runs entirely from RAM. The binary is likely compiled to run from RAM as a RAM-only binary (unfortunately named `no_flash` in SDK parlance), but subsequently post-processed for flash

residence. The bootrom **unpackages** the binary into RAM before execution.

As part of the packaging process, tooling like `picotool` adds a `LOAD_MAP` that tells the bootrom which parts of the flash-resident image it must load into RAM, and where to put them. This tooling may also hash or sign the binary in the same step. In this case, the bootrom hashes the data it loads as it unpackages the binary, as well as relevant metadata such as the `LOAD_MAP` itself. The bootrom compares the resulting hash to the precomputed hash or signature in the `IMAGE_DEF` to verify the unpackaged contents in RAM before running those contents.

Compare this with RP2040, where a flash-resident binary which executes from RAM (a `copy_to_ram` binary in SDK parlance) must begin by executing from flash, then copy itself to RAM before continuing from there. In the RP2040 case, the loader itself (or rather the SDK `crt0`) executes in-place in flash to perform the copy. This makes it impossible to perform any trustworthy level of verification, because the loader itself executes in untrusted memory.

5.1.11. Anti-rollback Protection

Anti-rollback on a secured RP2350 prevents booting an older binary which may have known vulnerabilities. It prevents this even if the binary is correctly signed and meets all other requirements for bootability.

Full `IMAGE_DEF` version information is of the form `(rollback).major.minor`, where the rollback part is optional. If a **rollback version** is present, it is accompanied by a list of OTP rows whose ordered values are used to form a **thermometer** of bits indicating the minimum rollback version that may run on the device.

A thermometer code is a base-1 (unary) number where the integer value is one plus the index of the most-significant set bit. For example, the bit string `00001111` encodes a value of four, and the all-zeroes bit pattern encodes a value of zero. The bootrom uses this encoding because:

- it allows OTP rows containing counters to be incremented, and
- it does not allow them to be decremented

On a secured RP2350, the bootrom compares the rollback version of the `IMAGE_DEF` against the thermometer-coded minimum rollback version stored in OTP. If the `IMAGE_DEF` value is lower, the bootrom refuses to boot the image.

The `IMAGE_DEF` rollback version is covered by the image's signature, thus cannot be modified by an adversary who does not know the signing key. The list of OTP rows which define the chip's minimum rollback version is also stored in the program image, and also covered by the image signature.

The list of OTP rows in the `IMAGE_DEF` must always have at least one bit spare beyond the `IMAGE_DEF`'s rollback version (enforced by `picotool`). As a result, older binaries always contain enough information for the bootrom to detect that the chip's minimum rollback version has been incremented past the rollback version in the `IMAGE_DEF`. You can append more rows to the list on newer binaries to accommodate higher rollback versions without ambiguity.

When an executable image with a non-zero rollback version is successfully booted, its rollback version is written to the OTP thermometer. The BOOT_FLAGS0.ROLLBACK_REQUIRED flag may be used to *require* an `IMAGE_DEF` have a rollback version on a secured RP2350. This flag is set automatically when updating the rollback version in OTP.

NOTE

An `IMAGE_DEF` with a rollback version of 0 will not automatically set the BOOT_FLAGS0.ROLLBACK_REQUIRED flag, so it is recommended that the minimum rollback version used is 1, unless the BOOT_FLAGS0.ROLLBACK_REQUIRED flag is manually set during provisioning.

5.1.12. Flash Image Boot

RP2350 is designed primarily to run code from a QSPI flash device, either in-package or soldered separately to the circuit board. Code runs either in-place in flash, or in SRAM after being loaded from flash. **Flash boot** is the process of discovering that code and preparing to run it. **Flash image boot** uses a program binary stored directly in flash rather than in a flash partition. Flash image boot requires the bootrom to discover a block loop starting within the first 4 kB of flash which contains a valid `IMAGE_DEF` (and no `PARTITION_TABLE`).

Flash image boot has no partition table, so it cannot be used with A/B version checking, which requires separate A/B partitions. The IMAGE_DEF will boot if it is valid (which includes requiring a signature on a secured RP2350).

For the non-signed case, the IMAGE_DEF can be as small as a 20-bytes; see Section 5.9.5.

TIP

A more complicated version of this scenario stores multiple IMAGE_DEFs in the block loop. In this case, the last IMAGE_DEF for the current architecture is booted, if valid. You can use this to implement universal binaries for various supported architectures, or to include multiple signatures for targeting devices with different keys.

5.1.13. Flash Partition Boot

If a PARTITION_TABLE, but no IMAGE_DEF, is found in the valid block loop that starts within the first 4 kB of flash, and it is valid (including signature if necessary on a secured RP2350), the bootrom searches that partition table's partitions for an executable image to boot. This process, when successful, is referred to as **flash partition boot**.

The partitions are searched in order, skipping those marked as ignored for the current architecture. The bootrom ignores partitions as an optimisation, or to prevent automatic architecture switching.

If the partition is not part of an A/B pair, the first 4 kB is searched for the start of a valid block loop. If a valid block loop is found, and it contains an executable image with a valid (including signature on a secured RP2350) IMAGE_DEF, then that executable image is chosen for boot.

If the partition is the A partition of an A/B pair, the bootrom searches both partitions as described above. If both partitions result in a bootable IMAGE_DEF, the IMAGE_DEF with the higher version number is chosen. Otherwise, the valid IMAGE_DEF is chosen. There are some exceptions to this rule in advanced scenarios; see Section 5.1.16 and Section 5.1.17 for details.

5.1.14. Partition-Table-in-Image Boot

If both a PARTITION_TABLE and an IMAGE_DEF block are found in the valid block loop that starts within the first 4 kB of flash, a third type of flash boot takes place. The IMAGE_DEF and PARTITION_TABLE must only be recognised, not necessarily valid or correctly signed. This stipulation prevents a causality loop.

This is known as **partition-table-in-image** boot, since the application contains the partition table (instead of vice versa). This partition table is referred to as an **embedded partition table**.

The PARTITION_TABLE is loaded as the current partition table, and the IMAGE_DEF is launched directly. The table defined by the PARTITION_TABLE is *not* searched for IMAGE_DEFs to boot.

The following common cases might use this scenario:

- You are only using the PARTITION_TABLE for flash permissions. You want to load that partition table, then boot as normal.
- The IMAGE_DEF contains a small bootloader stored alongside the partition table. In this case, the partition table will once again be loaded, and the associated image entered. The entered image will then likely pick a partition from the partition table, and launch an image from there itself.

5.1.15. Flash Boot Slots

The previous sections within this chapter discuss block loops starting within the first 4 kB of flash. Such a block loop contained either an IMAGE_DEF, a partition table (searched for IMAGE_DEFs), or an IMAGE_DEF *and* a PARTITION_TABLE (not searched).

All the previously mentioned cases discovered their block loop in **slot 0**. Under certain circumstances, the neighbouring **slot 1** is also searched.

Slot 0 starts at the beginning of flash, and has a size of n × 4 kB sectors. Slot 1 has the same size and follows immediately after slot 0. The value of n defaults to 1. Both slots are 4 kB in size, but you can override this value by specifying a value in FLASH_PARTITION_SLOT_SIZE and then setting BOOT_FLAGS0.OVERRIDE_FLASH_PARTITION_SLOT_SIZE.

Similarly to how a choice can be made between IMAGE_DEFs in A/B partitions, a choice can be made between A/B PARTITION_TABLEs via the two **boot slots**. This allows for versioning partition tables, targeted drag and drop of UF2s (Section 5.1.18) containing partition tables, etc. similar to the process used for images.

Slot 1 is only of use when potentially using partition tables. In the simple case of an IMAGE_DEF and no PARTITION_TABLE found in a block loop starting in slot 0, that image likely actually overlays the space where slot 1 would be, but in any case, slot 1 is ignored since there is no PARTITION_TABLE.

If slot 0 contains a PARTITION_TABLE or does not contain an IMAGE_DEF (including nothing/garbage in slot 0), slot 1 can be considered. As an optimisation, in the former case, the scanning of slot 1 can be prevented by setting the singleton flag in the PARTITION_TABLE.

NOTE

When IMAGE_DEFs are also present in the slots, the PARTITION_TABLE's VERSION item determines which of slot 0 and slot 1 to use. The IMAGE_DEF metadata is ignored for the purpose of version comparison.

5.1.16. Flash Update Boot and Version Downgrade

Normally the choice of slot 0 versus slot 1, and partition A versus partition B, is made based on the version of the valid PARTITION_TABLE or IMAGE_DEF in those slots or partitions respectively. The greater of the two versions wins.

It is however perfectly valid to downgrade to a lower-versioned IMAGE_DEF when using A/B partitions, provided this does not violate anti-rollback rules on a secured RP2350.

Downloading the new image (and its IMAGE_DEF) into the non-currently-booting partition and doing a normal reboot will not work in this case, as the newly downloaded image has a lower version.

For this purpose, you can enable a **flash update boot** boot by passing the FLASH_UPDATE boot type constant flag through the watchdog scratch registers and a pointer to the start of the region of flash that has just been updated.

The bootrom automatically performs a flash update boot after programming a flash UF2 written to the USB Mass Storage drive. You can also invoke a flash update boot programmatically via the reboot() API (see Section 5.4.8.24).

The flash address range passed through the reboot parameters is treated specially during a flash update boot. A PARTITION_TABLE in a slot, or IMAGE_DEF in a partition, will be chosen for boot irrespective of version, if the start of the region is the start of the respective slot or partition.

In order for the downgrade to persist, the first sector of the previously booting slot or partition must be erased so that the newly installed PARTITION_TABLE or IMAGE_DEF will continue to be chosen on subsequent boots. This erase is performed as follows during a FLASH_UPDATE boot.

1. When a PARTITION_TABLE is valid (and correctly signed if necessary) and its slot is chosen for boot, the first sector of the other slot is erased.
2. When a valid (and correctly signed if necessary) IMAGE_DEF is launched, the first sector of the other image is erased.
3. On explicit request by the image, after it is launched, the first sector of the other image is erased. This is an alternative to the standard behaviour in the previous bullet, and is selected by a special "Try Before You Buy" flag in the IMAGE_DEF. For more information about this feature, see Section 5.1.17.

NOTE

Flash update and version downgrade have no effect when using a single slot, or standalone (non A/B) partitions.

5.1.17. Try Before You Buy

Try before you buy (abbreviated **TBYB**) is an `IMAGE_DEF`-only feature that allows for a completely safe cycle of version upgrade:

1. An executable image is running from say partition B.
2. A new image is downloaded into partition A.
3. On download completion, a `FLASH_UPDATE` reboot is performed for the newly updated partition A.
4. The bootrom will preferentially try to boot partition A (due to the flash update). Note that a non TBYB image will always be chosen over a TBYB image in A/B partitions during a normal non-`FLASH_UPDATE` boot.
    - If the new image fails validation/signature then the old image in partition B will be used on subsequent (non-`FLASH_UPDATE`) boots, recovering from the failed upgrade.
5. If the new image is valid (and correctly signed if necessary), it is entered under a watchdog timer, and has 16.7 seconds to mark itself OK via the explicit_buy() function.
    - If the image calls back, the first 4 kB sector of the other partition (containing image B) is erased, and the TBYB flag of the current image is cleared, so that A becomes the preferred partition for subsequent boots.
    - If the image does not call back within the allotted time, then the system reboots, and will continue to boot partition B (containing the original image) as partition A is still marked as TBYB image.

The erase of the first sector of the opposite partition in the A/B pair severs its image's block loop, rendering it unbootable. This ensures the tentative image booted under TBYB becomes the preferred boot image going forward, even if the opposite image had a higher version.

The watchdog timeout is fixed at 16.7 seconds (24-bit count on a 1-microsecond timebase). This can be shortened after entering the target image, for example if it only needs a few hundred milliseconds for its self-test routine. It can also be extended by reloading the watchdog counter, at the risk of getting stuck in the tentative image if it fails in a way that repeatedly reloads the watchdog.

5.1.18. UF2 Targeting

Section 5.5 describes the USB Mass Storage drive, and the ability to download UF2 files to that drive to store and/or execute code/data on the RP2350.

Since RP2350 supports multiple processor architectures, and partition tables with multiple partitions, some information on the device must be used to determine what to do with a flash-addressed UF2. Depending on the context, the flash addresses in the UF2 may be absolute flash storage addresses (as was always the case on RP2040), or runtime addresses of code and data within a flash partition. **UF2 targeting** refers to the rules the bootrom applies to interpret flash addresses in a UF2 file.

UF2 supports a 32-bit family ID embedded in the file. This enables the device to recognise firmware that targets it specifically, as opposed to firmware intended for some other device. The RP2350 bootrom recognises some standard UF2 family IDs (`rp2040`, `rp2350-arm-s`, `rp2350-arm-ns`, `rp2350-riscv`, `data` and `absolute`) defined in Table 454. You may define your own family IDs in the partition table for more refined targeting.

The UF2 family ID is used as follows:

1. A UF2 with the `absolute` family ID is downloaded without regard to partition boundaries. A partition table (if present) or OTP configuration define whether `absolute` family ID downloads are allowed. The default factory settings do allow for `absolute` family ID downloads.

2. If there is no partition table, then the `data`, `rp2350-arm-s` (if Arm is enabled) and `rp2350-riscv` (if RISC-V is enabled) family IDs are allowed by default. The UF2 is always downloaded to the start of flash.
3. If there is a partition table, then non-`absolute` family IDs target a single partition under the control of the partition table:
    a. A UF2 will not be downloaded to a partition that doesn't have `BL`-write flash permissions
    b. Each partition lists which family IDs it accepts (both RP2350 standard and user defined)
    c. With A/B partitions; the A partition indicates the family IDs supported, and the UF2 goes to the partition that isn't the currently booting one (strictly the one that won't be the one chosen if the device were rebooted now).
    d. Further refinement with A/B is allowed to support secondary A/B partitions containing data/executables used (owned) by the main partitions; see Section 5.1.18.1 for detailed information.

For details of the exact rules used when picking a UF2 target partition, see Section 5.5.3.

NOTE

UF2 family ids are used for partition targeting when copying UF2s to the USB drive, or when using `picotool load -p`. When using `picotool load` without the `-p` flag images can be written anywhere in flash that has `BL`-write permissions.

5.1.18.1. Owned Partitions

An executable may require data from another partition (e.g. Wi-Fi firmware). When the main executable is stored in A/B partitions, for safe upgrades, it may be desirable to associate two other partitions C and D with the primary A and B partitions, such that:

- the data in partition C is used for executable in partition A, and
- the data in partition D is used for the executable in partition B.

In this scenario A is marked as the **owner** of C in the partition table, and C is A's **owned partition**. This affects UF2 image downloads which (due to their UF2 family ID) target partitions C and D.

When a UF2 download targets the C/D partition pair, the bootrom checks the state of the A and B owning partitions to determine which of the owned partitions (C and D) receives the download. By default:

- If B would be the target partition for a UF2 with an A/B-compatible family ID, then D is the target for a UF2 with the C/D compatible family ID.
- Conversely, when A is the target partition for A/B downloads, C is the target partition for C/D downloads.

The `FLAGS_UF2_DOWNLOAD_AB_NON_BOOTABLE_OWNER_AFFINITY` flag in the partition table reverses this mapping.

5.1.19. Address Translation

RP2040 required images to be stored at the beginning of flash (`0x10000000`). RP2350 supports storing executable images in a partitions at arbitrary locations, to support more robust upgrade cycles via A/B versions, among other uses. This presents the issue that the address an executable is linked at, and therefore the binary contents of the image, would have to depend on the address it is stored at. This can be worked around to an extent with position-independent code, at cost to code size and performance.

RP2350 avoids this pitfall with hardware and bootrom support for **address translation**. An image stored at any 4 kB-aligned location in flash can appear at flash address `0x10000000` at runtime. The SDK continues to assume an image base of `0x10000000` by default.

When launching an image from a partition, the bootrom initialises QMI registers ATRANS0 through ATRANS3 to map a flash **runtime address** of `0x10000000` (by default) to the flash **storage address** of the start of the partition. It sets the size of the mapped region to the size of the partition, with a maximum of 16 MB. Accessing flash addresses beyond the size of the booted partition (but below the `0x11000000` chip select watermark) returns a bus fault.

Mapping to a runtime address of 0x10000000 is the default behaviour, but you may choose a different address, with some restrictions. The bootrom allows for runtime address values of 0x10000000, 0x10400000, 0x10800000, and 0x10c00000 for the beginning of the mapped regions, with the choice specified by the IMAGE_DEF. You must link your binary to run at the correct, higher base address. One example where this is useful is an application which runs from a high flash address, and then remains mapped at this high address when launching a second application running at address 0x10000000. You might use this for a Secure image providing services to a Non-secure client image.

This custom address translation is enabled by a negative ROLLING_WINDOW_DELTA value (see Section 5.9.3.5). The above four runtime addresses translate to a ROLLING_WINDOW_DELTA of 0, -0x400000, -0x800000, or -0xc00000, which are the only supported non-positive values. The delta indicates the offset into the image which appears at a runtime address 0x10000000: for negative values this indicates the runtime flash address space starts *before* the start of the image.

Positive values are also useful, for example when prepending data to an already-linked image as a post-processing step. Positive deltas must be multiples of 4 kB. For example, a ROLLING_WINDOW_DELTA of 0x1000 will set up address translation such that the image data starting at offset 0x1000 is mapped to 0x10000000 at runtime, lopping off the first 4 kB of the image. The first 4 kB is inaccessible except via the untranslated XIP window (which defaults to Secure access only).

NOTE

Because address translation within the 0x100000000 → 0x11000000 and 0x11000000 → 0x12000000 windows is independent, it is only possible to boot from partitions which are entirely contained within the first 16 MB of flash.

This address translation is performed by hardware in the QMI. For more information, see Section 12.14.4.

5.1.20. Automatic Architecture Switching

If the bootrom encounters a valid and correctly signed IMAGE_DEF for the non-current architecture (i.e. RISC-V when booted in Arm mode, or Arm when booted in RISC-V), it performs an **automatic architecture switch**. The bootrom initiates a reboot into the correct architecture for the binary it discovered, which then boots successfully on the second attempt. Information passed in watchdog scratch registers (such as a RAM image boot type) is retained, so that the second boot makes the same decisions as the first, and arrives at the same preferred image to boot.

This happens only when:

- The architecture to be switched to is available according to OTP critical flags
- The architecture switch feature is not disabled by the BOOT_FLAGS0.DISABLE_AUTO_SWITCH_ARCH flag
- The bootrom found no valid binary for the *current* architecture before finding one for the *other* architecture

TIP

When storing executable images for both architectures in flash, it's usually preferable to boot an image for the *current* architecture. To do this, keep the images in different partitions, marking the partition for Arm as ignored during boot under RISC-V and vice versa. This avoids always picking the image in the first partition and auto-switching to run it under the other architecture.

For hardware support details for architecture switching, see Section 3.9.

5.2. Processor-Controlled Boot Sequence

The bootrom contains the first instructions the processors execute following a reset. Both processors enter the bootrom at the same time, and in the same location, but the boot sequence runs mostly on core 0.

Core 1 redirects very early in the boot sequence to a low-power state where it waits to be launched, after boot, by user software on core 0. If core 1 is unused, it remains in this low-power state.

Source Code Reference

The sequence described in this section is implemented on Arm by the source files `arm8_bootrom_rt0.S` and `varm_boot_path.c` in the bootrom source code repository. RISC-V cores instead begin from `riscv_bootrom_rt0.S`, but share the boot path implementation with Arm.

5.2.1. Boot Outcomes

The bootrom decides the **boot outcome** based on the following system state:

- The contents of the attached QSPI memory device on chip select 0, if any
- The contents of POWMAN registers BOOT0 through BOOT3
- The contents of watchdog registers SCRATCH4 through SCRATCH7
- The contents of OTP, particularly CRIT1, BOOT_FLAGS0 and BOOT_FLAGS1
- The QSPI `CSn` pin being driven low externally (to select BOOTSEL)
- The QSPI `SD1` pin being driven high externally (to select UART boot in BOOTSEL mode)

Based on these, the outcome of the boot sequence may be to:

- Call code via a **vector** specified in `SCRATCH` or `BOOT` registers prior to the most recent reboot
  - e.g. into code retained in RAM following a power-up from a low-power state.
- Run an image from external flash
  - either in-place, or loaded into RAM during the boot sequence
  - in-package flash on RP2354 is external for boot purposes: it is a separate silicon die, and the RP2350 die does not implicitly trust it
- Run an image preloaded into SRAM (distinct from the vector case)
- Load and run an image from OTP into SRAM
- Enter the USB bootloader
- Enter the UART bootloader
- Perform a one-shot operation requested via the reboot() API, such as a flash update boot
  - this may be requested by the user, or by the UART or UF2 bootloaders
- Refuse to boot, due to lack of suitable images and the UART and USB bootloaders being disabled via OTP

This section makes no distinction between the different types of flash boot (flash image boot, flash partition boot and flash partition-table-in-image boot). Likewise, it does not distinguish these types from packaged binaries, which are loaded into RAM at boot time, because these are just flash binaries with a special load map. This section just describes the sequence of decisions the bootrom makes to decide which medium to boot from.

5.2.2. Sequence

This section enumerates the steps of the processor-controlled boot sequence for Arm processors. There are some minor differences on Arm versus RISC-V, which are discussed in Section 5.2.2.2.

A **valid** image in Table 450 refers to one which contains a valid block loop, with one of those blocks being a valid image definition. On a secured RP2350 this image must be (correctly) signed, and must meet all other security requirements such as minimum rollback version.

Shaded cells in the Action column of Table 450 indicate a boot outcome as described in Section 5.2.1. Other cells are transitory states which continue through the sequence. Both cores start the sequence at `Entry`.

The main sequential steps in Table 450 are:

- Entry
- Core 1 Wait
- Boot Path Start
- Await Rescue
- Generate Boot Random
- Check POWMAN Vector
- Check Watchdog Vector
- Prepare for Image Boot
- Try RAM Image Boot
- Check BOOTSEL
- Try OTP Boot
- Try Flash Boot
- Prepare for BOOTSEL
- Enter USB Boot
- Enter UART Boot
- Boot Failure

See also Section 5.2.2.1 for a summary of Table 450 in pseudocode form.

Table 450. Processor-controlled boot sequence

Condition (If…)	Action (Then…)
Step: `Entry`	
Always	Check core number in CPUID or MHARTID.
Running on core 0	Clear boot RAM (except for core 1 region and the `always` region).
	Go to `Boot Path Start`.
Running on core 1	Go to `Core 1 Wait`.
Step: `Core 1 Wait`	
Always	Wait for RCP salt register to be marked valid by core 0.
	Wait for core 0 to provide an entry point through Secure SIO FIFO, using the protocol described in Section 5.3.
	Outcome: Set Secure main `sp` and `VTOR`, then jump into the entry point provided.
Step: `Boot Path Start`	
Always	Check rescue flag, CHIP_RESET.RESCUE_FLAG
Rescue flag set	Go to `Await Rescue`.
Rescue flag clear	Go to `Generate Boot Random`.
Step: `Await Rescue`	
Always	Clear the rescue flag to acknowledge the request.
	Outcome: Halt in place. The debugger attaches to give the processor further instruction.
Step: `Generate Boot Random`	

Condition (If…)	Action (Then…)
Always	Sample TRNG ROSC into the SHA-256 to generate a 256-bit per-boot random number.
	Store 128 bits in boot RAM for retrieval by get_sys_info(), and distribute the remainder to the RCP salt registers.
	Go to `Check POWMAN Vector`.
Step: `Check POWMAN Vector`	
Always	Read BOOT0 through BOOT3 to determine requested boot type.
Boot type parity is valid	Clear BOOT0 so this is ignored on subsequent boots.
A `BOOTDIS` flag is set	Go to `Check Watchdog Vector`.
Boot type is `VECTOR`	Outcome: Set Secure main `sp`, then call into the entry point provided.
(Return from `VECTOR`)	Go to `Check Watchdog Vector`.
Other or invalid boot type	Go to `Check Watchdog Vector`.
Step: `Check Watchdog Vector`	
Always	Read watchdog SCRATCH4 through SCRATCH7 to determine requested boot type.
Boot type parity is valid	Clear SCRATCH4 so this is ignored on subsequent boots.
Boot type is `BOOTSEL`	Make note for later: equivalent to selecting BOOTSEL by driving QSPI `CSn` low.
A `BOOTDIS` flag is set	Go to `Prepare for Image Boot` (so `BOOTSEL` is the only permitted type when the OTP BOOTDIS.NOW or POWMAN BOOTDIS.NOW flag is set).
Boot type is `VECTOR`	Outcome: Set Secure main `sp`, then call into the entry point provided.
(Return from `VECTOR`)	Go to `Prepare for Image Boot`.
Boot type is `RAM_IMAGE`	Make note for later: this requests a scan of a RAM region for a preloaded image.
Boot type is `FLASH_UPDATE`	Make note for later: modifies some flash boot behaviour, as described in Section 5.1.16.
Always	Go to `Prepare for Image Boot`.
Step: `Prepare for Image Boot`	
Always	Clear `BOOTDIS` flags (OTP BOOTDIS.NOW and POWMAN BOOTDIS.NOW).
	Power up SRAM0 and SRAM1 power domains (XIP RAM domain is already powered).
	Reset all PADS and IO registers, and remove isolation from QSPI pads.
	Release USB reset and clear upper 3 kB of USB RAM (for search workspace).
	Go to `Try RAM Image Boot`.
Step: `Try RAM Image Boot`	
Watchdog type is not `RAM_IMAGE`	Go to `Check BOOTSEL`.
BOOT_FLAGS0.DISABLE_SRAM_WINDOW_BOOT is set	Go to `Check BOOTSEL`.
Otherwise	Scan indicated RAM address range for a valid image (base in SCRATCH2, length in SCRATCH3). This is used to boot into a RAM image downloaded via UF2, for example.
RAM image is valid	Outcome: Enter RAM image in the manner specified by its image definition.
No valid image	Go to `Prepare for Bootsel` (skipping flash and OTP boot).
Step: `Check BOOTSEL`	

Condition (If…)	Action (Then…)
Always	Check BOOTSEL request: QSPI `CSn` is low (BOOTSEL button), watchdog type is `BOOTSEL`, or RUN pin double-tap was detected (enabled by BOOT_FLAGS1.DOUBLE_TAP).
BOOTSEL requested	Go to `Prepare for BOOTSEL` (skipping flash and OTP boot).
Otherwise	Go to `Try OTP Boot`.
Step: `Try OTP Boot`	
Always	Check BOOT_FLAGS0.DISABLE_OTP_BOOT and BOOT_FLAGS0.ENABLE_OTP_BOOT (the disable takes precedence).
OTP boot disabled	Go to `Try Flash Boot`.
OTP boot enabled	Load data from OTPBOOT_SRC (in OTP) to OTPBOOT_DST0/OTPBOOT_DST1 (in SRAM), with the length specified by OTPBOOT_LEN.
	Check validity of the image in-place in SRAM.
Image is valid	Outcome: Enter RAM image in the manner specified by its image definition.
No valid image	Go to `Try Flash Boot`.
Step: `Try Flash Boot`	
Flash boot disabled by BOOT_FLAGS0	Go to `Prepare for BOOTSEL`.
Always	Issue XIP exit sequence to chip select 0.
FLASH_DEVINFO has GPIO and size for chip select 1	Issue XIP exit sequence to chip select 1.
Always	Scan flash for a valid image (potentially in a partition) with a range of instructions (EBh, BBh, 0Bh, 03h) and SCK divisors (3 to 24)
Valid image found	Outcome: Enter flash image in the manner specified by its image definition. This may including loading some flash contents into RAM. Save the current flash read mode as an XIP setup function at the base of boot RAM, which can be called later to restore the current mode (e.g. following a serial programming operation).
No valid image	Go to `Prepare for BOOTSEL`.
Step: `Prepare for BOOTSEL`	
Always	Erase SRAM0 through SRAM9, XIP cache and USB RAM to all-zeroes before relinquishing memory and peripherals to Non-secure.
	Enable XOSC and configure PLL for 48 MHz, according to BOOTSEL_XOSC_CFG and BOOTSEL_PLL_CFG (default is to expect a 12 MHz crystal).
	Check QSPI `SD1` pin (with default pull-down resistor) for UART/USB boot select.
	Scan flash for a partition table (always using an 03h serial read command with an `SCK` divisor of 6). The USB bootloader may download UF2s to different flash addresses depending on partitions and their contents.
	Advance all OTP soft locks to the `BL` state from OTP, if more restrictive than their `S` state.
QSPI `SD1` pulled low	Go to `Enter USB Boot`.
QSPI `SD1` driven high	Go to `Enter UART boot`.
Step: `Enter USB Boot`	

Condition (If…)	Action (Then…)
Always	Check BOOT_FLAGS0.DISABLE_BOOTSEL_USB_PICOBOOT_IFC and BOOT_FLAGS0.DISABLE_BOOTSEL_USB_MSD_IFC to see which USB interfaces are permitted.
Both USB interfaces disabled	Go to `Boot Failure`.
Otherwise	Outcome: Enter USB bootloader. The bootloader reboots if a UF2 image is downloaded, marking a `FLASH_UPDATE` in the watchdog scratch registers if applicable, and the boot path restarts from `Entry`. Valid images boot; invalid images usually end up back in the USB bootloader.
Step: `Enter UART Boot`	
Always	Check BOOT_FLAGS0.DISABLE_BOOTSEL_UART_BOOT to see if UART boot is permitted.
UART boot disabled	Go to `Boot Failure`.
Otherwise	Outcome: Enter UART bootloader. The bootloader reboots once an image has been downloaded, with a `RAM_IMAGE` boot type, and the boot path restarts from `Entry`. Valid images boot; invalid images usually end up back in the UART bootloader.
Step: `Boot Failure`	
Always	Outcome: Take no further action. No valid boot image was discovered, and the selected BOOTSEL interface was disabled. Attach the debugger to give the processor further instruction. See the boot reason in boot RAM for diagnostics on why the boot failed.

TIP

The bootrom internally refers to BOOTSEL mode as NSBOOT, because the USB and UART bootloaders run in the Non-secure state under Arm. This chapter may also occasionally refer to BOOTSEL as NSBOOT.

5.2.2.1. Boot Sequence Pseudocode

The following pseudocode summarises Table 450.

```
if (powman_vector_valid && powman_reboot_mode_is_pcsp) {
    // This call may return and continue the boot path
    if (correct_arch) powman_vector_pc(); else hang();
}
if (watchdog_vector_valid) {
    // Make note of RAM_IMAGE, FLASH_UPDATE, BOOTSEL reboot types
    check_special_reboot_mode();
    if (watchdog_reboot_mode_is_pcsp) {
        // This call may return and continue the boot path
        if (correct_arch) watchdog_vector_pc(); else hang();
    }
}

// RAM image window specified by watchdog_scratch, e.g. after a UF2 RAM
// download: either execute the RAM image or fall back to UART/USB boot.
if (watchdog_reboot_mode_is_ram_image && !ram_boot_disabled_in_otp) {
    // This only returns if there is no valid RAM image to enter.
    // You can't return from the RAM image.
    try_boot_ram_image(ram_image_window);
```

```
} else {
    // Otherwise try OTP and flash boot (unless there is a request to skip)
    skip_flash_and_otp_boot =
        bootsel_button_pressed() ||
        watchdog_reboot_mode_is_bootsel ||
        (double_tap_enabled_in_otp() && double_run_reset_detected());

    if (!skip_flash_and_otp_boot) {
        if (otp_boot_enabled_in_otp && !otp_boot_disabled_in_otp) {
            // This only returns if there is no valid OTP image to enter.
            // You can't return from the OTP image.
            try_otp_boot();
        }
        if (!flash_boot_disabled_in_otp) {
            // This only returns if there is no valid flash image to enter.
            // You can't return from the flash image.
            try_flash_boot();
        }
    }
}
// Failed to find an image, so drop down into one of the bootloaders
if (sd1_high_select_uart) {
    // Does not return except via reboot
    if (nsboot_uart_disabled) hang(); else nsboot(uart);
} else {
    // Does not return except via reboot
    if (nsboot_usb_disabled) hang(); else nsboot(usb);
}
```

5.2.2.2. Differences between Arm and RISC-V

The boot sequence outlined in Table 450 has the following differences on RISC-V:

- Secure boot is not supported (from any image source).
- Anti-rollback checking is not supported as it applies only to secure boot.
- Additional security checks such as the use of the RCP to validate booleans are disabled.
- The UART and USB bootloaders continue to run in Machine mode, rather than transitioning from the Arm Secure to Non-secure state, meaning there is no hardware-enforced security boundary between these boot phases.
- The XIP setup function written to boot RAM on a successful flash boot contains RISC-V rather than Arm instructions.

5.2.3. POWMAN Boot Vector

POWMAN contains scratch registers similar to the watchdog scratch registers, which persist over power-down of the switched core power domain, in addition to most system resets. These registers allow users to install their own boot handler, and divert control away from the main boot sequence on non-POR/BOR resets. It recognises the following values written to BOOT0 through BOOT3:

- BOOT0: magic number `0xb007c0d3`
- BOOT1: Entry point XORed with magic `-0xb007c0d3` (`0x4ff83f2d`)
- BOOT2: Stack pointer
- BOOT3: Entry point

Use this to vector into code preloaded in RAM which was retained during a low-power state.

If either of the magic numbers mismatch, POWMAN vector boot does not take place. If the numbers match, the bootrom zeroes BOOT0 before entering the vector, so that the behaviour does not persist over subsequent reboots.

The POWMAN boot vector is permitted to return. The boot sequence continues as normal after a return from POWMAN vector boot, as though the vector boot had not taken place. There is no requirement for the vector to preserve the global pointer (`gp`) register on RISC-V. Use this to perform any additional setup required for the boot path, such as issuing a power-up command to an external QSPI device that may have been powered down (e.g. via a B9h power-down command).

The entry point (`pc`) must have the LSB set on Arm (the Thumb bit) and clear on RISC-V. If this condition is not met, the bootrom assumes you have passed a RISC-V function pointer to an Arm processor (or vice versa) and hangs the core rather than continuing, since executing code for the wrong architecture has spectacularly undefined consequences.

The linker should automatically set the Thumb bit appropriately for a function pointer relocation, but this is something to be aware of if you pass hardcoded values such as the base of SRAM: this is correctly passed as `0x20000001` on Arm (Thumb bit set) and `0x20000000` on RISC-V (no Thumb bit, halfword-aligned).

5.2.4. Watchdog Boot Vector

Watchdog boot allows users to install their own boot handler, and divert control away from the main boot sequence on non-POR/BOR resets. It recognises the following values written to the watchdog's upper scratch registers:

- SCRATCH4: magic number `0xb007c0d3`
- SCRATCH5: entry point XORed with magic `-0xb007c0d3` (`0x4ff83f2d`)
- SCRATCH6: stack pointer
- SCRATCH7: entry point

If either of the magic numbers mismatch, watchdog boot does not take place. If the numbers match, the Bootrom zeroes SCRATCH4 before transferring control, so that the behaviour does not persist over subsequent reboots.

Watchdog boot can also be used to select the bootrom's special one-shot boot modes, described in Section 5.2.4.1. The term **one-shot** refers to the fact these only affect the next boot (and not subsequent ones) due to the bootrom clearing SCRATCH4 each boot. These boot types are encoded by setting a special entry point (`pc`) value of `0xb007c0d3`, which is otherwise not a valid entry address, and then setting the boot type in the stack pointer (`sp`) value. Section 5.2.4.1 lists the supported values.

The watchdog boot vector is permitted to return. The boot path continues as normal when it returns: use this to perform any additional setup required for the boot path, such as issuing additional commands to an external QSPI device. On RISC-V the vector is permitted to use its own global pointer (`gp`) value, as the bootrom only uses `gp` during USB boot, which installs its own value.

With the exception of the magic boot type entry point (`0xb007c0d3`), the vector entry point `pc` must have the LSB set on Arm (the Thumb bit) and clear on RISC-V. If this condition is not met, the bootrom assumes you have passed a RISC-V function pointer to an Arm processor (or vice versa) and hangs the core rather than continuing.

5.2.4.1. Special Watchdog Boot Types

The magic entry point `0xb007c0d3` indicates a special one-shot boot type, identified by the stack pointer value:

`BOOTSEL`

Selected by `sp` = `2`. Boot into BOOTSEL mode. This will be either UART or USB boot depending on whether QSPI `SD1` is driven high (default pull-down selects USB boot). See Section 5.2.8 for more details.

`RAM_IMAGE`

Selected by `sp` = `3`. Boot into an image stored in SRAM or XIP SRAM. BOOTSEL mode uses this to request execution of an image it loaded into RAM before rebooting. See Section 5.2.5 for more details.

FLASH_UPDATE

Selected by sp = 4. BOOTSEL selects this mode when rebooting following a flash download. Changes some flash boot behaviour, such as allowing older versions to boot in preference to newer ones. See Section 5.1.16 for more details.

Parameters to the one-shot boot type are passed in:

- SCRATCH2: Parameter 0
- SCRATCH3: Parameter 1

These directly correspond to the p0 and p1 boot parameters passed into the reboot() API. For example, on a RAM_IMAGE boot, this specifies the base and size of the RAM region to be searched for a valid IMAGE_DEF. See the API listing in Section 5.4.8.24 for more details. When not performing one of the listed boot types, SCRATCH2 and SCRATCH3 remain free for arbitrary user values, and the bootrom does not modify or interpret their contents.

5.2.5. RAM Image Boot

The bootrom is directed (via values in the watchdog registers) to boot into an image in SRAM or XIP SRAM. The two parameters indicate the start and size of the region to search for a block loop containing a valid (and correctly signed if necessary) IMAGE_DEF. These are passed as parameter 0/1, in watchdog scratch 2/3.

If the image to be booted is contained in XIP SRAM, the XIP SRAM must be pinned in place by the bootrom prior to launch. For this reason, if you are using XIP SRAM for your binaries, you must add a special entry to the LOAD_MAP item (see Section 5.9.3.2).

5.2.6. OTP Boot

If OTP boot is enabled, then code from OTP is executed in preference to code from flash. Note that the OTP code is free to "chain" into an executable stored in flash.

Code from OTP is copied into SRAM at the specified location, then execution proceeds similarly to RAM Image Boot. The SRAM with the data copied from OTP is searched for a valid (and correctly signed if necessary) IMAGE_DEF. If found, it is booted; otherwise OTP boot falls through to Flash Boot (if enabled).

OTP boot could, for example, be used to execute some hidden decryption code to decode a flash image on startup. The OTP boot code can hide itself (in OTP) even from Secure code, once it is done.

5.2.7. Flash Boot

The bootrom scans flash up to 16 times until it finds a valid IMAGE_DEF or PARTITION_TABLE. At this point, the flash settings are considered valid, and the flash boot proceeds if a valid bootable IMAGE_DEF is found with these settings. It uses the following combinations of flash read instruction and SCK divisor for the 16 attempts:

Table 451. QSPI read modes supported by the bootrom, in the order it attempts them.

Mode	Clock Divisor
EBh quad	3
BBh dual	3
0Bh serial	3
03h serial	3
EBh quad	6
BBh dual	6
0Bh serial	6

Mode	Clock Divisor
03h serial	6
EBh quad	12
BBh dual	12
0Bh serial	12
03h serial	12
EBh quad	24
BBh dual	24
0Bh serial	24
03h serial	24

QSPI does not provide a reliable method to detect whether a device is attached. However, this is not much of an issue for boot purposes: either there is a device with valid and bootable contents, or there are no such contents (either due to lack of a connected device, invalid device contents, or failure to communicate in the current QSPI mode).

When there is no device (or no recognisable contents), the bootrom tries all 16 modes in Table 451 before finally giving up. The size of the initial search region is limited to 4 kB to minimise the time spent scanning flash before falling through to USB or UART boot. This same 4 kB limit also applies to search within a flash partition, which allows the bootrom to reliably sever the contained image's block loop with a single 4 kB sector erase at the start of a partition, such as on a version downgrade.

There are three main ways that the bootrom locates flash images:

Flash image boot

A flash image can be written directly to flash storage address `0x0`, and the bootrom will find it from there. This is the most similar to flash boot on RP2040 (the main differences being the removal of a `boot2` in the first 256 bytes of the image, and the new requirement for a valid image definition anywhere within the first 4 kB of the image).

Flash partition boot

A flash image can be written into a partition of a partition table. The partition table is described by a `PARTITION_TABLE` block stored at the start of flash. The bootrom finds the partition table and scans its partitions to look for bootable images.

Partition-table-in-image boot

A flash image containing an `IMAGE_DEF` and `PARTITION_TABLE` block in a single block loop is written to the start of flash. The bootrom loads the embedded partition table, and enters the image in the same way as the flash image boot case.

Revisit the linked bootrom concepts sections to get the fullest understanding of each of these three forms of flash boot. For the purposes of this section, all that matters is whether the bootrom can discover a valid, bootable image or not. In all three cases, the image must have a valid `IMAGE_DEF`, and meet all relevant security requirements such as being correctly signed, and having a rollback version greater than or equal to the one stored in OTP.

The bootrom enters the flash image in whatever QSPI mode it discovered to work during flash programming. Any further setup (such as prefixless continuous read modes) is performed by the flash image itself. This setup code, referred to as an **XIP setup function**, is usually copied into RAM before execution to avoid running from flash whilst the XIP interface is being reconfigured.

TIP

The PICO_EMBED_XIP_SETUP=1 flag in the SDK enables inclusion and execution of an XIP setup function on RP2350 builds. In this case the function executes on the core 0 stack during early startup, so no additional static memory need be allocated. This is not the case for subsequent calls, because the stack is often not executable post-startup.

You should save your XIP setup function in the first 256 bytes of boot RAM to make it easily locatable when the XIP mode is re-initialised following a serial flash programming operation which had to drop out of XIP mode. The bootrom writes a default XIP setup function to this address before entering the flash image, which restores the mode the bootrom discovered during flash programming.

NOTE

You cannot execute an XIP setup function directly from boot RAM, because boot RAM is never executable. You must copy it into SRAM before execution.

XIP setup functions should be fully position-independent, and no more than 256 bytes in size. If you are unable to meet these requirements, you should install a stub function which calls your XIP setup function elsewhere in RAM.

5.2.8. BOOTSEL (USB/UART) Boot

The bootrom samples the state of QSPI CSn shortly after reset. Based on the result, the bootrom decides whether to enter BOOTSEL mode, which refers collectively to the USB and UART bootloaders.

The bootrom initialises the chip select to the following state:

- Output disabled
- Pulled high (note CSn is an active-low signal, so this deselects the external QSPI device if there is one)

If the chip select remains high, the bootrom continues with its normal, non-BOOTSEL sequence. By default on a blank device, this means driving the chip select low and attempting to boot from an external flash or PSRAM device.

If chip select is driven low externally, the bootrom enters BOOTSEL mode. You must drive the chip select low with a sufficiently low impedance to overcome the internal pull-up. A 4.7 kΩ resistance to ground is a good intermediate value which reliably creates a low input logic level, but will not affect the output levels when RP2350 drives the chip select.

The QSPI SD1 line, which RP2350 initially pulls low, selects which bootloader to enter:

- SD1 remains pulled low: enter USB bootloader
- SD1 driven high: enter UART bootloader

USB boot is a low-friction method for programming an RP2350 from a sophisticated host like a Linux PC. It also directly exposes more advanced options like OTP programming. See Section 5.5 for the drag-and-drop mass storage interface, or Section 5.6 for the PICOBOOT vendor interface.

UART boot is a minimal interface for bootstrapping a flashless RP2350 from another microcontroller. UART boot uses QSPI SD2 for UART TX, and QSPI SD3 for UART RX, at a fixed baud rate of 1 Mbaud. For more details about UART boot, see Section 5.8.

5.2.8.1. BOOTSEL Clock Requirements

BOOTSEL mode requires either a crystal attached across the XIN and XOUT pins, or a clock signal from an external oscillator driven into the XIN pin. See Table 1438 for the electrical specifications of these two XOSC pins.

The bootrom assumes a default XOSC frequency of 12 MHz. It configures the USB PLL to derive a fixed 48 MHz frequency from the XOSC reference. For USB, this must be a precise frequency. If you use a non-12 MHz crystal, and intend to use USB boot, program BOOTSEL_PLL_CFG and BOOTSEL_XOSC_CFG in OTP, and then set

BOOT_FLAGS0.ENABLE_BOOTSEL_NON_DEFAULT_PLL_XOSC_CFG. For details about calculating the correct PLL parameters for your crystal, see Section 8.6.3.

UART boot uses the same PLL configuration as USB boot. However, the permissible range of crystal frequencies under the default PLL configuration is wider. See Section 5.8.1.

5.2.9. Boot Configuration (OTP)

User configuration stored in OTP can be found in Section 13.9, starting at CRIT1.

The main controls for the bootrom are stored in BOOT_FLAGS0 and BOOT_FLAGS1. These are both in page 1 of OTP, which has the following default permissions on a blank device:

- Read-write for Secure (`S`)
- Read-write for bootloader (`BL`)
- Read-only for Non-secure (`NS`)

Boot key hashes are stored in page 2 of OTP, starting from BOOTKEY0_0. There is space for up to four boot key hashes in this page. See Section 5.10.1 for an example of how keys can be installed.

5.3. Launching Code On Processor Core 1

As described in Section 5.2, after reset, processor core 1 sleeps at start-up, and remains asleep until woken by core 0 via the SIO FIFOs.

If you are using the SDK then you can use the `multicore_launch_core1()` function to launch code on processor core 1. However this section describes the procedure to launch code on processor core 1 yourself.

The procedure to start running on processor core 1 involves both cores moving in lockstep through a state machine coordinated by passing messages over the inter-processor FIFOs. This state machine is designed to be robust enough to cope with a recently reset processor core 1 which may be anywhere in its boot code, up to and including going to sleep. As result, the procedure may be performed at any point after processor core 1 has been reset (either by system reset, or explicitly resetting just processor core 1).

The following C code describes the procedure:

```
// values to be sent in order over the FIFO from core 0 to core 1
//
// vector_table is value for VTOR register
// sp is initial stack pointer (SP)
// entry is the initial program counter (PC) (don't forget to set the thumb bit!)
const uint32_t cmd_sequence[] =
        {0, 0, 1, (uintptr_t) vector_table, (uintptr_t) sp, (uintptr_t) entry};

uint seq = 0;
do {
    uint cmd = cmd_sequence[seq];
    // always drain the READ FIFO (from core 1) before sending a 0
    if (!cmd) {
        // discard data from read FIFO until empty
        multicore_fifo_drain();
        // execute a SEV as core 1 may be waiting for FIFO space
        __sev();
    }
    // write 32 bit value to write FIFO
    multicore_fifo_push_blocking(cmd);
    // read 32 bit value from read FIFO once available
```

```
    uint32_t response = multicore_fifo_pop_blocking();
    // move to next state on correct response (echo-d value) otherwise start over
    seq = cmd == response ? seq + 1 : 0;
} while (seq < count_of(cmd_sequence));
```

5.4. Bootrom APIs

Whilst some ROM space is dedicated to the implementation of the boot sequence and USB/UART boot interfaces, the bootrom also contains public functions that provide useful RP2350 functionality that may be useful for any code or runtime running on the device.

A categorised list is available in Section 5.4.6.

The full alphabetical list is available in Section 5.4.7.

5.4.1. Locating The API Functions

The API functions are normally made available to the user by wrappers in the SDK. However, a lower level method is provided to locate them (since their locations may change with each bootrom release) for other runtimes, or those who wish to locate them directly.

Table 452 shows the fixed memory layout of certain words in the bootrom used to locate these functions when using the Arm architecture. Table 453 shows the additional entries for use when using the RISC-V architecture.

Table 452. Bootrom contents at fixed (well known) addresses for Arm code

Address	Contents	Description
0x00000000	32-bit pointer	Initial boot stack pointer
0x00000004	32-bit pointer	Pointer to boot reset handler function
0x00000008	32-bit pointer	Pointer to boot NMI handler function
0x0000000c	32-bit pointer	Pointer to boot Hard fault handler function
0x00000010	'M', 'u', 0x02	Magic
0x00000013	byte	Bootrom version
0x00000014	16-bit pointer	Pointer to ROM entry table (BOOTROM_ROMTABLE_START)
0x00000016	16-bit pointer	Pointer to a helper function (rom_table_lookup_val())
0x00000018	16-bit pointer	Pointer to a helper function (rom_table_lookup_entry())

Table 453. Bootrom contents at fixed (well known) addresses for RISC-V code

Address	Contents	Description
0x00007df6	16-bit pointer	Pointer to ROM entry table (BOOTROM_ROMTABLE_START)
0x00007df8	16-bit pointer	Pointer to a helper function (rom_table_lookup_val())
0x00007dfa	16-bit pointer	Pointer to a helper function (rom_table_lookup_entry())
0x00007dfc	32-bit instruction	RISC-V Entry Point

Assuming the three bytes starting at address 0x00000010 are ('M', 'u', 0x02), the other fixed location fields can be assumed to be valid and used to lookup bootrom functionality.

The version byte at offset 0x00000013 is informational, and should not be used to infer the exact location of any functions. It has the value 2 for A2 silicon.

The following code from the SDK shows how the SDK looks up a bootrom function:

```
static __force_inline void *rom_func_lookup_inline(uint32_t code) {
#ifdef __riscv
    // on RISC-V the code (a jmp) is actually embedded in the table
    rom_table_lookup_fn rom_table_lookup =
        (rom_table_lookup_fn) (uintptr_t)*(uint16_t*)(BOOTROM_TABLE_LOOKUP_ENTRY_OFFSET
        + rom_offset_adjust);
    return rom_table_lookup(code, RT_FLAG_FUNC_RISCV);
#else
    // on Arm the function pointer is stored in the table, so we dereference it
    // via lookup() rather than lookup_entry()
    rom_table_lookup_fn rom_table_lookup =
        (rom_table_lookup_fn) (uintptr_t)*(uint16_t*)(BOOTROM_TABLE_LOOKUP_OFFSET);
    if (pico_processor_state_is_nonsecure()) {
        return rom_table_lookup(code, RT_FLAG_FUNC_ARM_NONSEC);
    } else {
        return rom_table_lookup(code, RT_FLAG_FUNC_ARM_SEC);
    }
#endif
}
```

As well as API functions, there are a few data values that can be looked up. The following code demonstrates:

```
void *rom_data_lookup(uint32_t code) {
    rom_table_lookup_fn rom_table_lookup =
        (rom_table_lookup_fn) (uintptr_t)*(uint16_t*)(BOOTROM_TABLE_LOOKUP_OFFSET);
    return rom_table_lookup(code, RT_FLAG_DATA);
}
```

The `code` parameter correspond to the `CODE` values in the tables below, and is calculated as follows:

```
uint32_t rom_table_code(char c1, char c2) {
   return (c2 << 8) | c1;
}
```

These codes are also available in `bootrom.h` in the SDK as `#defines`.

5.4.2. API Function Availability

Some functions are not available under all architectures or security levels. The API listing in Section 5.4.6 uses the following terms to list the availability of each individual API entry point:

`Arm-S`

: The function is available to Secure Arm code. The majority of functions are available for `Arm-S` unless they deal specifically with RISC-V or Non-secure functionality.

`RISC-V`

: The function is available to RISC-V code. Most of the functions that are available under `Arm-S` are also exposed under RISC-V unless they deal specifically with Arm security states.

`Arm-NS`

: The function is available to Non-secure Arm code. The function in this case performs additional permission and argument checks to prevent Secure data from leaking or being corrupted.

Each individual `Arm-NS` API function must be explicitly enabled by Secure code before use, via set_ns_api_permission(). A

disabled Non-secure API returns BOOTROM_ERROR_NOT_PERMITTED if disabled by Secure code. All Non-secure APIs are disabled initially. There is no permission control on Non-secure code calling Secure-only Arm-S functions, but such a call will crash when it attempts to access Secure-only hardware.

The Arm-NS functions may escalate through a Secure Gateway (SG) instruction to allow Non-secure code to perform limited operations on nominally Secure-only hardware, such as QSPI direct-mode interface used for flash programming.

The RISC-V functions do not have separate entry points based on privilege level. Both M-mode and U-mode software can call bootrom APIs, assuming they have execute permissions on ROM addresses in the PMP. However, U-mode calls will crash if they attempt to access M-mode-only hardware.

5.4.3. API Function Return Codes

Some functions do not support returning any error, and are marked void. The remainder return either 0 (BOOTROM_OK) or a positive value (if data needs to be returned) for success. These bootrom error codes are identical to the error codes used by the SDK, so they can be used interchangeably. This explains the gaps in the numbering for SDK error codes that aren't used by the bootrom.

Name	Value	Description
value	>= 0	The function succeeded and returned the value
BOOTROM_OK	0	The function executed successfully
BOOTROM_ERROR_NOT_PERMITTED	-4	The operation was disallowed by a security constraint
BOOTROM_ERROR_INVALID_ARG	-5	One or more parameters passed to the function is outside the range of supported values; BOOTROM_ERROR_INVALID_ADDRESS and BOOTROM_ERROR_BAD_ALIGNMENT are more specific errors.
BOOTROM_ERROR_INVALID_ADDRESS	-10	An address argument was out-of-bounds or was determined to be an address that the caller may not access.
BOOTROM_ERROR_BAD_ALIGNMENT	-11	An address passed to the function was not correctly aligned.
BOOTROM_ERROR_INVALID_STATE	-12	Something happened or failed to happen in the past, and consequently the request cannot currently be serviced.
BOOTROM_ERROR_BUFFER_TOO_SMALL	-13	A user-allocated buffer was too small to hold the result or working state of the function.
BOOTROM_ERROR_PRECONDITION_NOT_MET	-14	The call failed because another bootrom function must be called first.
BOOTROM_ERROR_MODIFIED_DATA	-15	Cached data was determined to be inconsistent with the full version of the data it was copied from.
BOOTROM_ERROR_INVALID_DATA	-16	The contents of a data structure are invalid
BOOTROM_ERROR_NOT_FOUND	-17	An attempt was made to access something that does not exist; or, a search failed.
BOOTROM_ERROR_UNSUPPORTED_MODIFICATION	-18	Modification is impossible based on current state; e.g. attempted to clear an OTP bit.
BOOTROM_ERROR_LOCK_REQUIRED	-19	A required lock is not owned. See Section 5.4.4.

5.4.4. API Functions And Exclusive Access

Various bootrom functions require access to parts of the system which:

- cannot be safely accessed by both cores at once, or
- limit the functionality of other hardware when in use

For example:

- Programming OTP: it is not possible to read from the memory mapped OTP data regions at the same time as accessing its serial programming interface.
- Use of the SHA-256 block: only one SHA-256 sum can be in progress at a time.
- Using the QSPI direct-mode interface to program the flash causes XIP access to return a bus fault.

It is beyond the purview of the bootrom to implement a locking strategy, as the style and scope of the locking required is entirely up to how the application itself uses these resources.

Nevertheless, it is important that, say, a Non-secure call to a flash programming API can't cause a hard fault in other Secure code running from flash. There must be some way for user software to coordinate with bootrom APIs on such changes of state. The bootrom implements the **mechanism** but not the **policy** for mutual exclusion over bootrom API calls.

The solution the bootrom provides is to use the boot locks (boot RAM registers BOOTLOCK0 through BOOTLOCK7) to inform the bootrom which resources are currently owned by the caller and therefore safe for it to use.

To enable lock checking in bootrom APIs, set boot lock `7` (`LOCK_ENABLE`) to the claimed state. When enabled, bootrom functions which use certain hardware resources (listed below) will check the status of the boot lock assigned to that resource, and return `BOOTROM_ERROR_LOCK_REQUIRED` if that lock is not in the claimed state.

Before calling a bootrom function with locking enabled, you must claim the relevant locks. It may take multiple attempts to claim if the API is concurrently accessed from other contexts. Follow the same steps as the SIO spinlocks (Section 3.1.4) to claim a lock.

The following boot locks are assigned:

- `0x0` : `LOCK_SHA_256` - if owned, then a bootrom API is allowed to use the SHA-256 block
- `0x1` : `LOCK_FLASH_OP` - if owned, then a bootrom API is allowed to enter direct mode on the QSPI memory interface (Section 12.14.5) in order to perform low-level flash operations
- `0x2` : `LOCK_OTP` - if owned, then a bootrom API is allowed to access OTP via the serial interface
- `0x7` : `LOCK_ENABLE` - if owned, then bootrom API resource ownership checking is enabled. This is off by default, since the bootrom APIs aim to be usable by default without additional setup.

5.4.5. SDK Access To The API

Bootrom functions are exposed in the SDK via the `pico_bootrom` library (see pico_bootrom).

Each bootrom function has a `rom_` wrapper function that looks up the bootrom function address and calls it.

The SDK provides a simple implementation of exclusive access via `bootrom_acquire_lock_blocking(n)` and `bootrom_release_lock(n)`. When enabled, as it is by default (`PICO_BOOTROM_LOCKING_ENABLED=1` is defined) the SDK enables bootrom locking via `LOCK_ENABLE`, and these two functions use the other `SHA_256`/`FLASH_OP`/`OTP` boot locks to take ownership of/release ownership of the corresponding bootrom resource.

The `rom_` wrapper functions the SDK call `bootrom_acquire_lock_locking` and `bootrom_relead_lock` functions around bootrom calls that have locking requirements.

5.4.6. Categorised List Of API Functions and ROM Data

The terms in parentheses after each function name (`Arm-S`, `Arm-NS`, `RISC-V`) indicate the architecture and security state combinations where that API is available:

- `Arm-S`: Arm processors running in the Secure state
- `Arm-NS`: Arm processors running in the Non-secure state
- `RISC-V`: RISC-V processors

See Section 5.4.2 for the full definitions of these terms.

List entries ending with parentheses, such as flash_op(), are callable functions. List entries without parentheses, such as git_revision, are pointers to ROM data locations.

5.4.6.1. Low-Level Flash Access

These low-level (Secure-only) flash access functions are similar to the ones on RP2040:

- connect_internal_flash() (`Arm-S`, `RISC-V`)
- flash_enter_cmd_xip() (`Arm-S`, `RISC-V`)
- flash_exit_xip() (`Arm-S`, `RISC-V`)
- flash_flush_cache() (`Arm-S`, `RISC-V`)
- flash_range_erase() (`Arm-S`, `RISC-V`)
- flash_range_program() (`Arm-S`, `RISC-V`)

These are new with RP2350:

- flash_reset_address_trans() (`Arm-S`, `RISC-V`)
- flash_select_xip_read_mode() (`Arm-S`, `RISC-V`)

5.4.6.2. High-Level Flash Access

The higher level access functions, provide functionality that is safe to expose (with permissions) to Non-secure code as well.

- flash_op() (`Arm-S`, `Arm-NS`, `RISC-V`)
- flash_runtime_to_storage_addr() (`Arm-S`, `Arm-NS`, `RISC-V`)

5.4.6.3. System Information

- flash_devinfo16_ptr (`Arm-S`, `RISC-V`)
- get_partition_table_info() (`Arm-S`, `Arm-NS` `RISC-V`)
- get_sys_info() (`Arm-S`, `Arm-NS`, `RISC-V`)
- git_revision (`Arm-S`, `Arm-NS`, `RISC-V`)

5.4.6.4. Partition Tables

- get_b_partition() (`Arm-S`, `RISC-V`)
- get_uf2_target_partition() (`Arm-S`, `RISC-V`)

- pick_ab_partition() (Arm-S, RISC-V)
- partition_table_ptr (Arm-S`, RISC-V)
- load_partition_table() (Arm-S, RISC-V)

5.4.6.5. Bootrom Memory and State

- set_bootrom_stack() (RISC-V)
- xip_setup_func_ptr (Arm-S, RISC-V)
- bootrom_state_reset() (Arm-S, RISC-V)

5.4.6.6. Executable Image management

- chain_image() (Arm-S, RISC-V)
- (explicit_buy() (Arm-S, RISC-V)

5.4.6.7. Security

These Secure-only functions control access for Non-secure code:

- set_ns_api_permission() (Arm-S)
- set_rom_callback() (Arm-S, RISC-V)
- validate_ns_buffer() (Arm-S, RISC-V)

5.4.6.8. Miscellaneous

These functions are provided to all platforms and security levels, but perform additional checks when called from Non-secure Arm code:

- reboot() (Arm-S, Arm-NS, RISC-V)
- otp_access() (Arm-S, Arm-NS, RISC-V)

5.4.6.9. Non-secure Only

- secure_call() (Arm-NS)

5.4.6.10. Bit Manipulation

Unlike RP2040 the bootrom does not contain bit manipulation functions. Processors on RP2350 implement hardware instructions for these operations which are far faster than the software implementations in the RP2040 bootrom.

5.4.6.11. Memcpy and Memset

Unlike RP2040, the bootrom does not provide memory copy or clearing functions, as your language runtime is expected to already provide well-performing implementations of these on Cortex-M33 or Hazard3.

The bootrom does contain private implementations of standard C `memcpy()` and `memset()`, for both Arm and RISC-V, but these are optimised for size rather than performance. They are not exported in the ROM table.

5.4.6.12. Floating Point

Unlike RP2040 the bootrom does not contain functions for floating point arithmetic. On Arm there is standard processor support for single-precision arithmetic via the Cortex-M FPU, and RP2350 provides an Arm coprocessor which dramatically accelerates double-precision arithmetic (the DCP, Section 3.6.2). The SDK defaults to the most performant hardware or software implementation available.

5.4.7. Alphabetical List Of API Functions and ROM Data

- bootrom_state_reset() (Arm-S, RISC-V)
- chain_image() (Arm-S, RISC-V)
- connect_internal_flash() (Arm-S, RISC-V)
- flash_devinfo16_ptr (Arm-S, RISC-V)
- flash_enter_cmd_xip() (Arm-S, RISC-V)
- flash_exit_xip() (Arm-S, RISC-V)
- flash_flush_cache() (Arm-S, RISC-V)
- flash_op() (Arm-S, Arm-NS, RISC-V)
- flash_range_erase() (Arm-S, RISC-V)
- flash_range_program() (Arm-S, RISC-V)
- flash_reset_address_trans() (Arm-S, RISC-V)
- flash_runtime_to_storage_addr() (Arm-S, Arm-NS, RISC-V)
- flash_select_xip_read_mode() (Arm-S, RISC-V)
- get_b_partition() (Arm-S, RISC-V)
- get_partition_table_info() (Arm-S, Arm-NS RISC-V)
- get_sys_info() (Arm-S, Arm-NS, RISC-V)
- get_uf2_target_partition() (Arm-S, RISC-V)
- git_revision (Arm-S, Arm-NS, RISC-V)
- load_partition_table() (Arm-S, RISC-V)
- otp_access() (Arm-S, Arm-NS, RISC-V)
- partition_table_ptr (Arm-S`, RISC-V)
- pick_ab_partition() (Arm-S, RISC-V)
- reboot() (Arm-S, Arm-NS, RISC-V)
- secure_call() (Arm-NS)
- set_bootrom_stack() (RISC-V)
- set_ns_api_permission() (Arm-S)
- set_rom_callback() (Arm-S, RISC-V)
- validate_ns_buffer() (Arm-S, RISC-V)
- xip_setup_func_ptr (Arm-S, RISC-V)

5.4.8. API Function Listings

5.4.8.1. bootrom_state_reset

Code: 'S','R'

Signature: void bootrom_state_reset(uint32_t flags)

Supported architectures: Arm-S, RISC-V

Resets internal bootrom state, based on the following flags:

- 0x0001 : STATE_RESET_CURRENT_CORE - Resets any internal bootrom state for the current core to a known state. This method should be called prior to calling any other bootrom APIs on the current core, and is called automatically by the bootrom during normal boot of core 0 or launch of code on core 1.
- 0x0002 : STATE_RESET_OTHER_CORE - Resets any internal bootrom state for the other core into a clean state. This is generally called by a debugger when resetting the state of one core via code running on the other.
- 0x0004 : STATE_RESET_GLOBAL_STATE - Resets all non core-specific state, including:
  - Disables access to bootrom APIs from Arm-NS (see also set_ns_api_permission()).
  - Unlocks all boot locks (Section 5.4.4).
  - Clears any Secure code callbacks. (see also set_rom_callback())

Note that the SDK calls this method on runtime initialisation to put the bootrom into a known state. This allows the program to function correctly if it is entered via a debugger, or otherwise without taking the usual boot path through the bootrom, which itself would reset the state.

5.4.8.2. chain_image

Code: 'C','I'

Signature: int chain_image(uint8_t *workarea_base, uint32_t workarea_size, int32_t region_base, uint32_t region_size)

Supported architectures: Arm-S, RISC-V. Note on RISC-V this function may require additional stack; see Section 5.4.8.26.

Returns: BOOTROM_OK (0) on success, or a negative error code on error.

Searches a memory region for a launchable image, and executes it if possible.

The region_base and region_size specify a word-aligned, word-multiple-sized area of RAM, XIP RAM or flash to search. The first 4 kB of the region must contain the start of a block loop with an IMAGE_DEF. If the new image is launched, the call does not return otherwise an error is returned.

The region_base is signed, as a negative value can be passed, which indicates that the (negated back to positive value) is *both* the region_base *and* the base of the "flash update" region.

This method potentially requires similar complexity to the boot path in terms of picking amongst versions, checking signatures etc. As a result it requires a user provided memory buffer as a work area. The work area should be word aligned, and of sufficient size or BOOTROM_ERROR_BAD_ALIGNMENT / BOOTROM_ERROR_INSUFFICIENT_RESOURCES will be returned. The work area size currently required is 3064, so 3 kB is a good choice.

This method is primarily expected to be used when implementing bootloaders.

NOTE

When chaining into an image, the BOOT_FLAGS0.ROLLBACK_REQUIRED flag will not be set, to prevent invalidating a bootloader without a rollback version by booting a binary which has one (see Section 5.10.8).

5.4.8.3. connect_internal_flash

Code: 'I','F'

Signature: `void connect_internal_flash(void)`

Supported architectures: Arm-S, RISC-V

Restores all QSPI pad controls to their default state, and connects the QMI peripheral to the QSPI pads.

If a secondary flash chip select GPIO has been configured via OTP FLASH_DEVINFO, or by writing to the runtime copy of FLASH_DEVINFO in boot RAM, then this bank 0 GPIO is also initialised and the QMI peripheral is connected. Otherwise, bank 0 IOs are untouched.

5.4.8.4. explicit_buy

Code: 'E','B'

Signature: `int explicit_buy(uint8_t *buffer, uint32_t buffer_size)`

Supported architectures: Arm-S RISC-V

Returns: BOOTROM_OK (0) on success, negative error code on error.

Perform an "explicit buy" of an executable launched via an IMAGE_DEF which was TBYB (Section 5.1.17) flagged. A "flash update" boot of such an image is a way to have the image execute once, but only become the "current" image if it safely calls back into the bootrom via this call.

This call may perform the following:

- Erase and rewrite the part of flash containing the TBYB flag in order to clear said flag.
- Erase the first sector of the other partition in an A/B partition scenario, if this new IMAGE_DEF is a version downgrade (so this image will boot again when not doing a normal boot)
- Update the rollback version in OTP if the chip is secure, and a rollback version is present in the image.

NOTE

The device may reboot while updating the rollback version, if multiple rollback rows need to be written - this occurs when the version crosses a multiple of 24 (for example upgrading from version 23 to 25 requires a reboot, but 23 to 24 or 24 to 25 doesn't). The application should therefore be prepared to reboot when calling this function, if rollback versions are in use.

NOTE

The first of the above requires 4 kB of scratch space, so you should pass a word aligned buffer of at least 4 kB to this method in this case, or BOOTROM_ERROR_BAD_ALIGNMENT / BOOTROM_ERROR_INSUFFICIENT_RESOURCES will be returned.

5.4.8.5. flash_devinfo16_ptr

Code: 'F','D'

Type: `uint16_t *flash_devinfo16_ptr`

Pointer to the flash device info used by the flash APIs, e.g. for bounds checking against size of flash devices, and configuring the GPIO used for secondary QSPI chip select.

If BOOT_FLAGS0.FLASH_DEVINFO_ENABLE is set, this boot RAM location is initialised from FLASH_DEVINFO at startup, otherwise it is initialised to:

- Chip select 0 size: 16 MB
- Chip select 1 size: 0 bytes
- No chip select 1 GPIO
- No D8h erase command support

The flash APIs use this boot RAM copy of `FLASH_DEVINFO`, so flash device info can updated by Secure code at runtime by writing through this pointer.

5.4.8.6. `flash_enter_cmd_xip`

Code: `'C','X'`

Signature: `void flash_enter_cmd_xip(void)`

Supported architectures: `Arm-S`, `RISC-V`

Compatibility alias for `flash_select_xip_read_mode(0, 12);`.

Configure the QMI to generate a standard `03h` serial read command, with 24 address bits, upon each XIP access. This is a slow XIP configuration, but is widely supported. CLKDIV is set to 12. The debugger may call this function to ensure that flash is readable following a program/erase operation.

Note that the same setup is performed by flash_exit_xip(), and the RP2350 flash program/erase functions do not leave XIP in an inaccessible state, so calls to this function are largely redundant. It is provided for compatibility with RP2040.

5.4.8.7. `flash_exit_xip`

Code: `'E','X'`

Signature: `void flash_exit_xip(void)`

Supported architectures: `Arm-S`, `RISC-V`

Initialise the QMI for serial operations (direct mode), and also initialise a basic XIP mode, where the QMI will perform 03h serial read commands at low speed (CLKDIV=12) in response to XIP reads.

Then, issue a sequence to the QSPI device on chip select 0, designed to return it from continuous read mode ("XIP mode") and/or QPI mode to a state where it will accept serial commands. This is necessary after system reset to restore the QSPI device to a known state, because resetting RP2350 does not reset attached QSPI devices. It is also necessary when user code, having already performed some continuous-read-mode or QPI-mode accesses, wishes to return the QSPI device to a state where it will accept the serial erase and programming commands issued by the bootrom's flash access functions.

If a GPIO for the secondary chip select is configured via `FLASH_DEVINFO`, then the XIP exit sequence is also issued to chip select 1.

The QSPI device should be accessible for XIP reads after calling this function; the name `flash_exit_xip` refers to returning the *QSPI device* from its XIP state to a serial command state.

5.4.8.8. `flash_flush_cache`

Code: `'F','C'`

Signature: `void flash_flush_cache(void)`

Supported architectures: `Arm-S`, `RISC-V`

Flush the entire XIP cache, by issuing an invalidate by set/way maintenance operation to every cache line (Section 4.4.1). This ensures that flash program/erase operations are visible to subsequent cached XIP reads.

Note that this unpins pinned cache lines, which may interfere with cache-as-SRAM use of the XIP cache.

No other operations are performed.

5.4.8.9. `flash_op`

Code: `'F','O'`

Signature: `int flash_op(uint32_t flags, uint32_t addr, uint32_t size_bytes, uint8_t *buf)`

Supported architectures: `Arm-S`, `Arm-NS`, `RISC-V`

Returns: `BOOTROM_OK` (0) on success, negative error code on error.

Perform a flash read, erase, or program operation. Erase operations must be sector-aligned (4096 bytes) and sector-multiple-sized, and program operations must be page-aligned (256 bytes) and page-multiple-sized; misaligned erase and program operations will return `BOOTROM_ERROR_BAD_ALIGNMENT`. The operation — erase, read, program — is selected by the `CFLASH_OP_BITS` bitfield of the `flags` argument.

`addr` is the address of the first flash byte to be accessed, ranging from `XIP_BASE` to `XIP_BASE` + `0x1ffffff` inclusive. This may be a runtime or storage address. `buf` contains data to be written to flash, for program operations, and data read back from flash, for read operations. `buf` is never written by program operations, and is completely ignored for erase operations.

The flash operation is bounds-checked against the known flash devices specified by the runtime value of `FLASH_DEVINFO`, stored in boot RAM. This is initialised by the bootrom to the OTP value FLASH_DEVINFO, if BOOT_FLAGS0.FLASH_DEVINFO_ENABLE is set; otherwise it is initialised to 16 MB for chip select 0 and 0 bytes for chip select 1. `FLASH_DEVINFO` can be updated at runtime by writing to its location in boot RAM, the pointer to which can be looked up in the ROM table.

If a resident partition table is in effect, then the flash operation is also checked against the partition permissions. The Secure version of this function can specify the caller's effective security level (Secure, Non-secure, bootloader) using the `CFLASH_SECLEVEL_BITS` bitfield of the flags argument, whereas the Non-secure function is always checked against the Non-secure permissions for the partition. Flash operations which span two partitions are not allowed, and will fail address validation.

If FLASH_DEVINFO.D8H_ERASE_SUPPORTED is set, erase operations will use a D8h 64 kB block erase command where possible (without erasing outside the specified region), for faster erase time. Otherwise, only 20h 4 kB sector erase commands are used.

Optionally, this API can translate `addr` from flash runtime addresses to flash storage addresses, according to the translation currently configured by QMI address translation registers, ATRANS0 through ATRANS7. For example, an image stored at a +2 MB offset in flash (but mapped at XIP address 0 at runtime), writing to an offset of +1 MB into the image, will write to a physical flash storage address of 3 MB. Translation is enabled by setting the `CFLASH_ASPACE_BITS` bitfield in the `flags` argument.

When translation is enabled, flash operations which cross address holes in the XIP runtime address space (created by non-maximum `ATRANSx_SIZE`) will return an error response. This check may tear: the transfer may be partially performed before encountering an address hole and ultimately returning failure.

When translation is enabled, flash operations are permitted to cross chip select boundaries, provided this does not span an ATRANS address hole. When translation is disabled, the entire operation must target a single flash chip select (as determined by bits 24 and upward of the address), else address validation will fail.

A typical call sequence for erasing a flash sector in the runtime address space from Secure code would be:

- `connect_internal_flash();`
- `flash_exit_xip();`
- `flash_op((CFLASH_OP_VALUE_ERASE << CFLASH_OP_LSB) | (CFLASH_SECLEVEL_VALUE_SECURE << CFLASH_SECLEVEL_LSB) | (CFLASH_ASPACE_VALUE_RUNTIME << CFLASH_ASPACE_LSB), addr, 4096, NULL);`
- `flash_flush_cache();`
- Copy the XIP setup function from boot RAM to SRAM and execute it, to restore the original XIP mode
  - The bootrom will have written a default setup function which restores the mode/clkdiv parameters found during flash search; user code can overwrite this with its own custom setup function.

A similar sequence is required for program operations. Read operations can leave the current XIP mode in effect, so only the `flash_op(…);` call is required.

Note that the RP2350 bootrom leaves the flash in a basic XIP state in between program/erase operations. However, during a program/erase operation, the QMI is in direct mode (Section 12.14.5) and any attempted XIP access will return a bus error response.

5.4.8.10. `flash_range_erase`

Code: `'R','E'`

Signature: `void flash_range_erase(uint32_t addr, size_t count, uint32_t block_size, uint8_t block_cmd)`

Supported architectures: `Arm-S`, `RISC-V`

Erase `count` bytes, starting at `addr` (offset from start of flash). Optionally, pass a block erase command e.g. `D8h block erase`, and the size of the block erased by this command — this function will use the larger block erase where possible, for much higher erase speed. `addr` must be aligned to a 4096-byte sector, and `count` must be a multiple of 4096 bytes.

This is a low-level flash API, and no validation of the arguments is performed. See flash_op() for a higher-level API which checks alignment, flash bounds and partition permissions, and can transparently apply a runtime-to-storage address translation.

The QSPI device must be in a serial command state before calling this API, which can be achieved by calling `connect_internal_flash()` followed by `flash_exit_xip()`. After the erase, the flash cache should be flushed via `flash_flush_cache()` to ensure the modified flash data is visible to cached XIP accesses.

Finally, the original XIP mode should be restored by copying the saved XIP setup function from boot RAM into SRAM, and executing it: the bootrom provides a default function which restores the flash mode/clkdiv discovered during flash scanning, and user programs can override this with their own XIP setup function.

For the duration of the erase operation, QMI is in direct mode (Section 12.14.5) and attempting to access XIP from DMA, the debugger or the other core will return a bus fault. XIP becomes accessible again once the function returns.

5.4.8.11. `flash_range_program`

Code: `'R','P'`

Signature: `void flash_range_program(uint32_t addr, const uint8_t *data, size_t count)`

Supported architectures: `Arm-S`, `RISC-V`

Program `data` to a range of flash storage addresses starting at `addr` (offset from the start of flash) and `count` bytes in size. `addr` must be aligned to a 256-byte boundary, and `count` must be a multiple of 256.

This is a low-level flash API, and no validation of the arguments is performed. See flash_op() for a higher-level API which checks alignment, flash bounds and partition permissions, and can transparently apply a runtime-to-storage address translation.

The QSPI device must be in a serial command state before calling this API — see notes on flash_range_erase().

5.4.8.12. flash_reset_address_trans

Code: 'R','A'

Signature: void flash_reset_address_trans(void)

Supported architectures: Arm-S, RISC-V

Restore the QMI address translation registers, ATRANS0 through ATRANS7, to their reset state. This makes the runtime-to-storage address map an identity map, i.e. the mapped and unmapped address are equal, and the entire space is fully mapped. See Section 12.14.4.

5.4.8.13. flash_runtime_to_storage_addr

Code: 'F','A'

Signature: int flash_runtime_to_storage_addr(uint32_t addr)

Supported architectures: Arm-S, Arm-NS, RISC-V

Returns: A positive value on success (the translated address), or negative error code on error

Applies the address translation currently configured by QMI address translation registers, ATRANS0 through ATRANS7. See Section 12.14.4.

Translating an address outside of the XIP runtime address window, or beyond the bounds of an ATRANSx_SIZE field, returns BOOTROM_ERROR_INVALID_ADDRESS, which is not a valid flash storage address. Otherwise, return the storage address which QMI would access when presented with the runtime address addr. This is effectively a virtual-to-physical address translation for QMI.

5.4.8.14. flash_select_xip_read_mode

Code: 'X','M'

Signature: void flash_select_xip_read_mode(bootrom_xip_mode_t mode, uint8_t clkdiv)

Supported architectures: Arm-S, RISC-V

Configure QMI for one of a small menu of XIP read modes supported by the bootrom. This mode is configured for both memory windows (both chip selects), and the clock divisor is also applied to direct mode.

The available modes are:

- 0: 03h serial read: serial address, serial data, no wait cycles
- 1: 0Bh serial read: serial address, serial data, 8 wait cycles
- 2: BBh dual-IO read: dual address, dual data, 4 wait cycles (including MODE bits, which are driven to 0)
- 3: EBh quad-IO read: quad address, quad data, 6 wait cycles (including MODE bits, which are driven to 0)

The XIP write command/format are not configured by this function.

When booting from flash, the bootrom tries each of these modes in turn, from 3 down to 0. The first mode that is found to work is remembered, and a default XIP setup function is written into boot RAM that calls *this function* (flash_select_xip_read_mode) with the parameters discovered during flash scanning. This can be called at any time to restore the flash parameters discovered during flash boot.

All XIP modes configured by the bootrom have an 8-bit serial command prefix, so that the flash device can remain in a serial command state, meaning XIP accesses can be mixed more freely with program/erase serial operations. This has a performance penalty, so users can perform their own flash setup *after flash boot* using continuous read mode or QPI mode to avoid or alleviate the command prefix cost.

5.4.8.15. get_b_partition

Code: 'G','B'

Signature: int get_b_partition(uint partition_a)

Supported architectures: Arm-S RISC-V

Returns: The index of the B partition of partition A if a partition table is present and loaded, and there is a partition A with a corresponding B partition; otherwise returns BOOTROM_ERROR_NOT_FOUND.

5.4.8.16. get_partition_table_info

Code: 'G','P'

Signature: int get_partition_table_info(uint32_t *out_buffer, uint32_t out_buffer_word_size, uint32_t flags_and_partition)

Supported architectures: Arm-S, Arm-NS, RISC-V

Returns: >= 0 on success (the number of words filled in out_buffer), negative error code on error.

Fills a buffer with information from the partition table. Note that this API is also used to return information over the PICOBOOT interface.

On success, the buffer is filled, and the number of words filled in the buffer is returned. If the partition table has not been loaded (e.g. from a watchdog or RAM boot), then this method will return BOOTROM_ERROR_PRECONDITION_NOT_MET, and you should load the partition table via load_partition_table() first.

Note that not all data from the partition table is kept resident in memory by the bootrom due to size constraints. To protect against changes being made in flash after the bootrom has loaded the resident portion, the bootrom keeps a hash of the partition table as of the time it loaded it. If the hash has changed by the time this method is called, then it will return BOOTROM_ERROR_INVALID_STATE.

The information returned is chosen by the flags_and_partition parameter; the first word in the returned buffer, is the (sub)set of those flags that the API supports. You should always check this value before interpreting the buffer.

Following the first word, returns words of data for each present flag in order. With the exception of PT_INFO, all the flags select "per partition" information, so each field is returned in flag order for one partition after the next. The special SINGLE_PARTITION flag indicates that data for only a single partition is required. Flags include:

- 0x0001 - PT_INFO : information about the partition table as a whole. The second two words for unpartitioned space in the same form described in Section 5.9.4.2.
  - Word 0 : partition_count (low 8 bits), partition_table_present (bit 8)
  - Word 1 : unpartitioned_space_permissions_and_location
  - Word 2 : unpartitioned_space_permissions_and_flags
- 0x8000 - SINGLE_PARTITION : only return data for a single partition; the partition number is stored in the top 8 bits of flags_and_partition

Per-partition fields:

- 0x0010 - PARTITION_LOCATION_AND_FLAGS : the core information about a partition. The format of these fields is described in Section 5.9.4.2.
  - Word 0 - permissions_and_location
  - Word 1 - permissions_and_flags
- 0x0020 - PARTITION_ID : the optional 64-bit identifier for the partition. If the HAS_ID bit is set in the partition flags, then the 64 bit ID is returned:
  - Word 0 - first 32 bits

- Word 1 - second 32 bits

- `0x0040` - `PARTITION_FAMILY_IDS` : Any additional UF2 family IDs that the partition supports being downloaded into it via the MSD bootloader beyond the standard ones flagged in the `permissions_and_flags` field (see Section 5.9.4.2).
- `0x0080` - `PARTITION_NAME` : The optional name for the partition. If the `HAS_NAME` field bit in `permissions_and_flags` is not set, then no data is returned for this partition; otherwise the format is as follows:
  - Byte 0 : 7 bit length of the name (LEN); top bit reserved
  - Byte 1 : first character of name
  - …
  - Byte LEN : last character of name
  - … *(padded up to the next word boundary)*

NOTE

Unpartitioned space is always reported in Word 1 as having a base offset of `0x0` and a size of `0x2000` sectors (32 MB). The bootrom applies unpartitioned space permissions to *any* flash storage address that is not covered by a partition.

5.4.8.17. `get_sys_info`

Code: `'G','S'`

Signature: `int get_sys_info(uint32_t *out_buffer, uint32_t out_buffer_word_size, uint32_t flags)`

Supported architectures: `Arm-S`, `Arm-NS`, `RISC-V`

Returns: A positive value on success (the number of words filled in out_buffer), negative error code on error.

Fills a buffer with various system information. Note that this API is also used to return information over the PICOBOOT interface.

The information returned is chosen by the flags parameter; the first word in the returned buffer, is the (sub)set of those flags that the API supports. You should always check this value before interpreting the buffer.

Following the first word, returns words of data for each present flag in order:

- `0x0001` : `CHIP_INFO` - unique identifier for the chip (3 words)
  - Word 0 : Value of the `CHIP_INFO_PACKAGE_SEL` register
  - Word 1 : RP2350 device id
  - Word 2 : RP2350 wafer id
- `0x0002` : `CRITICAL` (1 word)
  - Word 0 : Value of the OTP CRITICAL register, containing critical boot flags read out on last OTP reset event
- `0x0004` : `CPU_INFO` (1 word)
  - Word 0 : Current CPU architecture
    - 0 - Arm
    - 1 - RISC-V
- `0x0008` : `FLASH_DEV_INFO` (1 word)
  - Word 0 : Flash device info in the format of OTP FLASH_DEVINFO
- `0x0010` : `BOOT_RANDOM` - a 128-bit random number generated on each boot (4 words)
  - Word 0 : Per boot random number 0

- Word 1 : Per boot random number 1
- Word 2 : Per boot random number 2
- Word 3 : Per boot random number 3

- `0x0020` : `NONCE` - not supported
- `0x0040` : `BOOT_INFO` (4 words)
  - Word 0 : `0xttppbbdd`
    - `tt` - recent boot TBYB and update info (updated on regular non BOOTSEL boots)
    - `pp` - recent boot partition (updated on regular not BOOTSEL boots)
    - `bb` - boot type of the most recent boot
    - `dd` - recent boot diagnostic "partition"
  - Word 1 : Recent boot diagnostic. Diagnostic information from a recent boot (with information from the partition (or slot) indicated by `dd` above. "partition" numbers here are:
    - 0-15 : a partition number
    - -1 : none
    - -2 : slot 0
    - -3 : slot 1
    - -4 : image (the diagnostic came from the launch of a RAM image, OTP boot image or user `chain_image()` call).
  - Word 2 : Last reboot param 0
  - Word 3 : Last reboot param 1

"Boot Diagnostic" information is intended to help identify the cause of a failed boot, or booting into an unexpected binary. This information can be retrieved via PICOBOOT after a watchdog reboot, however it will not survive a reset via the RUN pin or POWMAN reset.

There is only one word of diagnostic information. What it records is based on the `pp` selection above, which is itself set as a parameter when rebooting programmatically into a normal boot.

To get diagnostic info, `pp` must refer to a slot or an "A" partition; image diagnostics are automatically selected on boot from OTP or RAM image, or when `chain_image()` is called.)

The diagnostic word thus contains data for either slot 0 and slot 1, or the "A" partition (and its "B" partition if it has one). The low half word of the diagnostic word contains information from slot 0 or partition A; the high half word contains information from slot 1 or partition B.

The format of each half-word is as follows (using the word *region* to refer to slot or partition)

- `0x0001` : `REGION_SEARCHED` - The region was searched for a block loop.
- `0x0002` : `INVALID_BLOCK_LOOP` - A block loop was found but it was invalid
- `0x0004` : `VALID_BLOCK_LOOP` - A valid block loop was found (Blocks from a loop wholly contained within the region, and the blocks have the correct structure. Each block consists of items whose sizes sum to the size of the block)
- `0x0008` : `VALID_IMAGE_DEF` - A valid `IMAGE_DEF` was found in the region. A valid `IMAGE_DEF` must parse correctly and must be executable.
- `0x0010` : `HAS_PARTITION_TABLE` - Whether a partition table is present. This partition table must have a correct structure formed if `VALID_BLOCK_LOOP` is set. If the partition table turns out to be invalid, then `INVALID_BLOCK_LOOP` is set too (thus both `VALID_BLOCK_LOOP` and `INVALID_BLOCK_LOOP` will both be set).
- `0x0020` : CONSIDERED - There was a choice of partition/slot and this one was considered. The first slot/partition is chosen based on a number of factors. If the first choice fails verification, then the other choice will be considered.

- the version of the PARTITION_TABLE/IMAGE_DEF present in the slot/partition respectively.
- whether the slot/partition is the "update region" as per a FLASH_UPDATE reboot.
- whether an IMAGE_DEF is marked as "explicit buy"

- 0x0040 : CHOSEN - This slot/partition was chosen (or was the only choice)
- 0x0080 : PARTITION_TABLE_MATCHING_KEY_FOR_VERIFY - if a signature is required for the PARTITION_TABLE (via OTP setting), then whether the PARTITION_TABLE is signed with a key matching one of the four stored in OTP
- 0x0100 : PARTITION_TABLE_HASH_FOR_VERIFY - set if a hash value check could be performed. In the case a signature is required, this value is identical to PARTITION_TABLE_MATCHING_KEY_FOR_VERIFY
- 0x0200 : PARTITION_TABLE_VERIFIED_OK - whether the PARTITION_TABLE passed verification (signature/hash if present/required)
- 0x0400 : IMAGE_DEF_MATCHING_KEY_FOR_VERIFY - if a signature is required for the IMAGE_DEF due to secure boot, then whether the IMAGE_DEF is signed with a key matching one of the four stored in OTP.
- 0x0800 : IMAGE_DEF_HASH_FOR_VERIFY - set if a hash value check could be performed. In the case a signature is required, this value is identical to IMAGE_DEF_MATCHING_KEY_FOR_VERIFY
- 0x1000 : IMAGE_DEF_VERIFIED_OK - whether the PARTITION_TABLE passed verification (signature/hash if present/required) and any LOAD_MAP is valid
- 0x2000 : LOAD_MAP_ENTRIES_LOADED - whether any code was copied into RAM due to a LOAD_MAP
- 0x4000 : IMAGE_LAUNCHED - whether an IMAGE_DEF from this region was launched
- 0x8000 : IMAGE_CONDITION_FAILURE - whether the IMAGE_DEF failed final checks before launching; these checks include:
  - verification failed (if it hasn't been verified earlier in the CONSIDERED phase).
  - a problem occurred setting up any rolling window.
  - the rollback version could not be set in OTP (if required in Secure mode)
  - the image was marked as Non-secure
  - the image was marked as "explicit buy", and this was a flash boot, but then region was not the "flash update" region
  - the image has the wrong architecture, but architecture auto-switch is disabled (or the correct architecture is disabled)

NOTE

The non-sensical combination of BOOT_DIAGNOSTIC_INVALID_BLOCK_LOOP and BOOT_DIAGNOSTIC_VALID_BLOCK_LOOP both being set is used to flag a PARTITION_TABLE which passed the initial verification (and hash/sig), but was later discovered to have invalid contents when it was fully parsed.

To get a full picture of a failed boot involving slots and multiple partitions, the device can be rebooted multiple times to gather the information.

5.4.8.18. get_uf2_target_partition

Code: 'G','U'

Signature: `int get_uf2_target_partition(uint8_t *workarea_base, uint32_t workarea_size, uint32_t family_id, resident_partition_t *partition_out)`

Supported architectures: Arm-S RISC-V. Note on RISC-V this function requires additional stack; see Section 5.4.8.26.

Returns: >= 0 on success (the target partition index), or a negative error code on error.

This method performs the same operation to decide on a taget partition for a UF2 family ID as when a UF2 is dragged

onto the USB drive in BOOTSEL mode.

This method potentially requires similar complexity to the boot path in terms of picking amongst versions, checking signatures etc. As a result it requires a user provided memory buffer as a work area. The work area should byte word-aligned and of sufficient size or `BOOTROM_ERROR_INSUFFICIENT_RESOURCES` will be returned. The work area size currently required is 3064, so 3K is a good choice.

If the partition table has not been loaded (e.g. from a watchdog or RAM boot), then this method will return `BOOTROM_ERROR_PRECONDITION_NOT_MET`, and you should load the partition table via load_partition_table() first.

5.4.8.19. `git_revision`

Code: `'G','R'`

Type: `const uint32_t git_revision`

The 8 most significant hex digits of the bootrom git revision. Uniquely identifies this version of the bootrom.

NOTE

This is the git revision built at chip tapeout; the git hash in the public repository is different due to squashed history, even though the contents are identical. The contents can be verified by building the public bootrom source and comparing the resulting binary with one binary dumped from the chip.

5.4.8.20. `load_partition_table`

Code: `'L','P'`

Signature: `int load_partition_table(uint8_t *workarea_base, uint32_t workarea_size, bool force_reload)`

Supported architectures: `Arm-S`, `RISC-V`. Note on `RISC-V` this function requires additional stack; see Section 5.4.8.26.

Returns: `BOOTROM_OK` (0) on success, or a negative error code on error.

Loads the current partition table from flash, if present.

This method potentially requires similar complexity to the boot path in terms of picking amongst versions, checking signatures etc. As a result it requires a user provided memory buffer as a work area. The work area should byte word-aligned and of sufficient size or `BOOTROM_ERROR_INSUFFICIENT_RESOURCES` will be returned. The work area size currently required is 3064, so 3K is a good choice.

If `force_reload` is false, then this method will return `BOOTROM_OK` immediately if the bootrom is loaded, otherwise it will reload the partition table if it has been loaded already, allowing for the partition table to be updated in a running program.

5.4.8.21. `otp_access`

Code: `'O','A'`

Signature: `int otp_access(uint8_t *buf, uint32_t buf_len, uint32_t row_and_flags)`

Supported architectures: `Arm-S`, `Arm-NS`, `RISC-V`

Returns: `BOOTROM_OK` (0) on success, or a negative error code on error.

Writes data from a buffer into OTP, or reads data from OTP into a buffer.

- `0x0000ffff` - `ROW_NUMBER`: 16 low bits are row number (0-4095)
- `0x00010000` - `IS_WRITE`: if set, do a write (not a read)

- 0x00020000 - IS_ECC: if this bit is set, each value in the buffer is 2 bytes and ECC is used when read/writing from 24 bit value in OTP. If this bit is not set, each value in the buffer is 4 bytes, the low 24-bits of which are written to or read from OTP.

The buffer must be aligned to 2 bytes or 4 bytes according to the IS_ECC flag.

This method will read and write rows until the first row it encounters that fails a key or permission check at which it will return BOOTROM_ERROR_NOT_PERMITTED.

Writing will also stop at the first row where an attempt is made to set an OTP bit from a 1 to a 0, and BOOTROM_ERROR_UNSUPPORTED_MODIFICATION will be returned.

If all rows are read/written successfully, then BOOTROM_OK will be returned.

5.4.8.22. partition_table_ptr

Code: 'P','T'

Type: resident_partition_table **partition_table_ptr

A pointer to the pointer to the resident partition table info. The resident partition table is the subset of the full partition table that is kept in memory, and used for flash permissions.

The public part of the resident partition table info is of the form:

Word	Bytes	Value
0	1	*partition_count* (0-16)
	1	*partition_count_with_permissions* (0-16). Set this to > *partition_count* when adding extra permission regions at runtime (do not modify the original partitions)
	1	*loaded* (0x01 if a partition table has been loaded from flash)
	1	0x00 (pad)
1	1	*unpartitioned_space_permissions_and_flags*
2-3	Partition 0	
	1	*permissions_and_location* for partition 0
	1	*permissions_and_flags* for partition 0
4-5	Partition 1	
	1	*permissions_and_location* for partition 1
	1	*permissions_and_flags* for partition 1
…	…	…
32-33	Partition 15	
	1	*permissions_and_location* for partition 15
	1	*permissions_and_flags* for partition 15

Details of the fields *permissions_and_location* and *permissions_and_flags* can be found in Section 5.9.4.

5.4.8.23. pick_ab_partition

Code: 'A','B'

Signature: int pick_ab_partition(uint8_t *workarea_base, uint32_t workarea_size, uint partition_a_num)

Supported architectures: Arm-S, RISC-V. Note on RISC-V this function requires additional stack; see Section 5.4.8.26.

Returns: >= 0 on success (the partition index), or a negative error code on error.

Determines which of the partitions has the "better" IMAGE_DEF. In the case of executable images, this is the one that would be booted

This method potentially requires similar complexity to the boot path in terms of picking amongst versions, checking signatures etc. As a result it requires a user provided memory buffer as a work area. The work area should bye word aligned, and of sufficient size or BOOTROM_ERROR_INSUFFICIENT_RESOURCES will be returned. The work area size currently required is 3064, so 3K is a good choice.

The passed partition number can be any valid partition number other than the "B" partition of an A/B pair.

This method returns a negative error code, or the partition number of the picked partition if (i.e. partition_a_num or the number of its "B" partition if any).

NOTE

This method does not look at owner partitions, only the A partition passed and its corresponding B partition.

5.4.8.24. reboot

Code: 'R','B'

Signature: int reboot(uint32_t flags, uint32_t delay_ms, uint32_t p0, uint32_t p1)

Supported architectures: Arm-S, Arm-NS, RISC-V

Returns: BOOTROM_OK (or doesn't return) on success, a negative error code on error.

Resets the RP2350 and uses the watchdog facility to restart.

The *delay_ms* is the millisecond delay before the reboot occurs. Note: by default this method is asynchronous (unless NO_RETURN_ON_SUCCESS is set - see below), so the method will return and the reboot will happen this many milliseconds later.

The *flags* field contains one of the following values:

- 0x0000 : REBOOT_TYPE_NORMAL - reboot into the normal boot path.
  - p0 - the boot diagnostic "partition" (low 8 bits only)
- 0x0002 : REBOOT_TYPE_BOOTSEL - reboot into BOOTSEL mode.
  - p0 - the GPIO number to use as an activity indicator (enabled by flag in p1).
  - p1 - a set of flags:
    - 0x01 : DISABLE_MSD_INTERFACE - Disable the BOOTSEL USB drive (see Section 5.5)
    - 0x02 : DISABLE_PICOBOOT_INTERFACE - Disable the PICOBOOT interface (see Section 5.6).
    - 0x10 : GPIO_PIN_ACTIVE_LOW - The GPIO in p0 is active low.
    - 0x20 : GPIO_PIN_ENABLED - Enable the activity indicator on the specified GPIO.
- 0x0003 : REBOOT_TYPE_RAM_IMAGE - reboot into an image in RAM. The region of RAM or XIP RAM is searched for an image to run. This is the type of reboot used when a RAM UF2 is dragged onto the BOOTSEL USB drive.
  - p0 - the region start address (word-aligned).
  - p1 - the region size (word-aligned).
- 0x0004 : REBOOT_TYPE_FLASH_UPDATE - variant of REBOOT_TYPE_NORMAL to use when flash has been updated. This is the type of reboot used after dragging a flash UF2 onto the BOOTSEL USB drive.

- `p0` - the address of the start of the region of flash that was updated. If this address matches the start address of a partition or slot, then that partition or slot is treated preferentially during boot (when there is a choice). This type of boot facilitates TBYB (Section 5.1.17) and version downgrades.

- `0x000d` : `REBOOT_TYPE_PC_SP` - reboot to a specific PC and SP. Note: this is not allowed in the `Arm-NS` variant.
  - `p0` - the initial program counter (PC) to start executing at. This must have the lowest bit set for Arm and clear for RISC-V
  - `p1` - the initial stack pointer (SP).

All of the above, can have optional flags ORed in:

- `0x0010` : `REBOOT_TO_ARM` - switch both cores to the Arm architecture (rather than leaving them as is). The call will fail with `BOOTROM_ERROR_INVALID_STATE` if the Arm architecture is not supported.
- `0x0020` : `REBOOT_TO_RISCV` - switch both cores to the RISC-V architecture (rather than leaving them as is). The call will fail with `BOOTROM_ERROR_INVALID_STATE` if the RISC-V architecture is not supported.
- `0x0100` : `NO_RETURN_ON_SUCCESS` - the watchdog hardware is asynchronous. Setting this bit forces this method not to return *if* the reboot is successfully initiated.

NOTE

The `p0` and `p1` parameters are generally written to watchdog scratch registers 2 & 3, and are interpreted post-reboot by the boot path code. The exception is `REBOOT_TYPE_NORMAL` where this API handles the `p0` value before rebooting; the boot path itself does not accept any parameters for `REBOOT_TYPE_NORMAL`.

5.4.8.25. `secure_call`

Code: `'S','C'`

Signature: `int secure_call(…)`

Supported architectures: `Arm-NS`

Returns: >= 0 on success, a negative error code on error.

Call a Secure method from Non-secure code, passing the method to be called in the register `r4` (other arguments passed as normal).

This method provides the ability to decouple the Non-secure code from the Secure code, allowing the former to call methods in the latter without needing to know the location of the methods.

This call will always return `BOOTROM_ERROR_INVALID_STATE` unless Secure Arm code has provided a handler function via set_rom_callback(); if there is a handler function, this method will return the return code that the handler returns, with the convention that `BOOTROM_ERROR_INVALID_ARG` if the "function selector" (in `r4`) is not supported.

Certain well-known "function selectors" will be pre-defined to facilitate interaction between Secure and Non-secure SDK code, or indeed with other environments (e.g. logging to secure UART/USB CDC, launch of core 1 from NS code, watchdog reboot from NS code back into NS code, etc.)

To avoid conflicts the following bit patterns are used for "function selectors":

- `0b0xxx xxxx xxxx xxxx xxxx xxxx xxxx xxxx` - a "well known" function selector; do not use for your own methods
- `0b10xx xxxx xxxx xxxx xxxx xxxx xxxx xxxx` - a "unique" function selector intended to be unlikely to clash with others'. The lower 30 bits should be chosen at random
- `0b11xx xxxx xxxx xxxx xxxx xxxx xxxx xxxx` - a "private" function selector intended for use by tightly coupled NS and S code

5.4.8.26. set_bootrom_stack

Code: 'S','S'

Signature: int set_bootrom_stack(uint32_t base_size[2])

Supported architectures: RISC-V

Returns: BOOTROM_OK (0) on success, a negative error code on error.

Most bootrom functions are written just once, in Arm code, to save space. As a result these functions are emulated when running under the RISC-V architecture. This is largely transparent to the user, however the stack used by the Arm emulation is separate from the calling user's stack, and is stored in boot RAM and is of quite limited size. When using certain of the more complex APIs or if nesting bootrom calls from within IRQs, you may need to provide a larger stack.

This method allows the caller to specify a region of RAM to use as the stack *for the current core* by passing a pointer to two values: the word aligned base address, and the size in bytes (multiple of 4).

The method fills in the previous base/size values into the passed array before returning.

5.4.8.27. set_ns_api_permission

Code: 'S','P'

Signature: int set_ns_api_permission(uint ns_api_num, bool allowed)

Supported architectures: Arm-S

Returns: BOOTROM_OK (0) on success, a negative error code on error.

Allow or disallow the specific NS API (note all NS APIs default to disabled).

ns_api_num is one of the following, configuring Arm-NS access to the given API. When an NS API is disabled, calling it will return BOOTROM_ERROR_NOT_PERMITTED.

- 0x0: get_sys_info
- 0x1: flash_op
- 0x2: flash_runtime_to_storage_addr
- 0x3: get_partition_table_info
- 0x4: secure_call
- 0x5: otp_access
- 0x6: reboot
- 0x7: get_b_partition

NOTE

All permissions default to disallowed after a reset (see also bootrom_state_reset()).

5.4.8.28. set_rom_callback

Code: 'R','C'

Signature: int set_rom_callback(uint callback_number, int (*callback)(…))

Supported architectures: Arm-S, RISC-V

Returns: >= 0 (the old callback pointer) on success, a negative error code on error.

The only currently supported *callback_number* is 0 which sets the callback used for the secure_call API.

A callback pointer of `0` deletes the callback function, a positive callback pointer (all valid function pointers are on RP2350) sets the callback function, but a negative callback pointer can be passed to get the old value without setting a new value.

If successful, returns >=0 (the existing value of the function pointer on entry to the function).

5.4.8.29. `validate_ns_buffer`

Code: `'V','B'`

Signature: `void *validate_ns_buffer(const void *addr, uint32_t size, uint32_t write, uint32_t *ok)`

Supported architectures: `Arm-S`

Returns: `addr` on success, a negative error code on error.

Utility method that can be used by Secure Arm code to validate a buffer passed to it from Non-secure code.

Both the `write` parameter and the (out) result parameter `ok` are RCP booleans, so `0xa500a500` for true, and `0x00c300c3` for false. This enables hardening of this function, and indeed the `write` parameter must be one of these values or the RCP will hang the system.

For success, the entire buffer must fit in range `XIP_BASE` → `SRAM_END`, and must be accessible by the Non-secure caller according to SAU + NS MPU (privileged or not based on current processor IPSR and NS CONTROL flag). Buffers in USB RAM are also allowed if access is granted to NS via ACCESSCTRL.

5.4.8.30. `xip_setup_func_ptr`

Code: `'X','F'`

Type: `void *(xip_setup_func_ptr)(void)'`

5.5. USB Mass Storage Interface

The Bootrom provides a standard USB bootloader that makes a writeable drive available for copying code to the RP2350 using UF2 files (see Section 5.5.2).

A suitable UF2 file copied to the drive is downloaded and written to Flash or RAM, and the device is automatically rebooted, making it trivial to download and run code on the RP2350 using only a USB connection.

5.5.1. The RP2350 Drive

RP2350 appears as a standard 128MB flash drive named `RP2350` formatted as a single partition with FAT16. There are only ever two actual files visible on the drive specified.

- `INFO_UF2.TXT` - contains a string description of the UF2 bootloader and version.
- `INDEX.HTM` - redirects to information about the RP2350 device.

The default `INDEX.HTM` for RP2350 A2 is https://raspberrypi.com/device/RP2?version=5A09D5312E22. The contents of these files and the name of the drive may be customised, see Section 5.7.

Any type of files may be written to the USB drive from the host, however in general these are not stored, and only *appear* to be so because of caching on the host side.

When a suitable UF2 file is written to the device however, the special contents are recognised and data is written to specified locations in RAM or Flash.

Where flash targeted UF2s are written on RP2350 is determined by the *family id* of the UF2 contents and the partition table.

If there is no partition table, then UF2s are stored at the address they specify; otherwise they (with the exception of the special *ABSOLUTE* family id) are stored into a single partition, with UF2 flash address `0x10000000` mapping to the start of the partition.

It is possible, based on the partition table or family id, that the UF2 is not downloadable anywhere in flash, in which case it will be ignored. Further detail can be discovered via GET_INFO - UF2_STATUS On the completed download of an entire valid UF2 file, RP2350 automatically reboots to run the newly downloaded code.

NOTE

Invalid UF2 files may not write at all or only write partially to RP2350 before failing. Not all operating systems notify you of disk write errors after a failed write. You can use `picotool verify` to verify that a UF2 file wrote correctly to RP2350.

5.5.2. UF2 Format Details

This section describes the constraints on a UF2 file to be valid for download.

TIP

To generate UF2 files, you can use the `picootol uf2 convert` functionality in picotool.

All data destined for the device must be in UF2 blocks with:

- A `familyID` present, with a value in the reserved range `0xe48bff58` through `0xe48bff5b` or a user family ID configured in a partition table (see table in Section 5.5.3)
- A `payload_size` of `256`

All data must be destined for (and fit entirely within) the following memory ranges (depending on the type of binary being downloaded which is determined by the address of the first UF2 block encountered):

- A regular flash image
  - `0x10000000`-`0x12000000` *flash:* All blocks must be targeted at 256 byte alignments. Writes beyond the end of physical flash will wrap back to the beginning of flash.
- A *RAM only* image
  - `0x20000000`-`0x20082000` *main RAM:* Blocks can be positioned with byte alignment.
  - `0x13ffc000`-`0x14000000` *XIP RAM:* (since flash is not being targeted, the flash cache is available for use as RAM with same properties as *main RAM*).

  NOTE

  Traditionally UF2 has only been used to write to flash, but this is more a limitation of using the metadata-free `.BIN` file as the source to generate the UF2 file. RP2350 takes full advantage of the inherent flexibility of UF2 to support the full range of binaries in the richer `.ELF` format produced by the build to be used as the source for the UF2 file.

- The `numBlocks` must specify a total size of the binary that fits in the regions specified above
- A change of `numBlocks` or the binary type (determined by *UF2* block target address) will discard the current transfer in progress.
- A change in the `familyID` will discard the current transfer in progress.

- All device destined data must be in blocks without the `UF2_FLAG_NOT_MAIN_FLASH` marking which relates to content to be ignored rather than flash vs RAM.

NOTE

When targeting flash, the UF2 block target addresses are interpreted to be in the content of a flash binary that *starts at* `0x10000000`. The UF2 image may be downloaded into a partition that starts somewhere else in flash, so the actual storage address is `uf2_image_target_base + uf2_block_target_addr - 0x10000000`.

The flash is always erased a 4 kB sector at a time, so including data for only a subset of the 256-byte pages within a sector in a flash-binary UF2 will leave the remaining 256-byte pages of the sector erased but undefined.

A binary is considered "downloaded" when each of the `numBlocks` blocks has been seen at least once in the course of a single valid transfer. The data for a block is only written the first time in case of the host resending duplicate blocks.

After a UF2 is completely downloaded, the RP2350 reboots, ostensibly to run the new binary. Since RP2350 supports downloading a variety of executable and non-executable UF2s into partitions, the partition contains a flag which can be used to turn off this reboot behaviour on a case by case basis.

NOTE

When rebooting after a flash download, a flash update boot is performed. As a result, the newly written partition will be preferred when considered in an A/B choice, but it will not be booted if another bootable image is found in an earlier partition. When rebooting after a RAM download, then the image search starts at the lowest address of a downloaded block (with main RAM considered lower than flash cache if both are present, and the search only spanning one of either main RAM or the flash cache)

It is possible for host software to temporarily disable UF2 writes via the PICOBOOT interface to prevent interference with operations being performed via that interface (see below), in which case any UF2 file write in progress will be aborted.

NOTE

If a problem is encountered downloading the UF2, then it will appear as if *nothing has happened* since the device will not reboot. The `picotool` command `uf2 info` can be used to determine the status of the last download in this case (see also GET_INFO - UF2_STATUS).

5.5.3. UF2 Targeting Rules

When the first block of a UF2 is downloaded, a choice is made where to store the UF2 in flash based on the *family ID* of the UF2. This choice is performed by the same code as the `get_uf2_target_partition()` API (see Section 5.4.8.18).

The following family IDs are defined by the bootrom, however the user is free to use their own for more specific targeting:

Table 454. Table of standard UF2 family IDs understood by the RP2350 bootrom

Name	Value	Description
`absolute`	`0xe48bff57`	Special family ID for content intended to be written directly to flash, ignoring partitions
`rp2040`	`0xe48bff56`	RP2040 executable image
`data`	`0xe48bff58`	Generic catch-all for data UF2s
`rp2350_arm_s`	`0xe48bff59`	RP2350 Arm Secure image (i.e. one intended to be booted by the bootrom)
`rp2350_riscv`	`0xe48bff5a`	RP2350 RISC-V image

Name	Value	Description
rp2350_arm_ns	0xe48bff5b	RP2350 Arm Non-secure image. Not directly bootable by the bootrom, however Secure user code is likely to want to be able to locate binaries of this type

NOTE

The only information available to the algorithm that makes the choice of where to store the UF2, is the UF2 family ID; the algorithm cannot look inside at the UF2 contents as UF2 data sectors may appear at the device in any order.

A UF2 with the absolute family ID is downloaded without regard to partition boundaries. A partition table (if present) or OTP configuration can define whether absolute family ID downloads are allowed, and download to the start of flash. The default factory settings allow for absolute family ID downloads

If there is a partition table present, any other family IDs download to a single partition; if there is no partition table present then the data, rp2350-arm-s (if Arm architecture is enabled) and rp2350-riscv (if RISC-V architecture is enabled) family IDs are allowed by default, and the UF2 is always downloaded to the start of flash.

If a partition table is present, then up to four passes are made over the partition table (from first to last partition encountered) until a matching partition is found; Each pass has different selection criteria:

1. Look for an (unowned) A partition, ignoring those marked NOT_BOOTABLE for the current CPU architecture

   Use of the NOT_BOOTABLE_ flags allows you to have separate boot partitions for each CPU architecture (Arm or RISC-V); were you not to use NOT_BOOTABLE_ flags in this scenario, and say the first encountered partition has an Arm IMAGE_DEF, then, when booting under the RISC-V architecture with auto architecture switching enabled, the bootrom would just switch back into the Arm architecture to boot the Arm binary. Marking the first partition as NOT_BOOTABLE_RISCV in the partition table solves this problem.

   The correct CPU architecture refers to a match between the architecture of the UF2 (determined by family ID of rp2350_arm_s or rp2350_riscv) and the current CPU architecture.

   This pass allows the user to drop either Arm or RISC-V UF2s, and have them stored as you'd want for the NOT_BOOTABLE_ flag scenario.

2. If auto architecture switching is enabled and the other architecture is available, look for an (unowned) A partition, ignoring those marked NOT_BOOTABLE for that CPU architecture.

   This pass is designed to match the boot use case of booting images from the other architecture as a fallback. If there is a partition that would be booted as a result auto architecture switching then this a reasonable place to store this UF2 for the alternative architecture.

3. Look for any unowned A partition that accepts the family ID

   This pass provides a way to target any UF2s to a partitions based on family ID, but assumes that you'd prefer a UF2 to go into a matching top-level partition vs an owned partition.

4. Finally, look for any A partition that accepts the family ID

   This pass implicitly only looks at owned partitions, since unowned partitions would have been matched in the previous pass.

If none of the passes find a match, then the UF2 contents will not be downloaded. The picotool command uf2 info can be used to determine the status of the last download in this case (see also GET_INFO - UF2_STATUS).

5.5.3.1. A/B Partitions And Ownership

You will note that each of the above passes refers to finding an A partition (remember, any partition that isn't a B partition is an A partition; i.e. an unpaired partition is classed as an A partition).

If the found A partition does not have a B partition paired with it, then the A partition *is* the UF2 target partition.

If however, the A partition has a B partition, then a further choice must be made as to which of the A/B partitions should be targeted.

1. If the A partition is unowned, then the partition choice is made based on any current valid IMAGE_DEF in those partitions. The valid partition with the higher version number is not chosen; in the case of executable IMAGE_DEFs, this is the opposite of what would happen during boot; this makes sense as you want to drop the UF2 on the partition which isn't currently booting.
2. If the A partition is marked owned, then the contents of the A partition and B partition are assumed not to contain IMAGE_DEFs which can be used to make a version based choice. Therefore, the owner of the A partition (A_{owner}) and its B partition (B_{owner}) are used to make the choice

   It is however dependent on the use case whether you would want a UF2 that is destined for partition A / partition B to go into partition A when partition A_{owner} has an IMAGE_DEF with the higher version (i.e. would boot if the IMAGE_DEF was executable) or when B_{owner} has an IMAGE_DEF with the higher version. By default, the bootrom picks partition A when partition A_{owner} has the higher versioned IMAGE_DEF, however this can be changed by setting the UF2_DOWNLOAD_AB_NON_BOOTABLE_OWNER_AFFINITY flag in partition A.

5.5.3.2. Multiple UF2 families

It is possible to include sectors targeting different family IDs in the same UF2 file. The intention in the UF2 specification is to allow one file to be shipped for multiple different devices, but the expectation is that each device only accepts one UF2 family ID.

Similarly on RP2350, it is only supported to download a UF2 file containing multiple family IDs if *only one* of those family IDs is acceptable for download to the device according to the above rules.

5.6. USB PICOBOOT Interface

The PICOBOOT interface is a low level USB protocol for interacting with the RP2350 while it is in BOOTSEL mode. This interface may be used concurrently with the USB Mass Storage Interface.

It provides for flexible reading from and writing to RAM or Flash, rebooting, executing code on the device and a handful of other management functions.

Constants and structures related to the interface can be found in the SDK header picoboot.h in the SDK

5.6.1. Identifying The Device

A RP2350 device can recognised by the Vendor ID and Product ID in its device descriptor (shown in Table 455), unless different values have been set in OTP (see Section 5.7)

Table 455. RP2350 Boot Device Descriptor

Field	Value
bLength	18
bDescriptorType	1
bcdUSB	2.10
bDeviceClass	0
bDeviceSubClass	0
bDeviceProtocol	0
bMaxPacketSize0	64
idVendor	0x2e8a - this value may be overridden in OTP

Field	Value
idProduct	0x000f - this value may be overridden in OTP
bcdDevice	1.00 - this value may be overridden in OTP
iManufacturer	1
iProduct	2
iSerial	3
bNumConfigurations	1

5.6.2. Identifying The Interface

The PICOBOOT interface is recognised by the vendor-specific Interface Class, the zero Interface Sub Class, and Interface Protocol (shown in Table 456).

Do not rely on the interface number, as that is dependent on whether the device is currently exposing the Mass Storage Interface. The device may not be currently exposing the PICOBOOT interface at all, so you should not assume it is present.

Table 456. PICOBOOT Interface Descriptor

Field	Value
bLength	9
bDescriptorType	4
bInterfaceNumber	varies
bAlternateSetting	0
bNumEndpoints	2
bInterfaceClass	0xff (vendor specific)
bInterfaceSubClass	0
bInterfaceProtocol	0
iInterface	0

5.6.3. Identifying The Endpoints

The PICOBOOT interface provides a single BULK_OUT and a single BULK_IN endpoint. These can be identified by their direction and type. You should not rely on endpoint numbers.

5.6.4. PICOBOOT Commands

The two bulk endpoints are used for sending commands and retrieved successful command results. All commands are exactly 32 bytes (see Table 457) and sent to the BULK_OUT endpoint.

Table 457. PICOBOOT Command Definition

Offset	Name	Description
0x00	dMagic	The value 0x431fd10b
0x04	dToken	A user provided token to identify this request by
0x08	bCmdId	The ID of the command. Note that the top bit indicates data transfer direction (0x80 = IN)

Offset	Name	Description
0x09	bCmdSize	Number of bytes of valid data in the args field
0x0a	reserved	0x0000
0x0c	dTransferLength	The number of bytes the host expects to send or receive over the bulk channel
0x10	args	16 bytes of command-specific data padded with zeros

If a command sent is invalid or not recognised, the bulk endpoints will be stalled. Further information will be available via the GET_COMMAND_STATUS request (see Section 5.6.5.2).

Following the initial 32 byte packet, if dTransferLength is non-zero, then that many bytes are transferred over the bulk pipe and the command is completed with an empty packet in the opposite direction. If dTransferLength is zero then command success is indicated by an empty IN packet.

The following commands are supported (note common fields dMagic, dToken, and reserved are omitted for clarity)

5.6.4.1. EXCLUSIVE_ACCESS (0x01)

Claim or release exclusive access for writing to the RP2350 over USB (versus the Mass Storage Interface)

Table 458. PICOBOOT EXCLUSIVE_ACCESS command structure

Offset	Name	Value / Description	
0x08	bCmdId	0x01 (EXCLUSIVE_ACCESS)	
0x09	bCmdSize	0x01	
0x0c	dTransferLength	0x00000000	
0x10	bExclusive	NOT_EXCLUSIVE (0)	No restriction on USB Mass Storage operation
		EXCLUSIVE (1)	Disable USB Mass Storage writes (the host should see them as write protect failures, but in any case any active *UF2* download will be aborted)
		EXCLUSIVE_AND_EJECT (2)	Lock the USB Mass Storage Interface out by marking the drive media as not present (ejecting the drive)

5.6.4.2. REBOOT (0x02)

Not supported on RP2350.

Use Section 5.6.4.10 instead.

5.6.4.3. FLASH_ERASE (0x03)

Erases a contiguous range of flash sectors.

Table 459. PICOBOOT FLASH_ERASE command structure

Offset	Name	Value / Description
0x08	bCmdId	0x03 (FLASH_ERASE)
0x09	bCmdSize	0x08
0x0c	dTransferLength	0x00000000
0x10	dAddr	The address in flash to erase, starting at this location. This must be sector (4 kB) aligned

Offset	Name	Value / Description
0x14	dSize	The number of bytes to erase. This must an exact multiple number of sectors (4 kB)

5.6.4.4. READ (0x84)

Read a contiguous memory (Flash or RAM or ROM) range from the RP2350

Table 460. PICOBOOT Read memory command (Flash, RAM, ROM) structure

Offset	Name	Value / Description
0x08	bCmdId	0x84 (READ)
0x09	bCmdSize	0x08
0x0c	dTransferLength	Must be the same as dSize
0x10	dAddr	The address to read from. May be in Flash or RAM or ROM
0x14	dSize	The number of bytes to read

5.6.4.5. WRITE (0x05)

Writes a contiguous memory range of memory (Flash or RAM) on the RP2350.

Table 461. PICOBOOT Write memory command (Flash, RAM) structure

Offset	Name	Value / Description
0x08	bCmdId	0x05 (WRITE)
0x09	bCmdSize	0x08
0x0c	dTransferLength	Must be the same as dSize
0x10	dAddr	The address to write from. May be in Flash or RAM, however must be page (256 byte) aligned if in Flash. Note the flash must be erased first or the results are undefined.
0x14	dSize	The number of bytes to write. If writing to flash and the size is not an exact multiple of pages (256 bytes) then the last page is zero-filled to the end.

5.6.4.6. EXIT_XIP (0x06)

A no-op provided for compatibility with RP2040. An XIP exit sequence (flash_exit_xip()) is issued once before entering the USB bootloader, which returns the external QSPI device from whatever XIP state it was in to a serial command state, and the external QSPI device then remains in this state until reboot.

Table 462. PICOBOOT EXIT_XIP command structure

Offset	Name	Value / Description
0x08	bCmdId	0x06 (EXIT_XIP)
0x09	bCmdSize	0x00
0x0c	dTransferLength	0x00000000

5.6.4.7. ENTER_XIP (0x07)

A no-op provided for compatibility with RP2040. Note that, unlike RP2040, the low-level bootrom flash operations do not leave the QSPI interface in a state where XIP is inaccessible, therefore there is no need to reinitialise the interface each time. XIP setup is performed once before entering the USB bootloader, using an 03h command with a fixed clock divisor

of 6.

Table 463. PICOBOOT Enter Execute in place (XIP) command

Offset	Name	Value / Description
0x08	bCmdId	0x07 (ENTER_XIP)
0x09	bCmdSize	0x00
0x0c	dTransferLength	0x00000000

5.6.4.8. EXEC (0x08)

Not supported on RP2350.

5.6.4.9. VECTORIZE_FLASH (0x09)

Not supported on RP2350.

5.6.4.10. REBOOT2 (0x0a)

Reboots the RP2350 out of BOOTSEL mode. Note that BOOTSEL mode may be re-entered if no valid bootable image is found.

The parameters *flags*, *delay_ms*, *p0*, *p1* are the same as for api_reboot()

Table 464. PICOBOOT REBOOT2 command structure

Offset	Name	Value / Description
0x08	bCmdId	0x0a (REBOOT2)
0x09	bCmdSize	0x10
0x0c	dTransferLength	0x00000000
0x10	dAddr	flags
0x14	dSize	delay_ms
0x18	dSize	p0
0x1c	dSize	p1

5.6.4.11. GET_INFO (0x8b)

Generic conduit for retrieving information from the device.

The transfer length indicates the maximum number of bytes to be retrieved. The fist word returned indicates the number of significant words of data that follow. A full "transfer length" is always returned, padding with zeroes as necessary.

Note *"Word 0"* below refers to the first word of the actual response (i.e. the word after the count word).

Table 465. PICOBOOT GET_INFO command structure

Offset	Name	Value / Description
0x08	bCmdId	0x0b (GET_INFO)
0x09	bCmdSize	0x10
0x0c	dTransferLength	the size of data to be received. Note this must be a multiple of 4, and less than 256

Offset	Name	Value / Description
0x10	bType	the type of information being retrieved: • 0x1 - INFO_SYS : Retrieves information from get_sys_info(); the flag parameter for that function comes from *dParam0*. • 0x2 - PARTITION : Retrieves information from get_partition_table_info(); the flags_and_partition parameter for that function comes from *dParam0*. • 0x03 - UF2_TARGET_PARTITION : Retrieves the partition that a given UF2 family_id would be downloaded into (if it were dragged on the USB drive in BOOTSEL mode). The family id is passed in *dParam0*. ◦ Word 0 : Target partition number: ▪ 0-15 : the partition number the family would be downloaded to ▪ 0xff : if the family would be downloaded at an absolute location ▪ -1 : if there is nowhere to download the family ◦ Word 1 : Target partition Section 5.9.4.2 if the partition number is not -1 ◦ Word 2 : Target partition Section 5.9.4.2 if the partition number is not -1 • 0x04 - UF2_STATUS : Retrieves information about the current/recent UF2 download ◦ Word 0 - 0xnnrr00af ▪ 'n' - no reboot flag; if 0x01, there is no reboot when the UF2 download completes ▪ 'r' - if 0x01, the UF2 being download is a RAM UF2 ▪ 'a' - UF2 download abort reason flags ▪ 0x1 EXCLUSIVELY_LOCKED ▪ 0x2 BAD_ADDRESS ▪ 0x4 WRITE_ERROR ▪ 0x8 REBOOT_FAILURE // if the UF2 targeted a disabled architecture ▪ 'f' - UF2 download status flags ▪ 0x1 IGNORED_FAMILY ◦ Word 1 - the current family id ◦ Word 2 - the number of 256 byte blocks successfully downloaded ◦ Word 3 - the total number of 256 byte blocks in the UF2 to download

5.6.4.12. OTP_READ (0x8c)

Reads data out of OTP. (see also otp_access() which provides the data). Data returned is subject to the "BL" OTP permissions, which define bootloader OTP access permissions.

Table 466. PICOBOOT OTP_READ command structure

Offset	Name	Value / Description
0x08	bCmdId	0x8c (OTP_READ)
0x09	bCmdSize	0x05
0x0c	dTransferLength	$wRowCount \times (bEcc ? 2 : 4)$
0x10	wRow	the first row number to read
0x12	wRowCount	the number of rows to read
0x14	bEcc	• 0 - if reading raw rows (32 bits are returned per row, the top 8 of which are zero) • 1 - if reading rows as ECC rows (16 bits per row are returned)

5.6.4.13. OTP_WRITE (0x0d)

Reads data out of OTP. (see also otp_access() which performs the operation). Writing is subject to the "BL" OTP permissions, which define bootloader OTP access permissions.

Table 467. PICOBOOT OTP_WRITE command structure

Offset	Name	Value / Description
0x08	bCmdId	0x0d (OTP_WRITE)
0x09	bCmdSize	0x05
0x0c	dTransferLength	$wRowCount \times (bEcc ? 2 : 4)$
0x10	wRow	the first row number to read
0x12	wRowCount	the number of rows to read
0x14	bEcc	• 0 - if writing raw rows (32 bits are provided per row, the top 8 of which are ignored) • 1 - if writing ECC rows (16 bits are provided per row, and are written with error correcting information to the OTP)

5.6.5. Control Requests

The following requests are sent to the interface via the default control pipe.

5.6.5.1. INTERFACE_RESET (0x41)

The host sends this control request to reset the PICOBOOT interface. This command:

- Clears the HALT condition (if set) on each of the bulk endpoints
- Aborts any in-process PICOBOOT or Mass Storage transfer and any flash write (this method is the only way to kill a stuck flash transfer).
- Clears the previous command result
- Removes EXCLUSIVE_ACCESS and remounts the Mass Storage drive if it was ejected due to exclusivity.

Table 468. PICOBOOT Reset PICOBOOT interface control

bmRequestType	bRequest	wValue	wIndex	wLength	Data
01000001b	01000001b	0000h	Interface	0000h	none

This command responds with an empty packet on success.

5.6.5.2. GET_COMMAND_STATUS (0x42)

Retrieve the status of the last command (which may be a command still in progress). Successful completion of a PICOBOOT Protocol Command is acknowledged over the bulk pipe, however if the operation is still in progress or has failed (stalling the bulk pipe), then this method can be used to determine the operation's status.

Table 469. PICOBOOT Get last command status control

bmRequestType	bRequest	wValue	wIndex	wLength	Data
11000001b	01000010b	0000h	Interface	0000h	none

The command responds with the following 16 byte response

Table 470. PICOBOOT Get last command status control response

Offset	Name	Description
0x00	dToken	The user token specified with the command

Offset	Name	Description	
0x04	dStatusCode	OK (0)	The command completed successfully (or is in still in progress)
		UNKNOWN_CMD (1)	The ID of the command was not recognised
		INVALID_CMD_LENGTH (2)	The length of the command request was incorrect
		INVALID_TRANSFER_LENGTH (3)	The data transfer length was incorrect given the command
		INVALID_ADDRESS (4)	The address specified was invalid for the command type; i.e. did not match the type Flash/RAM that the command was expecting
		BAD_ALIGNMENT (5)	The address specified was not correctly aligned according to the requirements of the command
		INTERLEAVED_WRITE (6)	A Mass Storage Interface UF2 write has interfered with the current operation. The command was abandoned with unknown status. Note this will not happen if you have *exclusive* access.
		REBOOTING (7)	The device is in the process of rebooting, so the command has been ignored.
		UNKNOWN_ERROR (8)	Some non-specific error occurred.
		INVALID_STATE (9)	Something happened or failed to happen in the past, and consequently the request can't (currently) be serviced.
		NOT_PERMITTED (10)	Permission violation e.g. write to read-only flash partition.
		INVALID_ARG (11)	Argument is outside of range of supported values.
		BUFFER_TOO_SMALL (12)	The provided buffer was too small to hold the result.
		PRECONDITION_NOT_MET (13)	The operation failed because another bootrom function must be called first.
		MODIFIED_DATA (14)	Cached data was determined to be inconsistent with the full version of the data it was calculated from.
		INVALID_DATA (15)	A data structure failed to validate.
		NOT_FOUND (16)	Attempted to access something that does not exist; or, a search failed.
		UNSUPPORTED_MODIFICATION (17)	Write is impossible based on previous writes; e.g. attempted to clear an OTP bit.
0x08	bCmdId	The ID of the command	
0x09	bInProgress	1 if the command is still in progress	0 otherwise
0x0a	reserved	(6 zero bytes)	

5.7. USB White-Labelling

To brand RP2350-based products, customers may replace identifying information exposed by USB interfaces. We call this **white-labelling**, and you can accomplish it in RP2350 by specifying values in OTP.

1. Write the OTP location of the white-label data structure via USB_WHITE_LABEL_ADDR (see that register description for the data structure contents).
2. Initialize the fields you wish to override in the white-label data structure and mark them valid.
3. Set USB_BOOT_FLAGS.WHITE_LABEL_ADDR_VALID to mark the white-labelling as valid.

The following fields can be modified:

5.7.1. USB Device Descriptor

The USB device descriptor includes the following 16-bit values:

- `VID` (default `0x2e8a`)
- `PID` (default `0x000f`)
- `BCD_DEVICE` (default `0x0100`)
- `LANG_ID` (default `0x0409`)

5.7.2. USB Device Strings

- `MANUFACTURER` (default "Raspberry Pi", max-length 30 UTF-16 or ASCII chars)
- `PRODUCT` (default "RP2350 Boot", max-length 30 UTF-16 or ASCII chars)
- `SERIAL_NUMBER` (default hex string of `device_id_lo`, `device_id_hi`, `wafer_id_lo`, `wafer_id_hi` - i.e. first 4 rows of OTP, max-length 30 UTF-16 or ASCII chars)

5.7.3. USB Configuration Descriptor

The USB Configuration Description is not strictly white-labelling, but is still helpful for users:

- `ATTRIBUTES_MAX_POWER_VALUES` (default `0xfa80` i.e. `bMaxPower` of `0xfa`, `bmAttributes=0x80`)

5.7.4. MSD Drive

- `VOLUME_LABEL` (default "RP2350", max-length 11 ASCII chars)

5.7.5. UF2 `INDEX.HTM` File

This is of the form:

```
<html>
    <head>
        <meta http-equiv="refresh" content="0;URL='*REDIRECT_URL*'"/>
    </head>
    <body>Redirecting to <a href='`*REDIRECT_URL*'>`*REDIRECT_NAME*</a></body>
</html>
```

- `REDIRECT_URL` (default "https://raspberrypi.com/device/RP2?version=5A09D5312E22", note the 12 hex digits are the first 6 of the `SYSINFO_GITREF_RP2350` and the first 6 of the bootrom `gitref`, max-length 127 ASCII chars)
- `REDIRECT_NAME` (default "raspberrypi.com", max-length 127 ASCII chars)

5.7.6. UF2 INFO_UF2.TXT File

This is of the form:

```
UF2 Bootloader v1.0
Model: MODEL
Board-ID: BOARD_ID
```

- `MODEL` (default "Raspberry Pi RP2350", max-length 127 ASCII chars)
- `BOARD_ID` (default "RP2350", max-length 127 ASCII chars)

5.7.7. SCSI Inquiry

Returned via the SCSI Inquiry command:

- `VENDOR` (default "RPI", max-length 8 ASCII chars)
- `PRODUCT` (default "RP2350", max-length 16 ASCII chars)
- `VERSION` (default "1", max-length 4 ASCII chars)

5.7.8. Volume Label Simple Example

Newer versions of `picotool` can load white-label data from a JSON file using the `picotool otp white-label -s <start row> <JSON filename>` command. An example JSON file to set the volume label to "SPOON" would be:

```
{
    "volume": {
        "label": "SPOON"
    }
}
```

The `<start row>` is the OTP row where the white-label structure will be written - for example `0x400`.

The full set of white-label fields which can be written using a JSON file are shown below. The `manufacturer`, `product` and `serial_number` fields support Unicode characters, if you need special characters or emoji in your product name, but this will take up twice as much room per character in the OTP for that field.

```
{
    "device": {
        "vid": "0x2e8b",
        "pid": "0x000e",
        "bcd": 2.15,
        "lang_id": "0x0c09",
        "manufacturer": "Test's Pis",
        "product": "Test RP2350?",
        "serial_number": "notnecessarilyanumber",
        "max_power": "0x20",
        "attributes": "0xe0"
    },
    "scsi": {
        "vendor": "TestPi",
        "product": "MyPi",
        "version": "v897"
    },
```

```
    "volume": {
        "label": "TestPi Boot",
        "redirect_url": "https://datasheets.raspberrypi.com/rp2350/rp2350-datasheet.pdf",
        "redirect_name": "The datasheet",
        "model": "My Test Pi",
        "board_id": "TPI-RP2350"
    }
}
```

5.7.9. Volume Label In-Depth Example

The following example demonstrates how to manually change the volume label using `picotool` to the value "SPOON":

1. First, define the row of white label structure to be `0x400`:

```
$ picotool otp set -e OTP_DATA_USB_WHITE_LABEL_ADDR 0x400
```

2. Next, because the volume label is located at index `0x8` within `OTP_DATA_USB_WHITE_LABEL_ADDR`, write to `0x408`. Define the location of the volume label string to be offset from `OTP_DATA_USB_WHITE_LABEL_ADDR` by `0x30`. For this example, "SPOON" has 5 characters, so we write `0x3005` to `0x408`:

```
$ picotool otp set -e 0x408 0x3005
```

3. Then, write the "S" and "P" characters:

```
$ picotool otp set -e 0x430 0x5053
```

4. Then, write the "O" and "O" characters:

```
$ picotool otp set -e 0x431 0x4f4f
```

5. Then, write the "N" character:

```
$ picotool otp set -e 0x432 0x4e
```

6. Finally, enable the valid override to use the new values (bit 8 marks the `VOLUME_LABEL` override as valid, and bit 22 marks the `OTP_DATA_USB_WHITE_LABEL_ADDR` override as valid):

```
$ picotool otp set -r OTP_DATA_USB_BOOT_FLAGS 0x400100
```

7. To put your changes into effect, reboot the device:

```
$ picotool reboot -u
```

5.8. UART Boot

UART boot is a minimal interface for bootstrapping a flashless RP2350 from a simple host, such as another microcontroller. It is available by default on a blank device, so it allows RP2350 to be deployed into the field on multi-device boards without loading firmware or programming OTP bits in advance.

To select UART boot, drive QSPI `CSn` low (BOOTSEL mode) and drive QSPI `SD1` high. The bootrom checks these signals shortly after device reset is released. UART `TX` appears on QSPI `SD2`, and UART `RX` appears on QSPI `SD3`.

The UART mode is `8n1`: one start bit, eight data bits, no parity, one stop bit. Data within each UART frame is sent and received LSB-first. The baud rate is fixed at 1 Mbaud.

5.8.1. Baud Rate and Clock Requirements

The nominal baud rate for UART boot is 1 Mbaud, divided from a nominal 48 MHz system clock frequency. UART boot uses the USB PLL to derive the system clock and UART baud clock, so you must either provide a crystal or drive a stable clock into the crystal oscillator XIN pad. The host baud rate must match the RP2350 baud rate within 3%.

By default the crystal is assumed to be 12 MHz, but the BOOTSEL_PLL_CFG and BOOTSEL_XOSC_CFG OTP locations override this to achieve a nominal 48 MHz system clock from any supported crystal. The same OTP configuration is used for both USB and UART boot.

TIP

You may drive a somewhat faster or slower clock into XIN without any OTP configuration, if you scale your UART baud rate appropriately. The permissible range is 7.5 to 16 MHz on XIN, limited by the PLL VCO frequency range.

5.8.2. UART Boot Shell Protocol

After the bootrom samples QSPI `CSn` and `SD1`, there will be a delay of several milliseconds as the bootrom goes through some necessary steps such as switching from the ring oscillator to the PLL, and erasing SRAM before releasing it to the Non-secure UART bootloader.

The UART bootloader signals it is ready to begin by printing the ASCII splash string `RP2350`. In bytes, this is `0x52`, `0x50`, `0x32`, `0x33`, `0x35`, `0x30`.

Before sending any commands, you must send a special knock sequence to unlock the interface. This is a measure to avoid transient effects due to noise on GPIOs and ensure the host and device are initially well-synchronised. The sequence is: `0x56`, `0xff`, `0x8b`, `0xe4`. This is the RP2040 UF2 family ID, chosen as a well-known magic number. Any sequence of bytes ending with this four-byte sequence is detected.

A UART boot shell command is always in the host-to-device direction (RP2350 receives), and consists of a single command byte, optionally followed by a 32-byte write payload. RP2350 responds with an optional 32-byte read payload followed by an echo of the command byte. You should wait for the command echo before sending the next command.

The supported commands are:

Command (ASCII)	Command (hex)	Description
`n`	`0x6e`	No-op. Do nothing, and report back when you've done it. Used to ping the interface when recovering lost synchronisation. Echoes the command byte, `'n'`.
`w`	`0x77`	Write a 32-byte payload to the current value of the read/write pointer. Increment the address pointer by 32. Echoes the command byte, `'w'`, once all 32 bytes are written to memory.
`r`	`0x72`	Read a 32-byte payload from the current value of the read/write pointer. Increment the address pointer by 32. Echoes the command byte, `'r'`, after transmitting the 32-byte read payload.

Command (ASCII)	Command (hex)	Description
c	0x63	Clear the read/write pointer. The pointer resets to the first location in SRAM 0x20000000, and you can begin a new read or write sequence from there. Echoes the command byte, 'c'.
x	0x78	Execute the payload that has been written to memory. Echoes the command byte, 'x', and then reboots, passing a RAM boot search window spanning all of main SRAM. If a valid binary was successfully written into SRAM before sending this command, it will execute.

Unrecognised commands are echoed with no other effect. More commands may be added in future versions.

5.8.3. UART Boot Programming Flow

1. Reset or power down the RP2350 device.
2. Drive CSn low to select BOOTSEL, and SD1 high to select UART.
3. Release the reset or power up the device.
4. Wait for the splash string to be transmitted on QSPI SD2 (TX).
5. Transmit the knock sequence 0x56, 0xff, 0x8b, 0xe4 on QSPI SD3 (RX)
6. Send a 'n' nop command to ensure the interface is awake; if there is no reply, send the knock sequence again.
7. Send 'w' commands until your entire write payload transfers.
8. (Optional) Send a 'c' clear command to reset the address pointer, and then send 'r' read commands to read back and verify the payload.
9. Send an 'x' execute command to attempt to run the payload.

There is no feedback from UART boot after echoing the final 'x' command. At this point the device reboots to attempt a RAM image boot on the data loaded by the Non-secure UART bootloader. If the RAM image boot fails, the bootrom falls through to the next boot source, continuing the normal boot flow. Maintaining CSn driven low and SD1 driven high will cause the bootrom to fall through back to UART boot a second time, re-sending the UART splash screen: this indicates the bootrom failed to recognise the UART boot binary.

5.8.4. Recovering from a Stuck Interface

Noise on the GPIOs may cause the UART boot shell to stop replying to commands, for example because it thinks the host is part way through a write payload, and the host thinks that it is not. To resynchronise to the start of the next command:

1. Wait 1 ms for the link to quiesce
2. Send 33 'n' NOP commands (size of longest command)
3. Wait 1 ms and flush your receive data
4. Send 1 'n' NOP command and confirm the device responds with an echoed NOP

If the interface fails to recover, reboot the device and try again. Failure may be caused by:

- Noise on GPIOs (particularly over long traces or wires)
- Incorrect baud rate matching
- An unstable frequency reference on XOSC XIN
- Mismatch of voltage levels (for example a QSPI_IOVDD of 1.8 V on RP2350, and a 3.3 V IO voltage on the host)

5.8.5. Requirements for UART Boot Binaries

A UART boot binary is a normal RAM binary. It must have a valid `IMAGE_DEF` in order for the boot path to recognise it as a bootable binary. The search window for the `IMAGE_DEF` is the whole of SRAM, but it's recommended to place it close to the beginning, because the bootrom searches linearly forward for the beginning of the `IMAGE_DEF`.

The maximum size for a UART boot binary is the entirety of main SRAM: 520 kB, or 532 480 bytes.

UART boot only supports loading to the start of SRAM, so your binary must be linked to run at address `0x20000000`. Sparse loading is not supported: your program must load as a single flat binary image.

All security requirements relating to RAM image boot apply to UART boot too. If secure boot is enabled, your binary must be signed. Likewise, if OTP anti-rollback versioning is in effect, your binary's rollback version must be no lower than the version number stored in OTP.

5.9. Metadata Block Details

5.9.1. Blocks And block loops

Blocks consist of a fixed 32-bit header, one or more items, a 32-bit relative offset to the next block, and a fixed 32-bit footer. All multi-byte values within a block are little-endian. Blocks must start on a word-aligned boundary, and the total size is always an exact number of words (a multiple of four bytes).

The final item in a block must be of type `PICOBIN_BLOCK_ITEM_LAST`, which encodes the total word count of the block's items.

The 32-bit relative link forms a linked list of blocks. To be valid, this linked list must eventually link back to the first block in the list, forming a closed block loop; failure to close the loop results in the entire linked list being ignored. The loop rule is used to avoid treating orphaned blocks from partially overwritten images being treated as valid.

Due to RAM restrictions in the boot path, size of blocks is limited to 640 bytes for `PARTITION_TABLE`s and 384 bytes for `IMAGE_DEF`s. Blocks larger than this are ignored.

The format of a simple block with two items is shown:

Item	Word	Bytes	Value
HEADER	`0`	`4`	`0xffffded3`
ITEM 0	`1`	`1`	*size_flag:1* (0 means 1-byte size, 1 means 2-byte size), *item_type:7*
		1	`s0 % 256`
		1	`s0 / 256` if *size_flag* == 1 or type-specific data for blocks that are never > 256 words
		1	Type-specific data
	…	…	…
ITEM 1	`1 + s0`	`1`	*size_flag:1* (0 means 1-byte size, 1 means 2-byte size), *item_type:7*
		1	`s1 % 256`
		1	`s1 / 256` if *size_flag* == 1 or type-specific data for blocks that are never > 256 words
		1	Type-specific data
	…	…	…

Item	Word	Bytes	Value
LAST_ITEM	1 + s0 + s1	1	0xff (*size_flag* == 1, *item type* == BLOCK_ITEM_LAST)
		2	s1 + s2 (other items' size)
		1	0x00 (pad)
LINK	2 + s0 + s1	4	Relative position in bytes of next block HEADER relative to this block's HEADER. this forms a loop, so a single block loop has 0 here.
FOOTER	3 + s0 + s1	4	0xab123579

IMAGE_DEF and PARTITION_TABLE blocks are recognised by their first item being an IMAGE_DEF or PARTITION_TABLE item.

Constants describing blocks can be found in the SDK in picobin.h in the SDK.

5.9.2. Common Block Items

The following items may appear in a IMAGE_DEF or a PARTITION_TABLE block.

5.9.2.1. VERSION item

A major/minor version number for the binary, 32 bits total, plus optionally a 16-bit rollback version and a list of OTP rows which can be read to determine the (thermometer-coded) minimum major rollback version which this device will allow to be installed. The major and minor are always present, whereas the rollback version and OTP row list are generally only included if rollback protection is required.

NOTE

The rollback version and OTP row list are only valid for IMAGE_DEFs, and are ignored on a RP2350 that has not been secured.

If the number of OTP row entries is zero, there is no rollback version for this block.

Word	Bytes	Value
0	1	0x48 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_1BS_VERSION)
	1	2 + ((num_otp_row_entries != 0) + num_row_entries + 1) / 2
	1	0x00 (pad)
	1	*num_otp_row_entries*
1	2	Minor Version
	2	Major Version
(2)	(2)	Rollback version (if *num_otp_entries* != 0)
	(2)	First 16-bit OTP Row index (if *num_otp_entries* != 0`)
...	...	Remaining 16-OTP Row indexes (padded with a zero to make a word boundary if necessary)

Each OTP row entry indicates the row number (1 through 4095 inclusive) of the first in a group of 3 OTP rows. The three OTP rows are each read as a 24 bit raw value, combined via a bitwise majority vote, and then the index of the most-significant 1 bit determines the version number. So, a single group of three rows can encode rollback versions from 0 to 23 inclusive, or, when all 24 bits are set, an indeterminate version of at least 24. Each additional OTP row index indicates a further group of 3 rows that increases the maximum version by 24.

There is no requirement for different OTP row entries to be contiguous in OTP. They should not overlap, though the

bootrom does not need to check this (the boot signing tool may).

NOTE

For this entry to be considered valid, the number of available bits in the indicated OTP rows must be *strictly greater than* the rollback version. This means that it is always possible to determine that the device's minimum rollback version is greater than the rollback version indicated in this block, even if we don't know the full list of OTP rows used by later major versions.

The major/minor version are used to disambiguate which is newer out of two binaries with the same major rollback version. For example, to select which A/B image to boot from. when no major rollback version is specified, A/B comparisons will treat the missing major version as zero, but no rollback check will be performed.

5.9.2.2. HASH_DEF item

Optional item with information about what how to hash:

Word	Bytes	Value
0	1	0x47 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_1BS_HASH_DEF)
	1	0x03 (*size_lo*)
	1	0x00 (pad)
	1	0x01 (PICOBIN_HASH_SHA-256)
1	2	Number of words of block hashed (not including HEADER word at the start of the block)
	2	0x0000 (pad)

block_words_hashed must include this item if using this item for a signature.

The most recent LOAD_MAP item (see Section 5.9.3.2) that defines *what* to hash.

5.9.2.3. HASH_VALUE item

Optional item containing a hash value that can be used by the bootrom to verify the hash of an image or partition table when not using signatures.

Word	Bytes	Value
0	1	0x09 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_HASH_VALUE)
	1	0x01 + n where *n* is the number of hash words included (1-8)
	2	0x0000 (pad)
1	4	Hash Value (lowest significant 32 bits)
…	…	…
n	4	Hash Value (highest significant 32 bits)

TIP

Whilst a SHA-256 hash is 8 words, you can include fewer (down to 1 word) to save space if you like, and only that many words will be compared against the full 8-word hash at runtime.

This HASH_VALUE item is paired with the most recent HASH_DEF item (Section 5.9.2.2) which defines what is being hashed.

5.9.2.4. SIGNATURE item

Optional item containing cryptographic signature that can be used by the bootrom to signature check the hashed contents of an image or partition table.

Word	Bytes	Value
0	1	0x4b (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_SIGNATURE)
	1	0x21 (Block size in words)
	1	0x00 (pad)
	1	0x01 (PICOBIN_SIGNATURE_SECP256K1)
1	4	Public Key (lowest significant 32 bits)
…	…	…
16	4	Public Key (highest significant 32 bits)
17	4	Signature of hash (lowest significant 32 bits)
…	…	…
32	4	Signature of hash (highest significant 32 bits)

This SIGNATURE item is paired with the most recent HASH_DEF item (Section 5.9.2.2) which defines what the hash value whose signature is checked.

5.9.3. Image Definition Items

5.9.3.1. IMAGE_DEF item

The IMAGE_DEF item must be the first item within an Image Definition:

Word	Bytes	Value
0	1	0x42 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_1BS_IMAGE_TYPE)
	1	0x01 (Block size in words)
	2	*image_type_flags*

The flags are defined in the SDK in picobin.h in the SDK, but are summarised here:

Bits	Field	Values
0-3	Image Type	0 IMAGE_TYPE_INVALID 1 IMAGE_TYPE_EXE : Image is executable 2 IMAGE_TYPE_DATA : Image is valid, but is not intended to be executed 3 *reserved*

Bits	Field	Values
The remaining bits are specific to the Image Type Values are only currently defined for the EXE Image Type:		
4-5	EXE Security	0 EXE_SECURITY_UNSPECIFIED 1 EXE_SECURITY_NS : Image runs in Non-secure mode 2 EXE_SECURITY_S : Image runs in Secure mode 3 *reserved*
6-7	*reserved*	0
8-10	EXE CPU	0 EXE_CPU_ARM : Image is for the Arm architecture 1 EXE_CPU_RISCV : Image is for the RISC-V architecture 2-7 *reserved*
11	*reserved*	0
12-14	EXE CHIP	0 EXE_CHIP_RP2040 1 EXE_CHIP_RP2350 2-7 *reserved*
15	EXE TBYB	0 *not set* 1 EXE_TBYB : Image is flagged for "Try Before You Buy"

5.9.3.2. LOAD_MAP item

Optional item with a similar representation to the ELF program header. This is used both to define the content to hash, and also to "load" data before image execution (e.g. a secure flash binary can be loaded into RAM prior to both signature check and execution).

The load map is a collection of runtime address, physical address, size and flags.

1. For a "packaged" binary, the information tells the bootrom where to load the code/data.
2. For a hashed or signed binary, the runtime addresses and size indicate code/data that must be included in the hash to be verified or signature checked.

NOTE

If the runtime_address is in equal to the storage_address, then data is never copied, it is just hashed in place.

Explanation of terms:

physical address

Where the data is stored in the logical address space of the image. For instance, the start of a flash image, even if stored in a partition, could have a physical address of `0x10000000`. The closest ELF concept is LMA.

runtime address

The address of the data at runtime. The closest ELF concept is VMA.

storage address

an absolute location where the data is stored in flash. Not necessarily the same as physical address for flash when partitions are in use.

RP2350 uses physical addresses in the `LOAD_MAP`, not storage addresses, since this data is written by a tool working on the ELF which will not necessarily know where the binary will finally be stored in flash.

This serves several purposes:

Word	Bytes	Value
0	1	0x06 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_LOAD_MAP)
	2	1 + num_entries * 3 (Block size in words)
	1	*absolute:1*, *num_entries:7*
1-3	Load Map Entry 0	
	4	• if *absolute* == 0 *storage_start_address_rel* (storage start address relative to the address of this word) • if *absolute* == 1 *storage_start_address* (absolute storage start address) Note: If this value is 0x00000000 irrespective of the value of the *absolute* flag, then the runtime address range is filled with zeros. In this case, the 32 bit *size* itself is hashed rather than *size* zero bytes.
	4	*runtime_start_address* (absolute runtime start address)
	4	• if *absolute* == 0 *size* (of memory range in bytes) • if *absolute* == 1 *storage_end_address* (absolute storage end address)
(4-6)	(Load Map Entry 1)	
	...	...

NOTE

All addresses must be word aligned, and sizes a multiple of 4

5.9.3.2.1. XIP Pinning via LOAD_MAP

Normally, when entering a binary, the XIP cache is un-pinned and flushed. This makes sense both for entering a flash binary, and for security purposes.

If, however, you have a non-flash binary with code or data in the XIP RAM address space, then you need to add a special LOAD_MAP entry to indicate to the bootrom that the XIP contents should be pinned.

Any load-map entry (with storage_address == runtime_address) and a valid size of greater than zero will suffice, as for example in this simple load map:

Word	Bytes	Value
0	1	0x06 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_LOAD_MAP)
	2	0x04 (Block size in words)
	1	0x81 (*absolute* == 1, *num_entries* == 1)
1-3	Load Map Entry 0	
	4	XIP_SRAM_BASE (*storage_start_address*)
	4	XIP_SRAM_BASE (*runtime_start_address*)
	4	0x04 (*size* in bytes)

5.9.3.3. VECTOR_TABLE item

Optional Arm only item for that specifies the location of the initial Arm vector table. The entry_point/initial_sp will be taken from here if present (unless there is also an ENTRY_POINT Item). Note if there is no ENTRY_POINT or VECTOR_TABLE, Item, then a VECTOR_TABLE at the start of the image is assumed.

Word	Bytes	Value
0	1	0x03 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_1BS_VECTOR_TABLE)
	1	0x02 (Block size in words)
	2	0x0000 (pad)
1	4	Vector table (runtime) address

NOTE

The VECTOR_TABLE Item is ignored on RISC-V.

5.9.3.4. ENTRY_POINT item

Optional item with info on initial PC, SP, and optionally the SP limit

Word	Bytes	Value
0	1	0x44 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_1BS_ENTRY_POINT)
	1	0x03 or 0x04 (Block size in words)
	2	0x0000 (pad)
1	4	Inital PC (runtime) address (aka entry point)
2	4	Initial SP address (aka stack pointer)
(3)	4	Optional SP limit address (aka stack limit)

5.9.3.5. ROLLING_WINDOW_DELTA item

Optional item that allow for binaries that aren't intended to be run at 0x10000000. Note that this delta is in addition to the roll resulting from the binary being stored in a partition in flash.

Word	Bytes	Value
0	1	0x05 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_1BS_ROLLING_WINDOW_DELTA)
	1	0x02 (Block size in words)
	2	0x0000 (pad)
1	4	signed 32 bit delta

The delta is the number of bytes into the image that 0x10000000 should be mapped.

If positive, the delta must be a multiple of 4 kB, and allows for "skipping over" other data before the start of the binary. If negative, the delta must be a multiple of 4 MB, and allows for running flash binaries linked to run at 0x10400000, 0x01080000 and 0x010c0000 as well as the standard 0x10000000

NOTE

The `ROLLING_WINDOW_DELTA` Item is ignored for non-flash binaries.

5.9.4. Partition Table Items

Partition tables allows dividing the 32 MB flash region (2 × 16 MB) into partitions. Permissions and other partition attributes may be specified for each partition, along with permissions for the un-partitioned space.

The permission specify read/write access for Secure code, Non-secure code, and "NSBoot" which refers to the boot loader (and PICOBOOT)

NOTE

These permissions are only advisory to Secure code, however they are respected by flash_op(), the PICOBOOT flash access commands, and UF2 downloads.

5.9.4.1. PARTITION_TABLE item

Word	Bytes	Value
0	1	0x44 (*size_flag* == 0, *item_type* == PICOBIN_BLOCK_ITEM_PARTITION_TABLE)
	2	Block size in words
	1	*singleton_flag:1*, *pad:3* , *partition_count:4*
1	4	*unpartitioned_space_permissions_and_flags*
Partition 0		
2	4	*permissions_and_location* for partition 0
3	4	*permissions_and_flags* for partition 0
	if _partition_0_has_id:	
3	4	*partition_0_ID_lo*
4	4	*partition_0_ID_hi*
	one word per additional family ID (can be none):	
x	4	*partition_0_family ID_0*
x + 1	4	*partition_0_family ID_1*
	…	…
	if _partition_0_has_name:	
y	1	*reserved:1* (0), *name_len_bytes:7*
	1	*partition_0_name_byte_0*
	1	*partition_0_name_byte_1*
	1	*partition_0_name_byte_2*

Word	Bytes	Value
y+1	1	*partition_0_name_byte_3*
	1	*partition_0_name_byte_4*
	1	*partition_0_name_byte_5*
	1	*partition_0_name_byte_6*
…	…	…
	?	*partition_0_name_byte_n_minus_x* to *partition_0_name_byte_n_minus_2*
	1	*partition_0_name_byte_n_minus_1*
	?	(padding zero bytes to reach word alignment)
(Partition 1)		
…	…	…

5.9.4.2. Partition location, permissions and flags

Two common words are stored in the partition table for both un-partitioned space and each partition. These common words describe the size/location, along with access permissions and various flags.

The permission fields are repeated in both words, hence the two words are `permissions_and_location` and `permissions_and_flags`.

Table 471. Permission Fields. `'P'` means the field applies to partitions, `'U'` means the field applies to un-partitioned space, however the word "partition" is always used in the description

Mask	AppliesTo	Description
`0x04000000u`	`'P' 'U'`	`PERMISSION_S_R_BITS`
		If set, the partition is readable by Secure code. See Section 5.1.3
`0x08000000u`	`'P' 'U'`	`PERMISSION_S_W_BITS`
		If set, the partition is writable by Secure code. See Section 5.1.3
`0x10000000u`	`'P' 'U'`	`PERMISSION_NS_R_BITS`
		If set, the partition is readable by Non-secure code. See Section 5.1.3
`0x20000000u`	`'P' 'U'`	`PERMISSION_NS_W_BITS`
		If set, the partition is writable by Non-secure code. See Section 5.1.3
`0x40000000u`	`'P' 'U'`	`PERMISSION_NSBOOT_R_BITS`
		If set, the partition is readable by NSBOOT (i.e. boot loader) Secure code. See Section 5.1.3
`0x80000000u`	`'P' 'U'`	`PERMISSION_NSBOOT_W_BITS`
		If set, the partition is writable by NSBOOT (i.e. boot loader) Secure code. See Section 5.1.3

Table 472. Location Fields. `'P'` means the field applies to partitions, `'U'` means the field applies to un-partitioned space, however the word "partition" is always used in the description

Mask	AppliesTo	Description
`0x00001fffu`	`'P' 'U'`	`LOCATION_FIRST_SECTOR_BITS`
		The sector number (0-4095) of the first sector in the partition (a sector is 4 kB)
`0x03ffe000u`	`'P' 'U'`	`LOCATION_LAST_SECTOR_BITS`
		The sector number (0-4095) of the last sector in the partition (a sector is 4 kB)

Table 473. Flags Fields. 'P' means the field applies to partitions, 'U' means the field applies to un-partitioned space, however the word "partition" is always used in the description

Mask	AppliesTo	Description
0x00000001u	'P'	FLAGS_HAS_ID_BITS
		If set, the partition has a 64 bit identifier
0x00000006u	'P'	FLAGS_LINK_TYPE_BITS
		The type of link stored in the partition: • 0x0 - None • 0x1 - A_PARTITION : This is a "B" partition, and The LINK_VALUE field stores the partition number of the corresponding "A" partition • 0x2 - OWNER : This is an "A" partition, and the LINK_VALUE field stores the partition number of the owning partition (which should also be an "A" partition).
0x00000078u	'P'	FLAGS_LINK_VALUE_BITS
		If LINK_TYPE is non zero, then this field holds the partition number of the linked partition.
0x00000180u	'P'	FLAGS_ACCEPTS_NUM_EXTRA_FAMILIES_BITS
		0-3 the number of extra non-standard UF2 family ids the partition accepts.
0x00000200u	'P'	FLAGS_NOT_BOOTABLE_ARM_BITS
		If set then this partition is marked non-bootable on Arm, and will be ignored during Arm boot. Setting this for non Arm bootable partitions can improve boot performance.
0x00000400u	'P'	FLAGS_NOT_BOOTABLE_RISCV_BITS
		If set then this partition is marked non-bootable on RISC-V, and will be ignored during RISC-V boot. Setting this for non RISC-V bootable partitions can improve boot performance.
0x00000800u	'P'	FLAGS_UF2_DOWNLOAD_AB_NON_BOOTABLE_OWNER_AFFINITY
0x00001000u	'P'	FLAGS_HAS_NAME_BITS
		If set, the partition has a name.
0x00002000u	'P' 'U'	FLAGS_UF2_DOWNLOAD_NO_REBOOT_BITS
		If set, the RP2350 will not reboot after dragging a UF2 into this partition.
0x00004000u	'P' 'U'	FLAGS_ACCEPTS_DEFAULT_FAMILY_RP2040_BITS
		If set, a UF2 with the *RP2040* family id 0xe48bff56 may be downloaded into this partition.
0x00008000u	'U'	FLAGS_ACCEPTS_DEFAULT_FAMILY_ABSOLUTE_BITS
		If set for un-partitioned spaced, a UF2 with the ABSOLUTE family id 0xe48bff57 may be downloaded onto the RP2350 and will be written at the addresses specified in the UF2 without regard to partition locations. Partition-defined flash access permissions will however still be respected (i.e. the UF2 download will fail if it needs to write over a read-only region of flash).
0x00010000u	'P' 'U'	FLAGS_ACCEPTS_DEFAULT_FAMILY_DATA_BITS
		If set, a UF2 with the DATA family id 0xe48bff58 may be downloaded into this partition.
0x00020000u	'P' 'U'	FLAGS_ACCEPTS_DEFAULT_FAMILY_RP2350_ARM_S_BITS
		If set, a UF2 with the RP2350_ARM_S family id 0xe48bff59 may be downloaded into this partition.
0x00040000u	'P' 'U'	FLAGS_ACCEPTS_DEFAULT_FAMILY_RP2350_RISCV_BITS
		If set, a UF2 with the RP2350_RISC_V family id 0xe48bff5a may be downloaded into this partition.

Mask	AppliesTo	Description
0x00080000u	'P' 'U'	FLAGS_ACCEPTS_DEFAULT_FAMILY_RP2350_ARM_NS_BITS
		If set, a UF2 with the RP2350_ARM_NS family id 0xe48bff5b may be downloaded into this partition.
0x03f00000u	'P' 'U'	reserved; should be 0

5.9.5. Minimum Viable Image Metadata

A minimum amount of metadata (i.e. a valid IMAGE_DEF block) must be embedded in any binary for the bootrom to recognise it as a valid program image, as opposed to, for example, blank flash contents or a disconnected flash device. This must appear within the first 4 kB of a flash image, or anywhere in a RAM or OTP image.

Unlike RP2040, there is no requirement for flash binaries to have a checksummed "boot2" flash setup function at flash address 0. The RP2350 bootrom performs a simple best-effort XIP setup during flash scanning, and a flash-resident program can continue executing in this state, or can choose to reconfigure the QSPI interface at a later time for best performance.

5.9.5.1. Minimum Arm IMAGE_DEF

Assuming CRIT1.SECURE_BOOT_ENABLE is clear, the minimum valid IMAGE_DEF is the following 20-byte sequence:

Word	LE Value	Bytes	Description
0	0xffffded3	4	PICOBIN_BLOCK_MARKER_START
1	0x10210142	1	0x42(*item_type == PICOBIN_BLOCK_ITEM_1BS_IMAGE_TYPE*)
		1	0x01 (Item is 1 word in size)
		2	0x1021 (PICOBIN_IMAGE_TYPE_IMAGE_TYPE_AS_BITS(EXE) | PICOBIN_IMAGE_TYPE_EXE_SECURITY_AS_BITS(S) | PICOBIN_IMAGE_TYPE_EXE_CPU_AS_BITS(Arm) | PICOBIN_IMAGE_TYPE_EXE_CHIP_AS_BITS(RP23500))
2	0x000001ff	1	0xff(*size_type* == 1, *item_type_* == PICOBIN_BLOCK_ITEM_2BS_LAST)
		2	0x0001 (*size*)
		1	0x00 (*pad*)
3	0x00000000	4	Relative pointer to next block in block loop - 0x00000000 means link to self, i.e. a loop containing just this block
4	0xab123579	4	PICOBIN_BLOCK_MARKER_END

The *LE Value* column indicates a 32-bit little-endian value that should appear verbatim in your program image.

Since the above block does not specify an explicit entry point, the bootrom will assume the binary starts with a Cortex-M vector table, and enter via the reset handler and initial stack pointer specified in that table (offsets +4 and +0 bytes into the table). An explicit vector table pointer can be provided by a PICOBIN_BLOCK_ITEM_1BS_VECTOR_TABLE item, or the entry point can be specified directly by a PICOBIN_BLOCK_ITEM_1BS_ENTRY_POINT item.

5.9.5.2. Minimum RISC-V IMAGE_DEF

The minimum valid IMAGE_DEF is the following 20-byte sequence:

Word	LE Value	Bytes	Description
0	0xffffded3	4	PICOBIN_BLOCK_MARKER_START
1	0x11010142	1	0x42(*item_type == PICOBIN_BLOCK_ITEM_1BS_IMAGE_TYPE*)
		1	0x01 (Item is 1 word in size)
		2	0x1101 (PICOBIN_IMAGE_TYPE_IMAGE_TYPE_AS_BITS(EXE) | PICOBIN_IMAGE_TYPE_EXE_CPU_AS_BITS(RISCV) | PICOBIN_IMAGE_TYPE_EXE_CHIP_AS_BITS(RP23500))
2	0x000001ff	1	0xff(*size_type* == 1, *item_type_* == PICOBIN_BLOCK_ITEM_2BS_LAST)
		2	0x0001 (*size*)
		1	0x00 (*pad*)
3	0x00000000	4	Relative pointer to next block in block loop - 0x00000000 means link to self, i.e. a loop containing just this block
4	0xab123579	4	PICOBIN_BLOCK_MARKER_END

The *LE Value* column indicates a 32-bit little-endian value that should appear verbatim in your program image.

Since the above block does not specify an explicit entry point, the bootrom will enter the binary at its lowest address, which is the default behaviour on RISC-V. This default entry point can be overridden by a PICOBIN_BLOCK_ITEM_1BS_ENTRY_POINT item. Note that PICOBIN_BLOCK_ITEM_1BS_VECTOR_TABLE is not valid on RISC-V, as unlike Cortex-M the RISC-V vector table does not define the program entry point.

5.10. Example Boot Scenarios

This section describes the setup and configuration steps for various different boot scenarios.

5.10.1. Secure Boot

To enable secure boot on RP2350, you must:

1. Set the SHA-256 hashes of the boot keys you will be using in BOOTKEY0_0 onwards
2. Set bits in BOOT_FLAGS1.KEY_VALID for the keys you will be using
3. Optionally set bits in BOOT_FLAGS1.KEY_INVALID for all unused keys — this is recommended to prevent a malicious actor installing their own boot keys at a later date
4. Set CRIT1.SECURE_BOOT_ENABLE to turn on secure boot.

NOTE

These steps are the minimum for enabling secure boot support *in the bootrom*. See Section 10.5 for additional steps you must take to fully secure your device, such as disabling hardware debug.

All of the above can be achieved with `picotool`. For example, when signing using `picotool seal` you can add an OTP JSON output file, to which it will add the relevant OTP field values to enable secure boot (BOOTKEY0_0, BOOT_FLAGS1.KEY_VALID and CRIT1.SECURE_BOOT_ENABLE):

```
$ picotool seal --sign unsigned.elf signed.elf private.pem /path/to/otp.json
```

To configure the SDK to output this OTP JSON file when signing, add the following command to your `CMakeLists.txt`:

```
pico_set_otp_key_output_file(target_name /path/to/otp.json)
```

You can then issue the following command to write this OTP JSON file to the device, thus enabling secure boot:

```
$ picotool otp load /path/to/otp.json
```

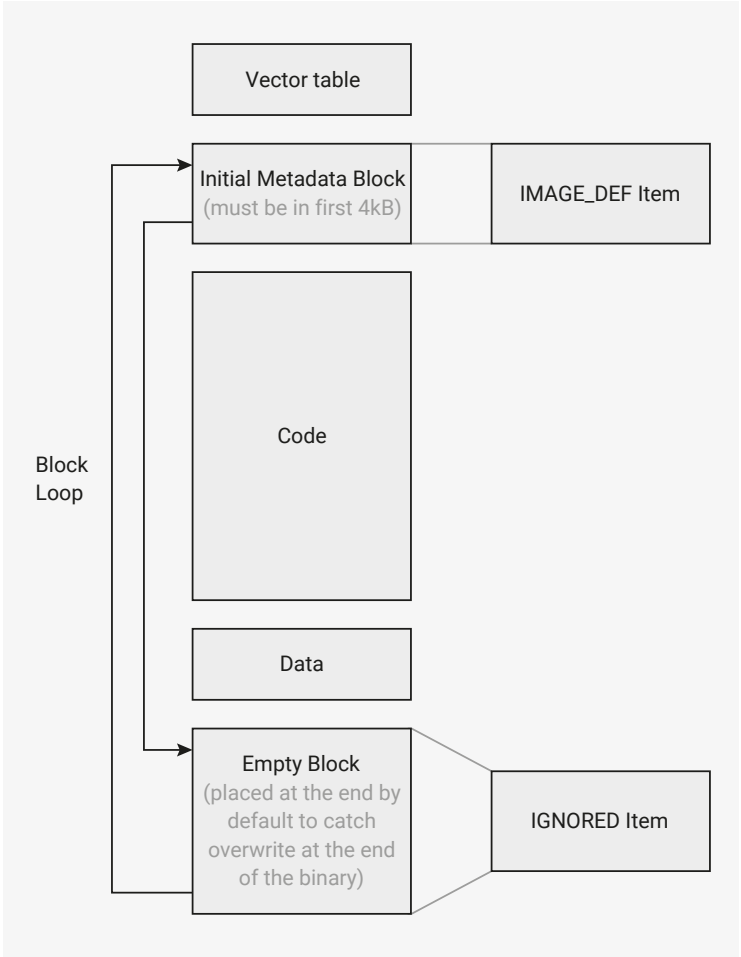
Once secure boot is enabled, the bootrom verifies signatures of images from all supported media: flash, OTP, and images preloaded into SRAM via the UART and USB bootloaders. At this point you lose the ability to run unsigned images; during development you may find it more convenient to leave secure boot disabled. The next section describes the generation of **signed images** to run on a secure-boot-enabled device.

5.10.2. Signed images

TIP

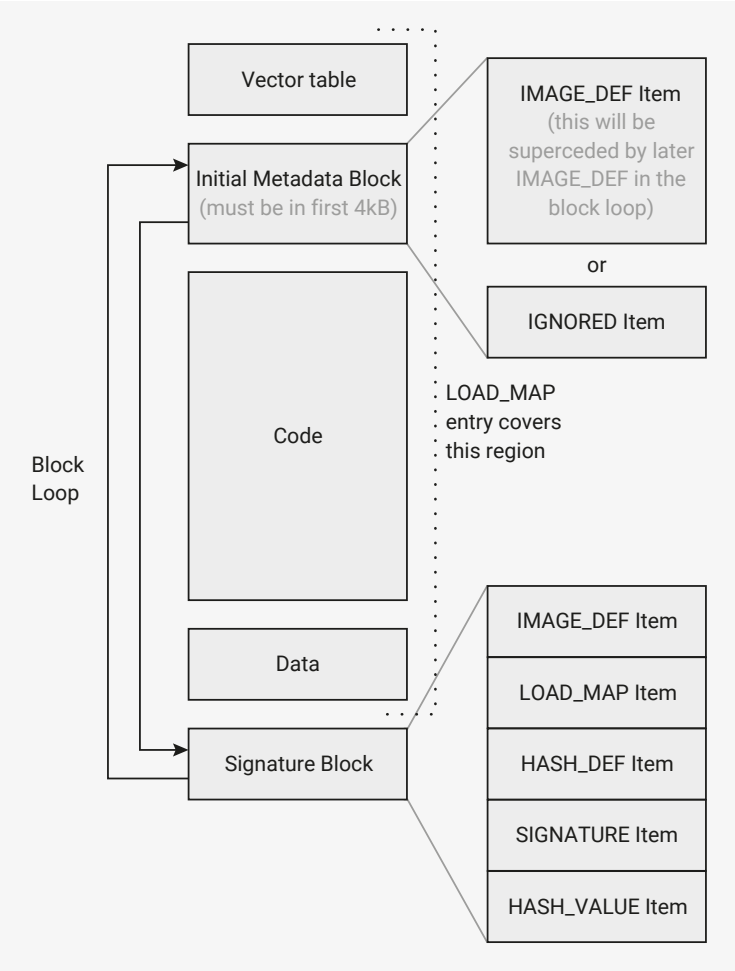
This section refers to the concepts of block loops and image definitions (and the associated `IMAGE_DEF` data structure) described in Section 5.1. You should read the bootrom concepts section before this one.

An example of an image (and its block loop) produced by the SDK is shown below.



The first block must be within the first 4 kB of the image, and is an `IMAGE_DEF` block describing the image. This block is linking to an empty block at the end of the image, that is included in the block loop to help detect partially written binaries. If the end of the image is missing or overwritten, then the block loop not be properly closed and will be considered invalid.

`picotool` can be used to sign a binary, in which case it modifies the image as follows:



Note that the marker block at the end of the image has been replaced with a new `IMAGE_DEF` block including the first block's information along with additional new information. The new information includes the signature (or hash value if hashing only), along with a `LOAD_MAP` entry indicating the regions of the image that are signed or hashed.

At runtime, the bootrom will pick the last valid `IMAGE_DEF` in the block loop as the one to boot.

Signing requires a SHA-256 hash of the data specified in the `LOAD_MAP`, along with the words of the block specified by the `HASH_VALUE` Item (which must include the first word of the SIGNATURE Item). This hash is then signed with an ECDSA secp256k1 private key, to produce the 64 byte signature stored in the SIGNATURE Item.

For secure boot, it is recommended to use packaged SRAM binaries instead of flash binaries, as the signature check is only performed during boot, so a malicious actor with physical access could replace the data on the external flash after the signature check to run unsigned code.

To sign and/or hash a binary in the SDK, you can add the following functions to your CMakeLists.txt file:

```
pico_sign_binary(target_name /path/to/keyfile.pem)
pico_hash_binary(target_name)
```

This will invoke the `picotool seal` command to sign and/or hash your binary when you call `pico_add_extra_outputs`. You can manually invoke `picotool seal` to sign and/or hash a binary using:

```
$ picotool seal --sign --hash unsigned.elf signed.elf private.pem
```

5.10.3. Packaged Binaries

A packaged binary is an SRAM/XIP RAM-only binary that has been post-processed for storage in flash. To create a packaged SRAM binary, you can take a binary compiled to run in SRAM (`no_flash` binary in the SDK) and add a relative `LOAD_MAP` Item into the `IMAGE_DEF` block, with the runtime address(es) in SRAM. The subsequent binary can then be run normally from RAM, or stored in flash to be loaded into RAM by the bootrom. This `LOAD_MAP` item will be added to all binaries when using `picotool seal`.

To package binaries in the SDK, add the following to your CMakeLists.txt file. This will target the UF2 file to the start of flash when dragged and dropped, and will invoke `picotool seal` to add an appropriate `LOAD_MAP`.

```
pico_package_uf2_output(target_name 0x10000000)
```

Alternatively you can use an absolute `LOAD_MAP`, with the storage_address in Flash and the runtime_address in SRAM, but these binaries will can only be run after storing in flash and cannot be booted directly in SRAM for debugging.

For example, if you have a binary compiled to run at `0x20000000` of length `0x8000`, and a metadata block at the end of the binary containing the `LOAD_MAP` as the second Item (after the `IMAGE_DEF`, which means the `LOAD_MAP` is 8 bytes into the block), then the relative `LOAD_MAP` would be:

Word	Bytes	Value
`0`	`1`	`0x06` (*size_flag* == 0, *item_type* == `PICOBIN_BLOCK_ITEM_LOAD_MAP`)
	`2`	`0x04` (Block size in words)
	`1`	`0x01` (*absolute* == 0, *num_entries* == 1)
`1-3`	Load Map Entry 0	
	`4`	`-0x8008` = `0xFFFF7FF8` (*storage_start_address_rel*)
	`4`	`0x20000000` (*runtime_start_address*)
	`4`	`0x8000` (*size* in bytes)

The absolute `LOAD_MAP` would be:

Word	Bytes	Value
`0`	`1`	`0x06` (*size_flag* == 0, *item_type* == `PICOBIN_BLOCK_ITEM_LOAD_MAP`)
	`2`	`0x04` (Block size in words)
	`1`	`0x81` (*absolute* == 1, *num_entries* == 1)
`1-3`	Load Map Entry 0	
	`4`	`0x10000000` (*storage_start_address*)
	`4`	`0x20000000` (*runtime_start_address*)
	`4`	`0x20008000` (*runtime_end_address*)

5.10.4. A/B Booting

This is a common boot scenario, to be able to update the software without overwriting it. A simple partition layout would be:

```
Partition 0
  Accepts UF2 Families: rp2350-arm-s, rp2350-riscv
Partition 1
  Accepts UF2 Families: rp2350-arm-s, rp2350-riscv
  Link Type: "A"
  Link Value: 0
```

This is a partition table with 2 partitions, where partition 0 is the A partition of partition 1 (which is thus the B partition).

NOTE

To avoid confusion, it is a recommended best practice to have the same permissions for both partitions, and for both partitions to accept the same UF2 families. The bootrom will only look at the UF2 families from the A partition when deciding if a given A/B pair accepts a particular family, and will not allow download to partition A even if it is writable if partition B isn't.

When dragging a UF2 to the device, it will target whichever partition is not currently booting. The bootrom will then perform a `FLASH_UPDATE` boot into the new binary (see Section 5.1.16)

NOTE

When starting with blank A/B partitions, the first download actually goes into partition B.

To create the partition table above with `picotool partition create`, the following json could be used:

```
{
  "version": [1, 0],
  "unpartitioned": {
    "families": ["absolute"],
    "permissions": {
      "secure": "rw",
      "nonsecure": "rw",
      "bootloader": "rw"
    }
  },
  "partitions": [
    {
      "name": "Example A",
      "id": 0,
      "size": "2044K",
      "families": ["rp2350-arm-s", "rp2350-riscv"],
      "permissions": {
        "secure": "rw",
        "nonsecure": "rw",
        "bootloader": "rw"
      }
    },
    {
      "name": "Example B",
      "id": 1,
      "size": "2044K",
      "families": ["rp2350-arm-s", "rp2350-riscv"],
      "permissions": {
        "secure": "rw",
        "nonsecure": "rw",
        "bootloader": "rw"
      },
      "link": ["a", 0]
```

```
    }
  ]
}
```

This can then be installed onto the device using `picotool load`, or UF2 drag and drop if you output the partition table as a UF2 file.

5.10.5. A/B Booting with Owned Partitions

The concept of owned partitions applies when:

- you require separate data partitions (which generally won't contain a block loop), but
- you would like these to be associated with a specific boot partition in an A/B pair

An example partition table for this scenario would be:

```
Partition 0
  Accepts Families: rp2350-arm-s, rp2350-riscv
Partition 1
  Accepts Families: rp2350-arm-s, rp2350-riscv
  Link Type: "A"
  Link Value: 0
Partition 2
  Accepts Families: data
  Link Type: "Owner"
  Link Value: 0
  ignored_during_arm_boot: true
  ignored_during_riscv_boot: true
Partition 3
  Accepts Families: data
  Link Type: "A"
  Link Value: 2
  ignored_during_arm_boot: true
  ignored_during_riscv_boot: true
```

This is a partition table with 4 partitions. As before, partition 0 is the A partition of partition 1 (which is thus a B partition). Additionally, partition 2 is the A partition of partition 3 (which is thus a B partition). Finally, partition 0 is the "owner" partition of partition 2.

As a result partitions 2 and 3 "belong to" partitions 0 and 1.

When downloading a UF2 into an owned partition, the bootloader will select which partition out of 2/3 it goes to target based on which partition out of 0/1 is currently booting. For example, if partition 1 is currently booting (due to having a higher version than partition 0), then any UF2 downloads with the data family ID will target partition 3.

TIP

There is a flag in each partition in the partition, that you can use to swap the "affinity" (i.e. have the data family ID target partition 2 instead of partition 3 in the scenario above.

NOTE

Only the get_uf2_target_partition() bootrom function considers owner partitions. The pick_ab_partition() function always pick solely based on the A/B partition it is passed, in other words if passed partition 2, it would not look at partitions 0 and 1.

To create this partition table using `picotool partition create`, the following JSON could be used:

```
{
  "version": [1, 0],
  "unpartitioned": {
    "families": ["absolute"],
    "permissions": {
      "secure": "rw",
      "nonsecure": "rw",
      "bootloader": "rw"
    }
  },
  "partitions": [
    {
      "name": "Example A",
      "id": 0,
      "size": 128k,
      "families": ["rp2350-arm-s", "rp2350-riscv"],
      "permissions": {
        "secure": "rw",
        "nonsecure": "rw",
        "bootloader": "rw"
      }
    },
    {
      "name": "Example B",
      "id": 1,
      "size": 128k,
      "families": ["rp2350-arm-s", "rp2350-riscv"],
      "permissions": {
        "secure": "rw",
        "nonsecure": "rw",
        "bootloader": "rw"
      },
      "link": ["a", 0]
    },
    {
      "name": "Example a",
      "id": 2,
      "size": 20k,
      "families": ["data"],
      "permissions": {
        "secure": "rw",
        "nonsecure": "rw",
        "bootloader": "rw"
      },
      "link": ["owner", 0],
      "ignored_during_arm_boot": true,
      "ignored_during_riscv_boot": true
    },
    {
      "name": "Example b",
      "id": 3,
      "size": 20k,
      "families": ["data"],
```

```
      "permissions": {
        "secure": "rw",
        "nonsecure": "rw",
        "bootloader": "rw"
      },
      "link": ["a", 2],
      "ignored_during_arm_boot": true,
      "ignored_during_riscv_boot": true
    }
  ]
}
```

This can then be installed onto the device using `picotool load`, or UF2 drag and drop if you output the partition table as a UF2 file.

5.10.6. Custom Bootloader

In this scenario, a bootloader is run before booting into an image. This could perform additional validation, or set up peripherals for use by the image. For this to work, the block loop must contain both an `IMAGE_DEF` for the bootloader and a `PARTITION_TABLE` to define the flash layout.

In this example, we want to have A / B versions of the bootloader, so we use both slot 0 and slot 1. See Section 5.1.15 for more details of this, as you may well need to increase the size of slot 0 in order to fit the bootloader.

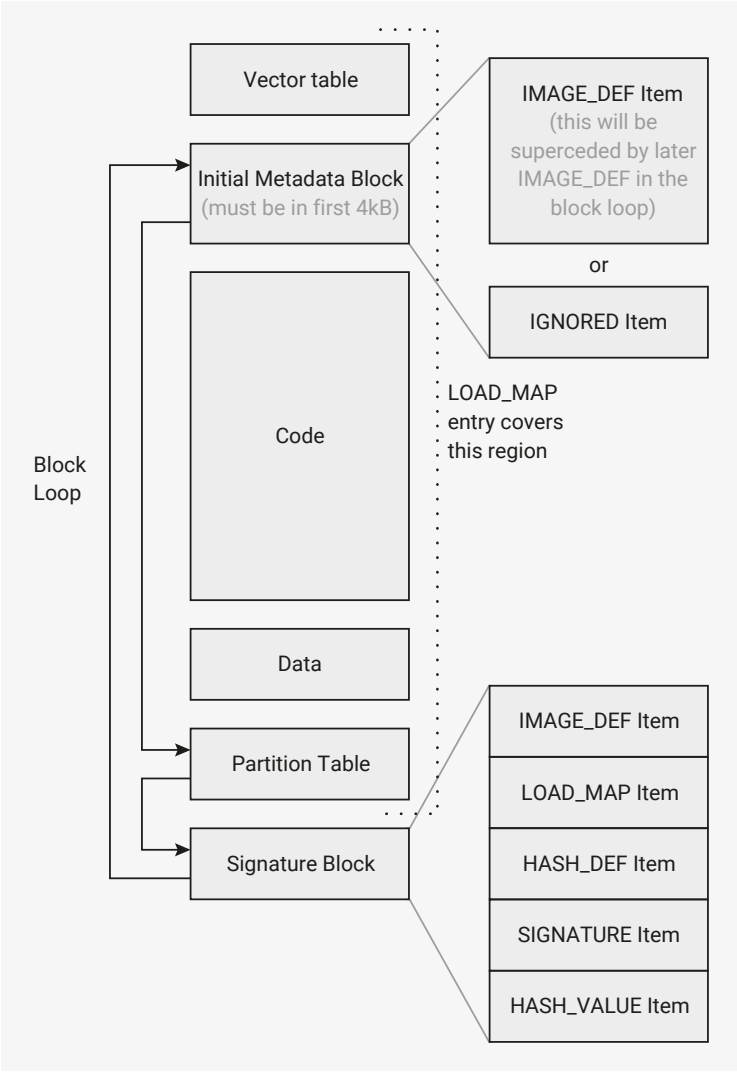
WARNING

Making a slot size change is not reversible, so feel free to leave out slot 1 if you try this in practice.

An example flash layout might resemble the following:

```
Slot 0 (0x00000000-0x00008000)
 Bootloader Image 0
 Partition Table 0
Slot 1 (0x00008000-0x00010000)
 Bootloader Image 1
 Partition Table 1
Partition 0 (0x00010000-0x00020000)
 Binary A
Partition 1 (0x00020000-0x00030000)
 Link Type: "A"
 Link Value: 0
 Binary B
```

The block loop with both `IMAGE_DEF` and `PARTITION_TABLE` might look like this (after signing in this case):



Note the 3 blocks in the block loop:

1. Original block in first 4 kB (contents doesn't matter, as it will be superseded by the later `IMAGE_DEF`)
2. `PARTITION_TABLE` at end of binary
3. Signed `IMAGE_DEF`

NOTE

It is possible to sign both the `PARTITION_TABLE` and the `IMAGE_DEF` separately, however for the fastest boot speed, the bootrom also allows you to use a "covering" `LOAD_MAP` in the `IMAGE_DEF`. As long as the `LOAD_MAP` defined area to be hashed/signed includes the entirety of the `PARTITION_TABLE` block, the "covering" signature is used to validate the `PARTITION_TABLE` too.

For the bootloader to find and launch a new image, it may wish to utilize various bootrom methods:

- get_partition_table_info() to get the full partition information.
- or get_partition_table_info() with `SINGLE_PARTITION`, the chosen partition number, and `PARTITION_LOCATION_AND_FLAGS`, to get the address of a single partition

```
uint32_t partition_info[3];
get_partition_table_info(partition_info, 3, PT_INFO_PARTITION_LOCATION_AND_FLAGS
    | PT_INFO_SINGLE_PARTITION | (boot_partition << 24));
uint16_t first_sector_number = (partition_info[1]
    & PICOBIN_PARTITION_LOCATION_FIRST_SECTOR_BITS)
    >> PICOBIN_PARTITION_LOCATION_FIRST_SECTOR_LSB;
uint16_t last_sector_number = (partition_info[1]
    & PICOBIN_PARTITION_LOCATION_LAST_SECTOR_BITS)
    >> PICOBIN_PARTITION_LOCATION_LAST_SECTOR_LSB;
uint32_t data_start_addr = first_sector_number * 0x1000;
uint32_t data_end_addr = (last_sector_number + 1) * 0x1000;
uint32_t data_size = data_end_addr - data_start_addr;
```

- get_sys_info() with `BOOT_INFO`, to get the `flash_update_boot_window_base` if any:

```
 uint32_t sys_info[5];
 get_sys_info(sys_info, 5*4, SYS_INFO_BOOT_INFO);
 uint32_t flash_update_boot_window_base = sys_info[3];
```

- pick_ab_partition() to pick the boot partition between A/B partitions if desired:

```
uint8_t boot_partition = pick_ab_partition(workarea, 0xC00, 0,
flash_update_boot_window_base);
```

- or get_b_partition() to find the other partition directly.
- chain_image() with `data_start_addr` and `data_size`, to boot a chosen image:

```
// note a negative 3rd parameter indicates to chain_image that the image is being chanined as
// part of a "flash update" boot, so TBYB and/or version downgrade may be in play
chain_image( workarea,
             0xc00,
            (XIP_BASE + data_start_addr) * (info.boot_type == BOOT_TYPE_FLASH_UPDATE ? -1 :
1),
            data_size
);
```

NOTE

The `workarea` used must not overlap the image being chained into, so beware SRAM or packaged binaries. If the binary overlaps the `workarea`, the results are undefined, but hardly likely to be good.

5.10.7. OTP Bootloader

This is similar to the custom bootloader scenario, but it will be stored in the OTP and will run in SRAM.

One possible use case could place decryption code into OTP which decrypts an executable image from a flash partition into RAM.

The entire bootloader will need to fit in the OTP rows from `0x0C0` to `0xF48` to avoid interfering with other reserved OTP functionality, giving a maximum size of 7440 bytes (2 bytes per ECC row). If some boot keys and OTP keys are unused, this region can extend slightly on either end.

The OTP bootloader itself should be stored in ECC format, starting from the row set in OTPBOOT_SRC with size set in OTPBOOT_LEN. When booting, it will be loaded into the address specified in OTPBOOT_DST0 and OTPBOOT_DST1, which must be in the main SRAM. The bootloader must fulfil the same criteria as a standard image: it must include an `IMAGE_DEF`, which must be signed if secure boot is enabled.

Once the OTP bootloader has been written to OTP, and the OTPBOOT_SRC, OTPBOOT_LEN, OTPBOOT_DST0 and OTPBOOT_DST1 set, OTP booting can be enabled by setting BOOT_FLAGS0.ENABLE_OTP_BOOT. If the OTP image fails the bootrom's launch checks, then, by default, boot continues along the normal flash boot path. You can prevent this by setting BOOT_FLAGS0.DISABLE_FLASH_BOOT.

WARNING

Take extreme care when writing an OTP bootloader. Once the ECC rows are written, they *cannot* be modified.

5.10.8. Rollback Versions And Bootloaders

WARNING

Ignoring the advice in this section could render your device unable to boot.

For bootloaders that need to chain into executable images with rollback versions on a secured RP2350, you *must* use separate OTP rows for:

- the bootloader rollback version
- the chained executable image's rollback version

Otherwise, bumping the version of the chained executable image renders the OTP bootloader and your device unable to boot.

You must also make sure that *both* the bootloader and the executable image have non-zero rollback versions, as the OTP flags relating to requiring rollback versions are global. Failure to do so will render your device unable to boot.

We recommend using the DEFAULT_BOOT_VERSION0 and DEFAULT_BOOT_VERSION1 rows for the binary's rollback version, and selecting some other unused rows in the OTP for the bootloader's rollback version.